From Linear Models to Neural Networks

Linear & logistic regression, neurons, MLPs, universal approximation

Prof. Nipun Batra

ES 667 · Deep Learning · IIT Gandhinagar

Lecture 1 chose the loss; Lecture 2 expands the prediction rule

Keep from Lecture 1

Regression output $\to$ Gaussian model $\to$ MSE

Class probabilities $\to$ Bernoulli / categorical model $\to$ cross-entropy

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$$x \to f_{\theta}(x) \to \text{output parameters} \to \text{same Lecture 1 loss}$$

We will not re-derive the losses; today is about making f_θ more expressive.

Outline

1. **Linear prediction** — weighted sums, batches, and output shapes
2. **One neuron** — add an activation to create a nonlinear feature
3. **The limitation** — see exactly why a linear model cannot solve XOR
4. **An MLP** — compose learned features across width and depth
5. **Practice** — trace one forward pass, count parameters, and make one update

One notation covers every prediction rule

Q QUESTION

Write $f_\theta(x)$ for the model's prediction at input x.

x: the input features
θ: all trainable weights and biases
f_θ: the rule that maps input to a score
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$f_{\boldsymbol{\theta}}$: the rule that maps input to a score or prediction

Starting point — one weighted sum plus a bias:

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Starting point — one weighted sum plus a bias:

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We will build the network gradually; the full nested formula comes only after each part is concrete.

The linear starting point

Start with one weighted sum

For one output, every input node sends a weighted value into one summation node.

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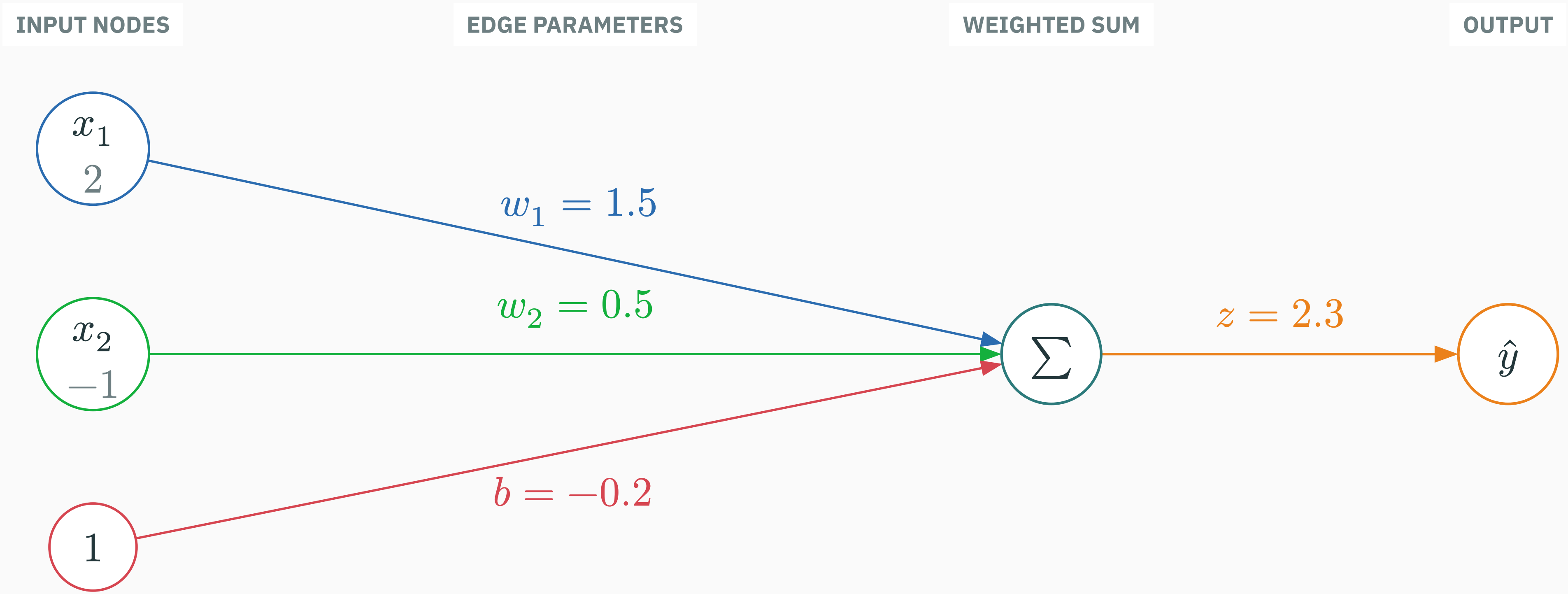

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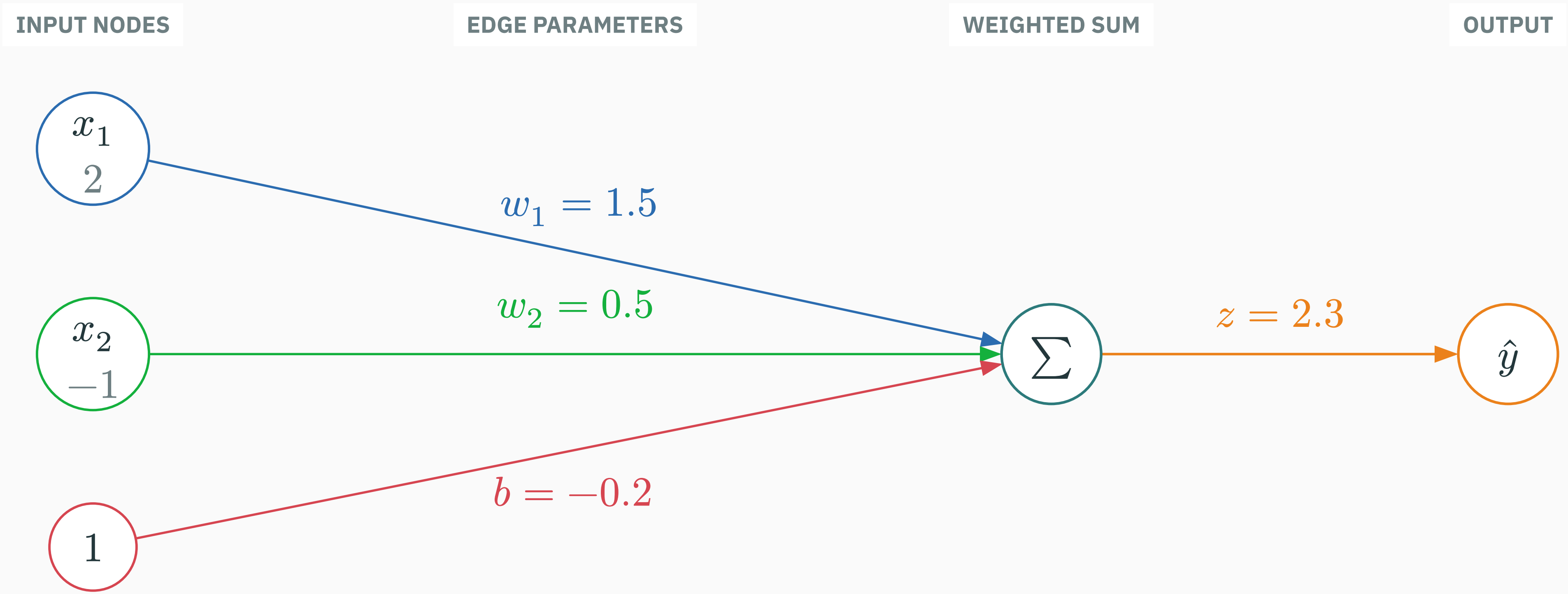

The constant node 1 turns the bias into one more incoming edge: $b \cdot 1$.

Calculate every contribution entering the node

D DERIVATION

BLUE EDGE

$$w_1 x_1 = 1.5(2) = 3.0$$

→

GREEN EDGE

$$w_2 x_2 = 0.5(-1) = -0.5$$

→

BIAS EDGE

$$b \cdot 1 = -0.2(1) = -0.2$$

→

SUM AT NODE

$$3.0 - 0.5 - 0.2 = 2.3$$

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$$\hat{y} = \underbrace{1.5(2)}_{3.0} + \underbrace{0.5(-1)}_{-0.5} + \underbrace{(-0.2)(1)}_{-0.2} = 2.3.$$

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Each edge contributes “weight × incoming value”; the node adds all contributions.

A batch is a labelled data matrix

V VISUAL

example	feature x_1	feature x_2	target y
1	2	-1	2.4
2	0	3	1.0
3	-1	1	-1.1

DATA MATRIX X

$$\boldsymbol{X} = \begin{pmatrix} 2 & -1 \\ 0 & 3 \\ -1 & 1 \end{pmatrix}$$

3 rows $=$ examples

2 columns $=$ features

TARGET VECTOR Y

$$\boldsymbol{y} = \begin{pmatrix} 2.4 \\ 1.0 \\ -1.1 \end{pmatrix}$$

one observed target per example

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Here $n = 3$ examples and $d = 2$ input features, so $\boldsymbol{X} \in \mathbb{R}^{3\times 2}$ and $\boldsymbol{y} \in \mathbb{R}^3$.

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one observed target per example

Row i contains the input $x^{(i)}$ and its observed answer $y^{(i)}$.

One matrix multiplication predicts all three rows

D DERIVATION

Reuse $\boldsymbol{w} = \begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix}$ and $b = -0.2$ for every example.

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Reuse $\boldsymbol{w} = \begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix}$ and $b = -0.2$ for every example.

$$\boldsymbol{X}\boldsymbol{w} + b\mathbf{1} = \begin{pmatrix} 2 & -1 \\ 0 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1.5 \\ 0.5 \end{pmatrix} + (-0.2) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

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ROW 1

$$1.5(2) + 0.5(-1) - 0.2 = 2.3$$

ROW 2

$$1.5(0) + 0.5(3) - 0.2 = 1.3$$

ROW 3

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$\hat{y}$ contains predictions; y contains observed targets. They are not the same object.

The bias node becomes a column of ones

V VISUAL

AUGMENT THE DATA MATRIX

$$\tilde{X} = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 3 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

STACK THE PARAMETERS

$$\boldsymbol{\theta} = \begin{pmatrix} w_1 \\ w_2 \\ b \end{pmatrix} = \begin{pmatrix} 1.5 \\ 0.5 \\ -0.2 \end{pmatrix}$$

$\tilde{X}$

original feature columns
$+$ one constant column

$\boldsymbol{\theta}$

all weights
$+$ the bias

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$$\hat{\boldsymbol{y}} = \tilde{\boldsymbol{X}}\boldsymbol{\theta}, \quad (n \times (d+1))((d+1) \times 1) \to (n \times 1).$$

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The column of ones is the batch version of the constant bias node on the previous network diagram.

For regression, one weighted sum gives one flat trend

V VISUAL

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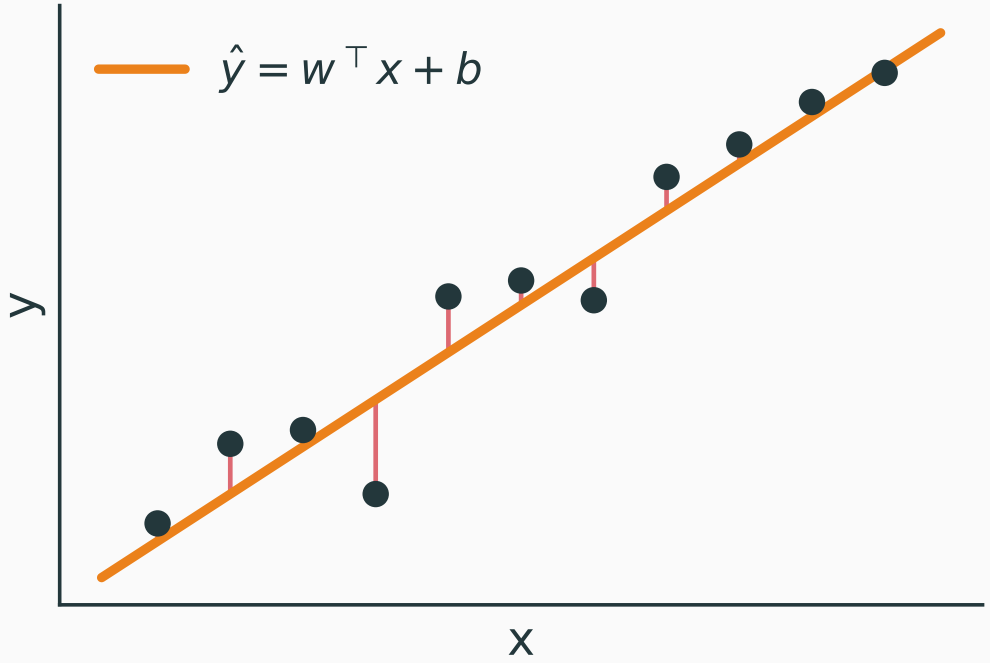

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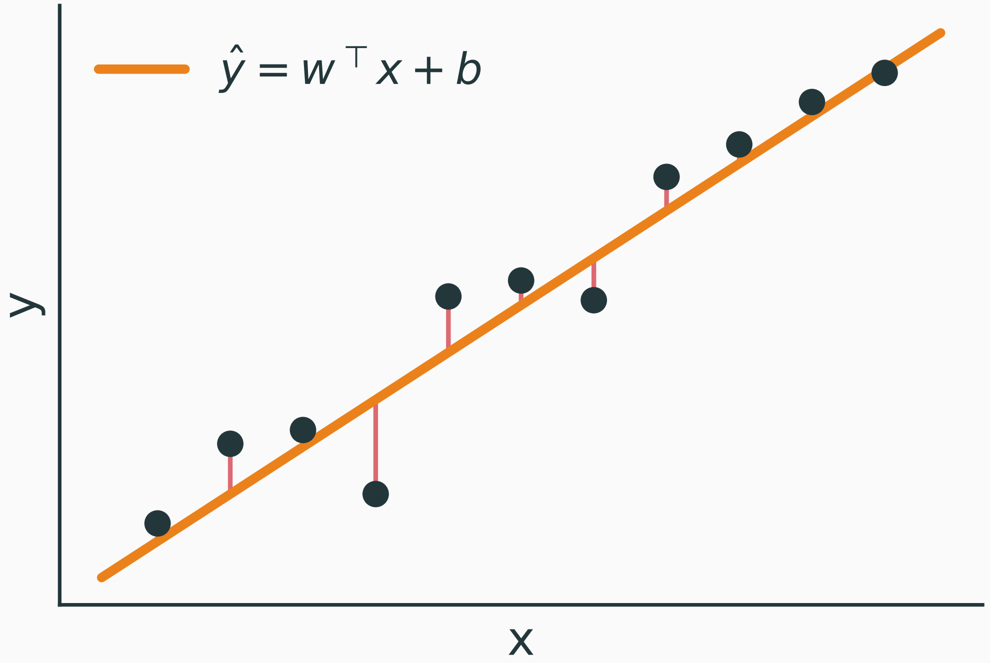

The model can tilt and shift a line or plane, but it cannot bend to follow nonlinear structure.

From a linear score to a neuron

A neuron has two stages

First compute a weighted sum plus bias; then pass that value through an activation.

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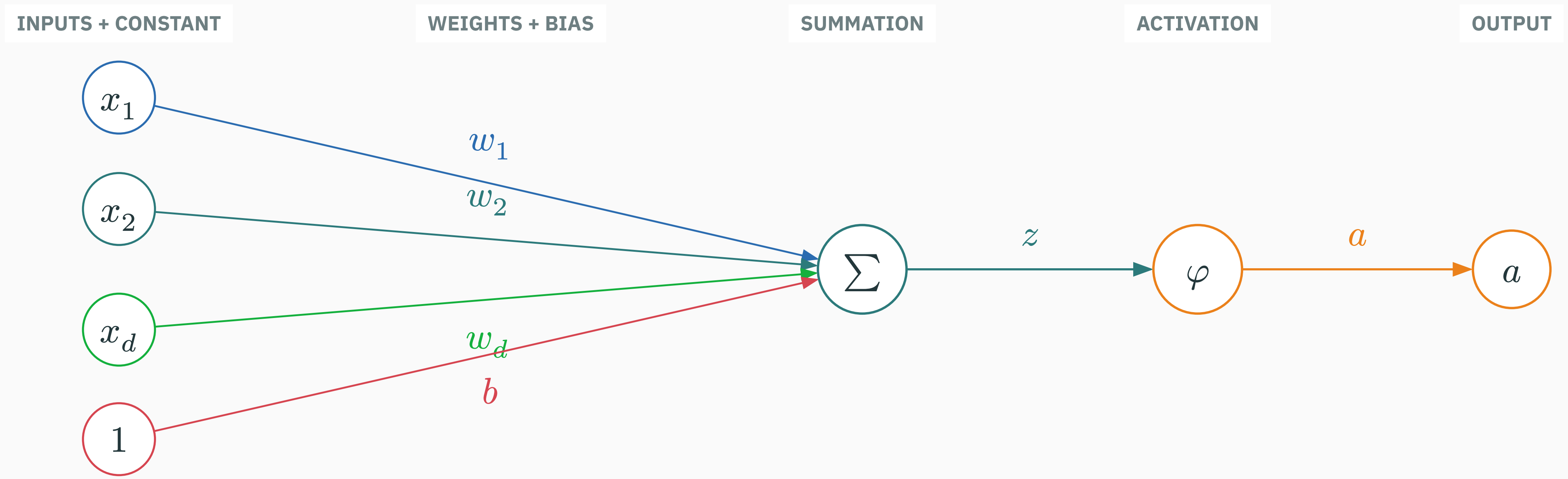

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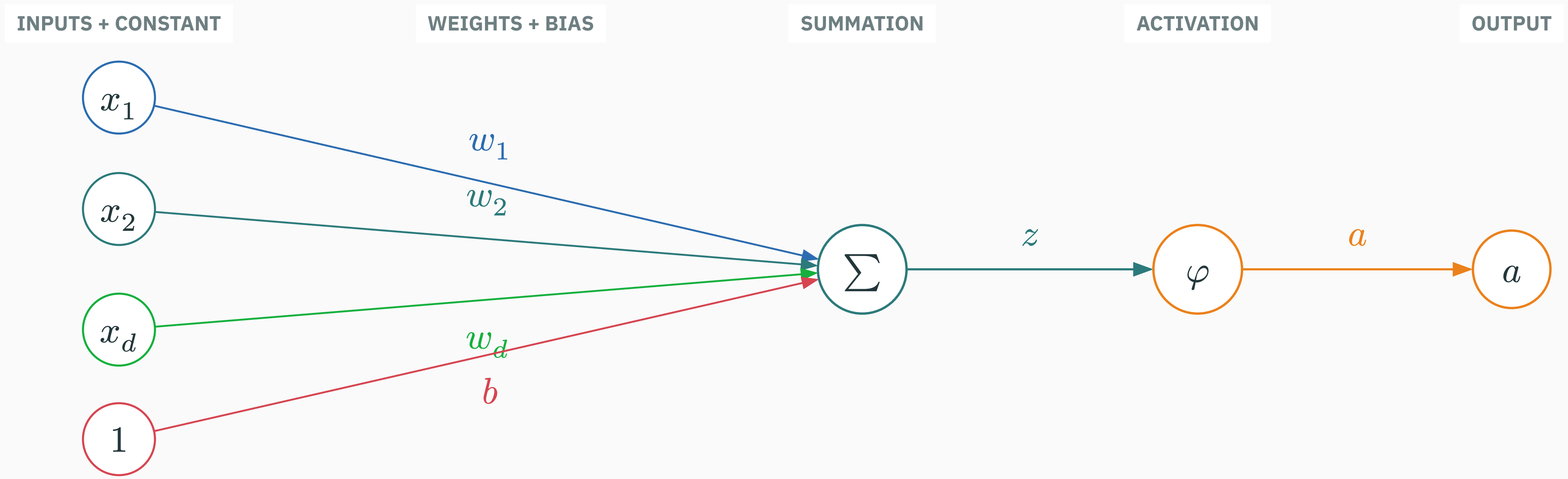

Weights and bias create z at the $\sum$ node; the separate φ node reshapes z into a.

Trace one concrete neuron through the two stages

D DERIVATION

INPUT

$$x = (2, -1)$$

$\rightarrow$

STAGE 1 · SUMMATION

$$w = (1, -1), \quad b = 0$$
$$z = 1(2) + (-1)(-1) + 0 = 3$$

$\rightarrow$

STAGE 2 · ACTIVATION

$$a = \mathrm{ReLU}(3) = 3$$

Trace one concrete neuron through the two stages

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Changing w or b changes which input patterns make z large.

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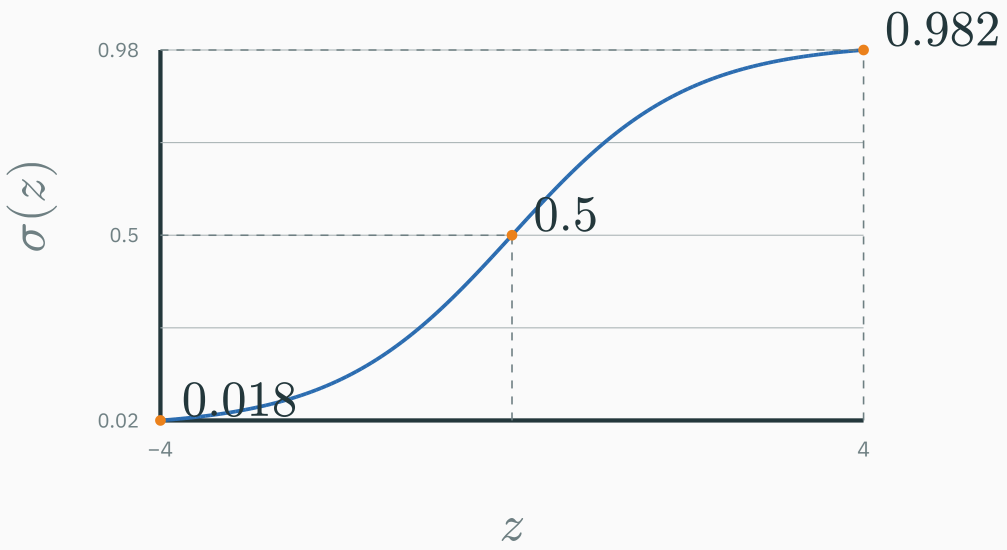
Topology tells us where values flow; this calculation tells us the number carried along each stage.

Bounded activations behave like smooth switches

V VISUAL

$$\sigma(z) = \frac{1}{1+\exp(-z)}$$

sigmoid: off $\longrightarrow$ on



Bounded activations behave like smooth switches

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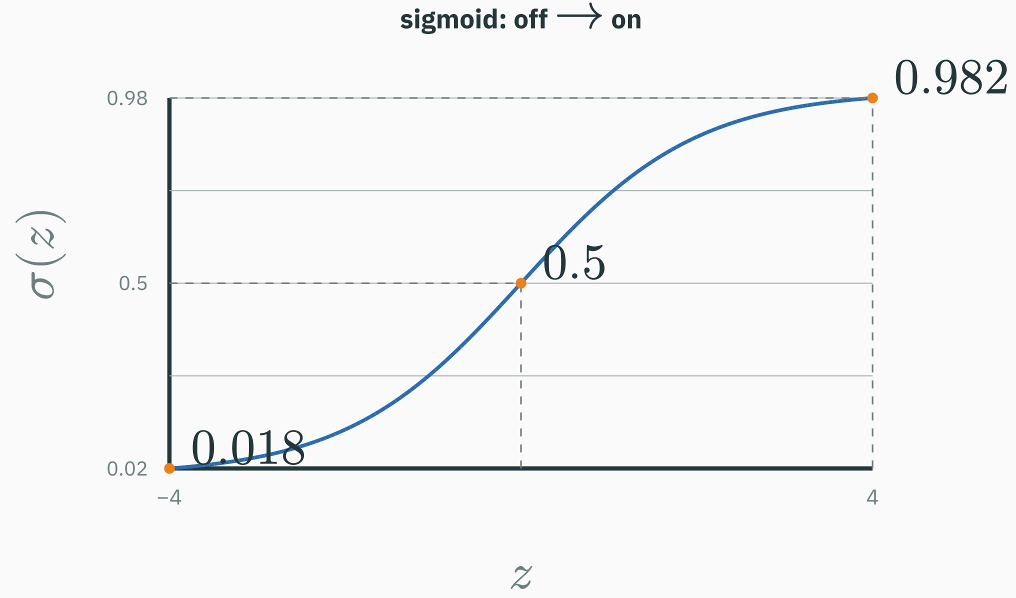
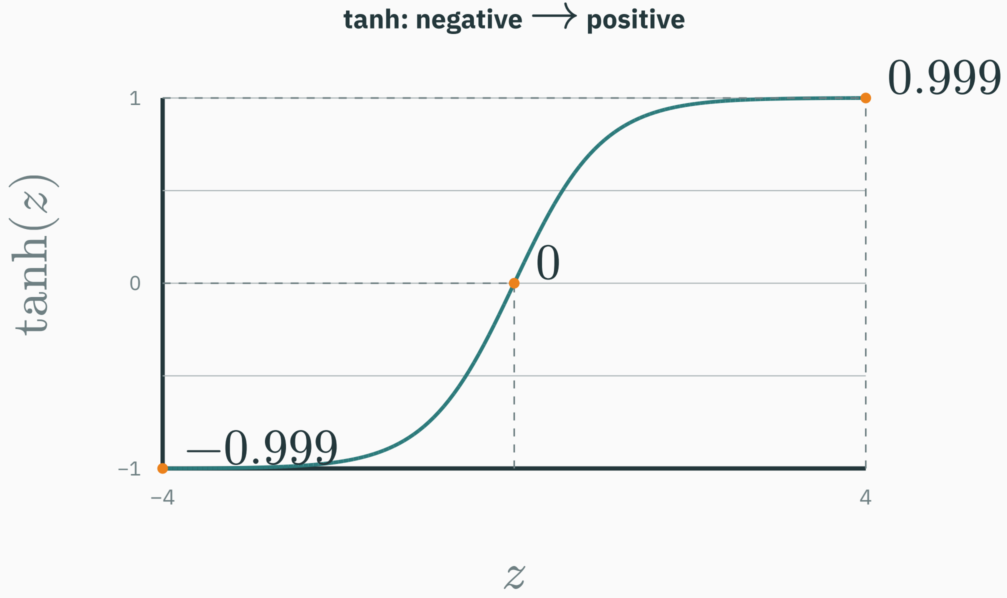
$$\sigma(z) = \frac{1}{1+\exp(-z)}$$

$$\tanh(z) = \frac{\exp(z)-\exp(-z)}{\exp(z)+\exp(-z)}$$





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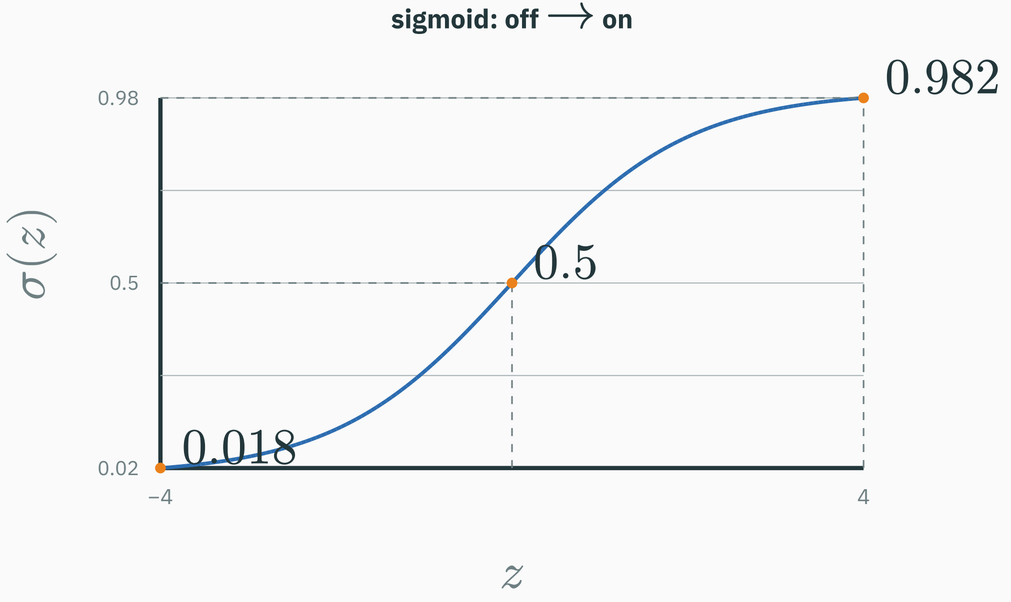
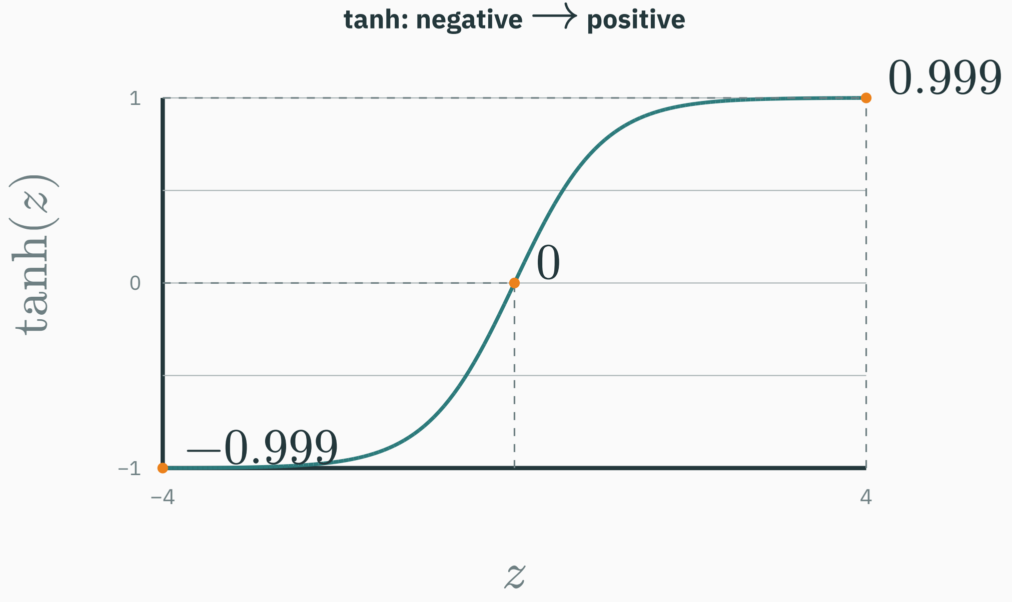


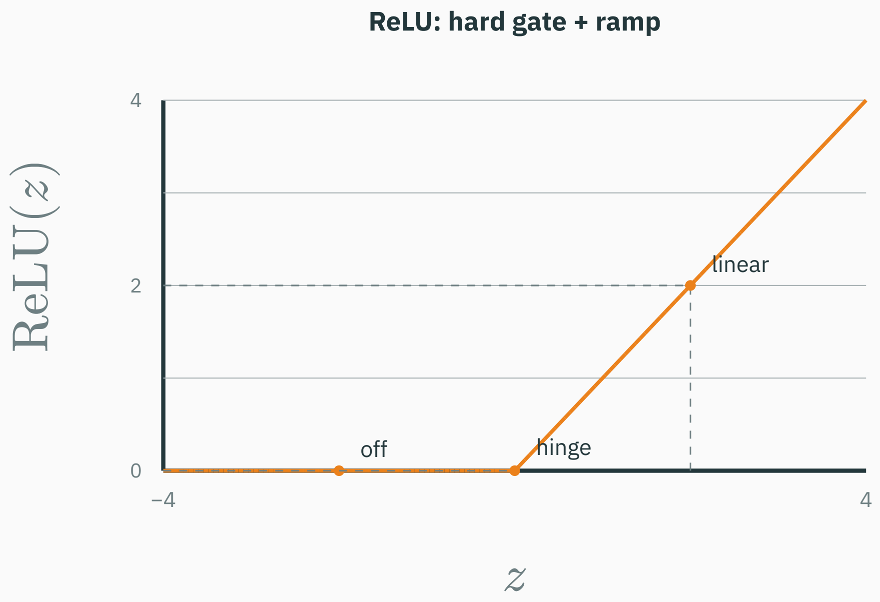


Sigmoid answers “how on?”; tanh answers “negative or positive, and how strongly?”

ReLU and GELU behave like ramps

V VISUAL

$$\mathrm{ReLU}(z) = \max(0, z)$$



ReLU and GELU behave like ramps

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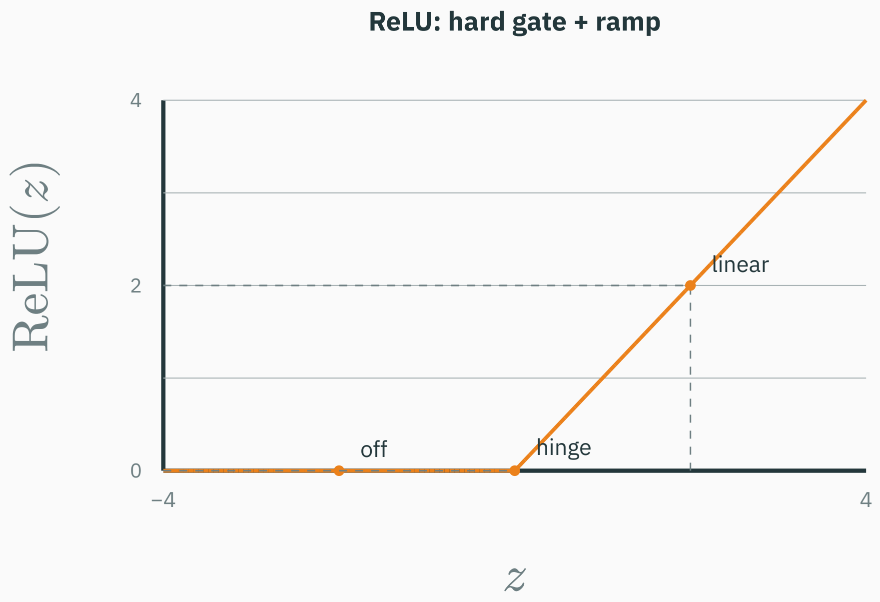
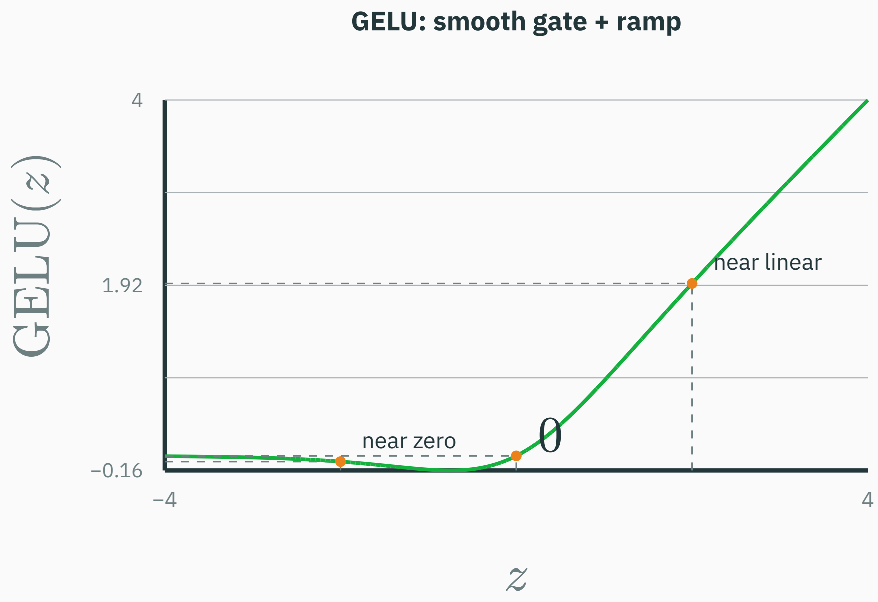
$$\mathrm{GELU}(z) \approx z\sigma(1.702z)$$





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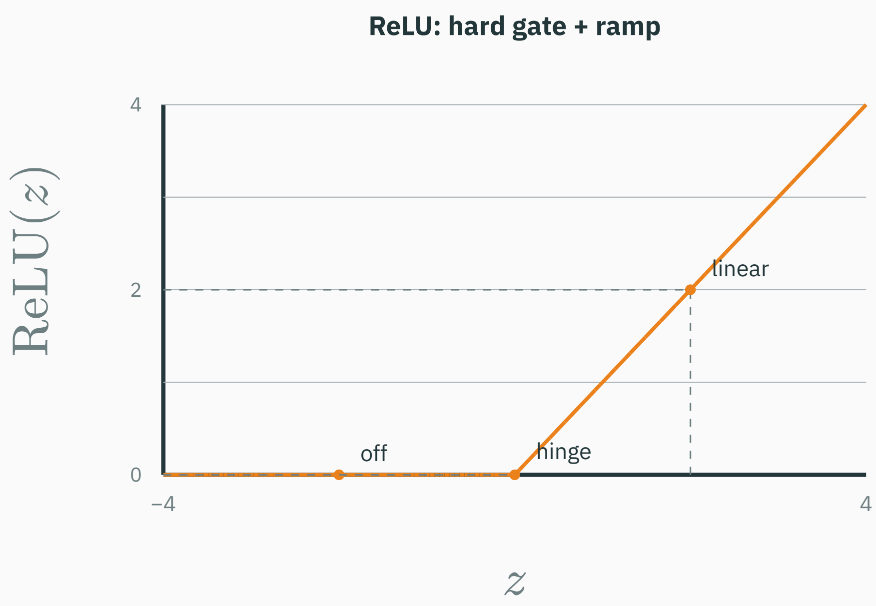
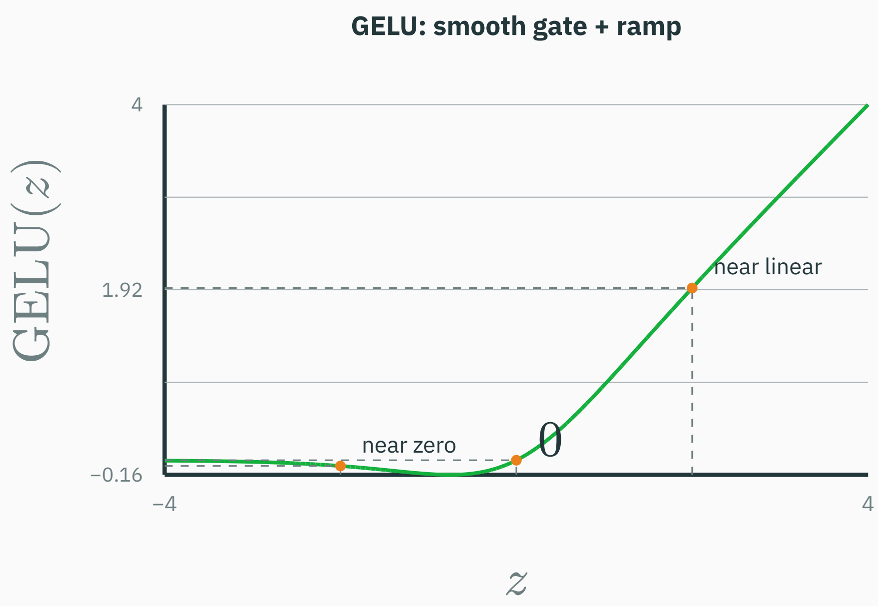
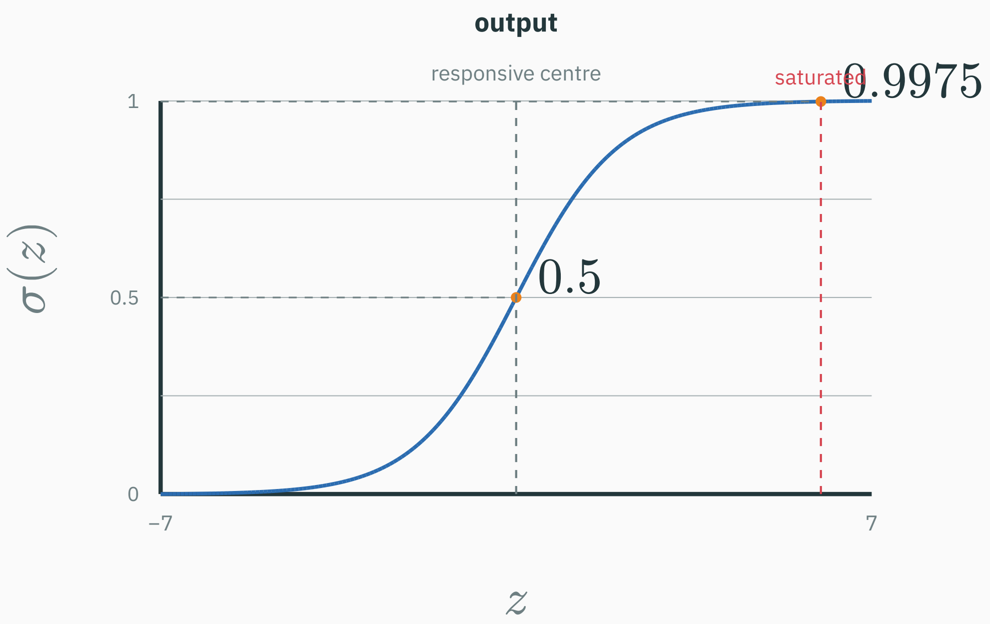




Both build ramp-like hidden features; ReLU has a sharp hinge, while GELU eases through the gate.

Saturation means the curve has become locally flat

VISUAL



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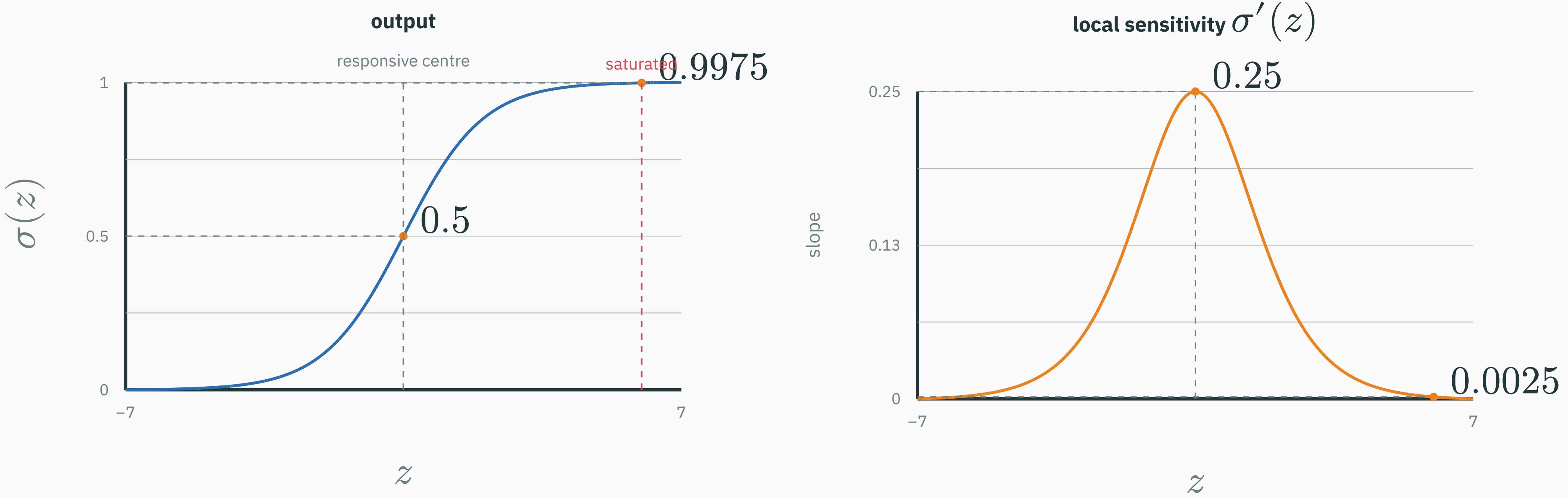

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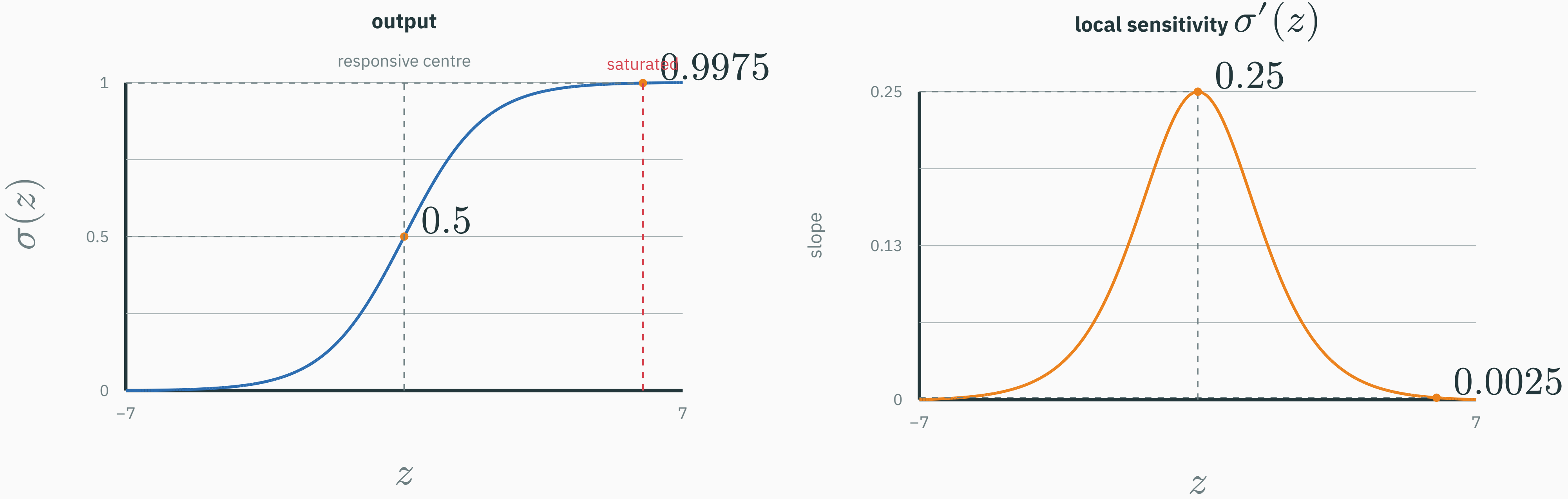

Near $z = 0$, the curve is steep; near $z = 6$, it is almost horizontal.

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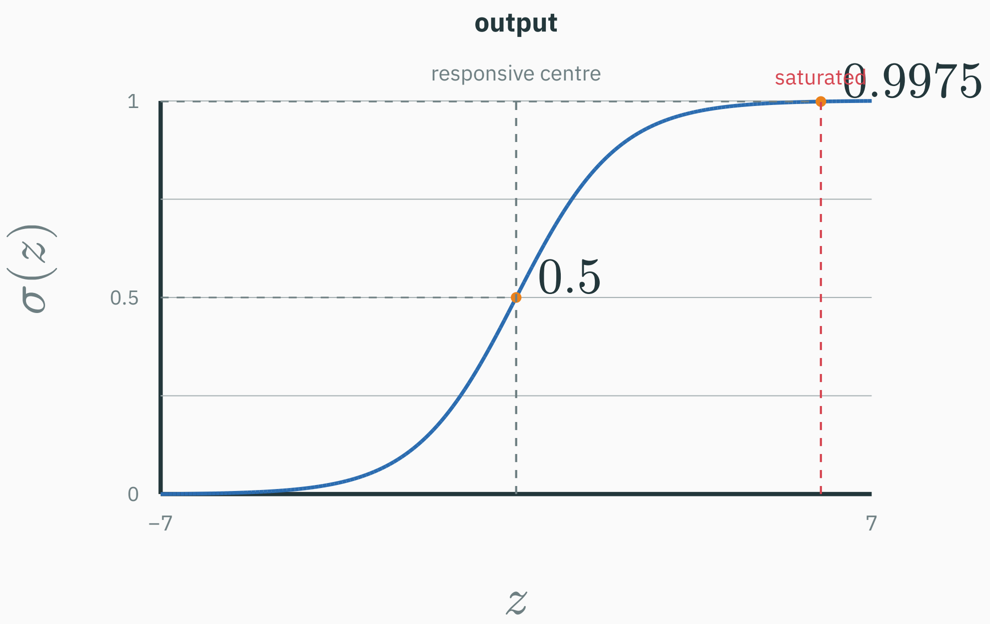

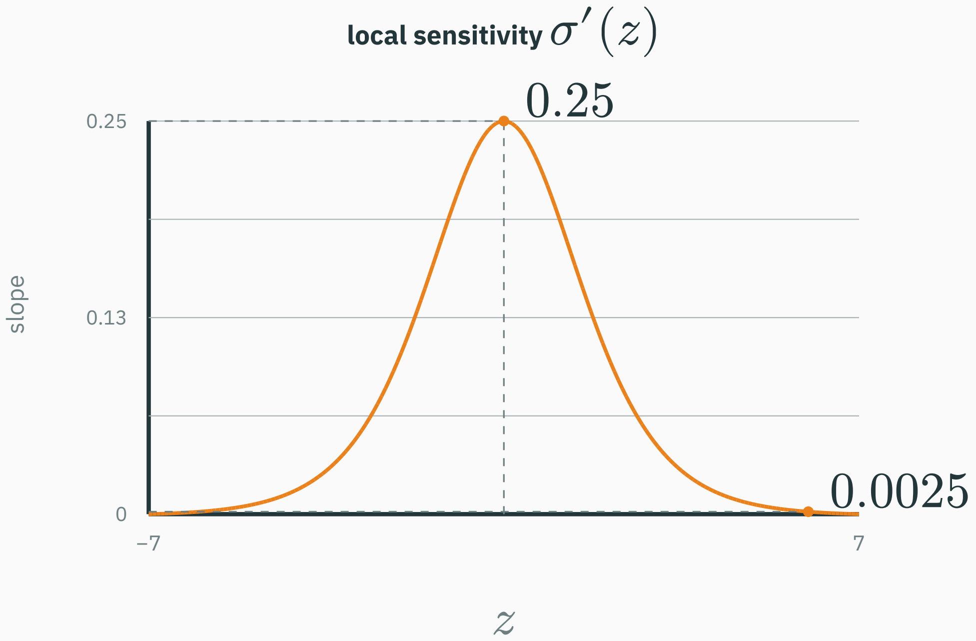

Near $z = 0$, the curve is steep; near $z = 6$, it is almost horizontal.

A saturated unit still outputs a number, but changing its input barely changes that output.

Calculate the effect of saturation

D DERIVATION

CENTRE: Z = 0

$$\sigma'(0) = 0.25$$

for $\Delta z = 0.1$:

$$\Delta h \approx 0.25(0.1) = 0.025$$

same Δz $\rightarrow$

SATURATED: Z = 6

$$\sigma'(6) \approx 0.0025$$

for $\Delta z = 0.1$:

$$\Delta h \approx 0.0025(0.1) = 0.00025$$

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$\frac{0.025}{0.00025} = 100 \rightarrow$ **the saturated unit responds 100× less.**

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For a positive ReLU input, $\mathrm{ReLU}'(z) = 1$, so the same $\Delta z = 0.1$ gives $\Delta h \approx 0.1$.

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For a positive ReLU input, $\mathrm{ReLU}'(z) = 1$, so the same $\Delta z = 0.1$ gives $\Delta h \approx 0.1$.

The derivative is a local responsiveness meter; Lecture 3 will connect it to the learning signal.

Question: do more linear layers create new shapes?

Q QUESTION

Remove every activation. What can a stack of “weighted sum + bias” layers represent?

A. One equivalent weighted sum + bias

B. Any curved decision boundary

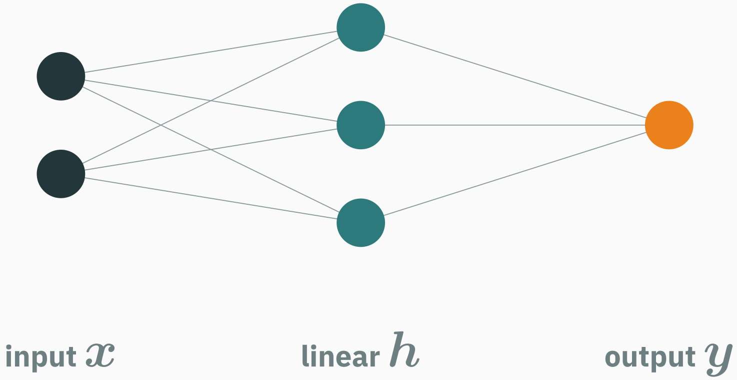
C. Only a constant function

D. A probability distribution automatically

Answer in nodes: two linear layers collapse into one

A ANSWER

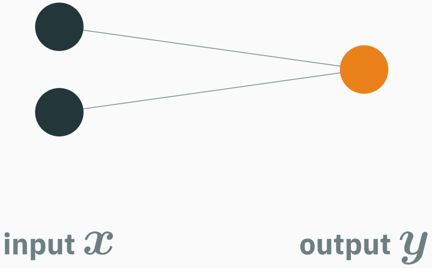
TWO LINEAR LAYERS



$$h = W_1 x + b_1$$
$$y = W_2 h + b_2$$

$\equiv$

ONE EQUIVALENT LAYER



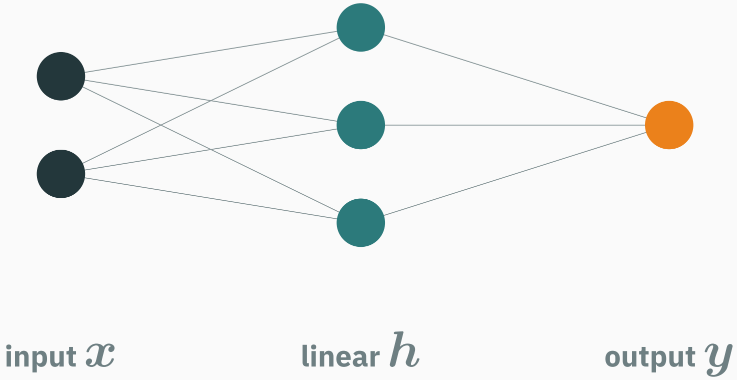
$$y = W_{\text{eq}} x + b_{\text{eq}}$$

no hidden representation is needed

Answer in nodes: two linear layers collapse into one

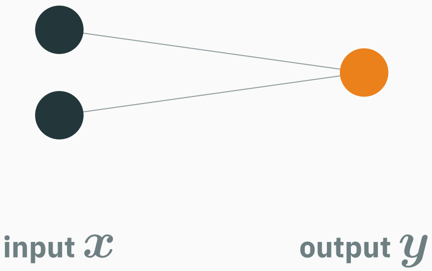
A ANSWER

TWO LINEAR LAYERS



$\equiv$

ONE EQUIVALENT LAYER



$$\boldsymbol{h} = \boldsymbol{W}_1 \boldsymbol{x} + \boldsymbol{b}_1$$
$$\boldsymbol{y} = \boldsymbol{W}_2 \boldsymbol{h} + \boldsymbol{b}_2$$

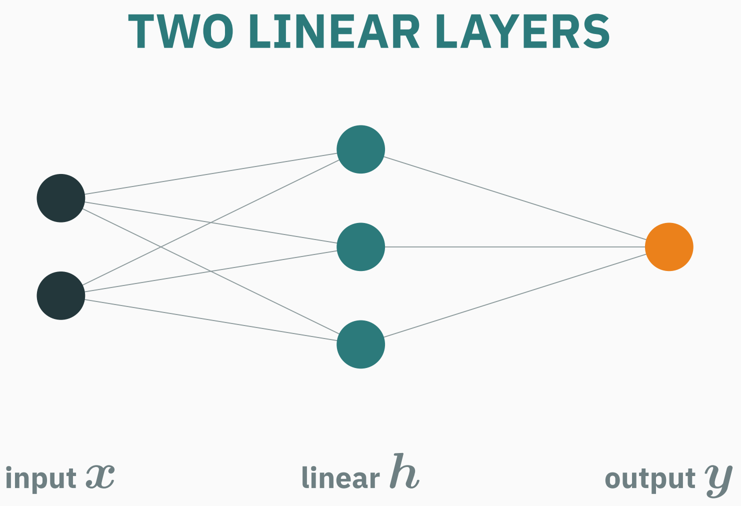
$$\boldsymbol{y} = \boldsymbol{W}_{\text{eq}} \boldsymbol{x} + \boldsymbol{b}_{\text{eq}}$$

no hidden representation is needed

$$\boldsymbol{W}_{\text{eq}} = \boldsymbol{W}_2 \boldsymbol{W}_1, \quad \boldsymbol{b}_{\text{eq}} = \boldsymbol{W}_2 \boldsymbol{b}_1 + \boldsymbol{b}_2$$

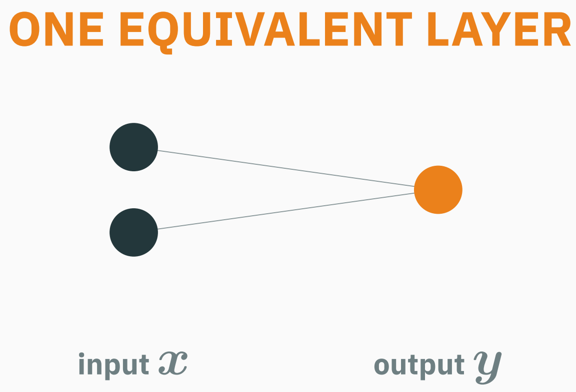
Answer in nodes: two linear layers collapse into one

A ANSWER



$$h = W_1 x + b_1$$
$$y = W_2 h + b_2$$

$\equiv$



$$y = W_{\text{eq}} x + b_{\text{eq}}$$

no hidden representation is needed

$$W_{\text{eq}} = W_2 W_1, \quad b_{\text{eq}} = W_2 b_1 + b_2$$

Without a φ node, the middle column adds parameters but no new bend or feature shape.

Interactive: activation functions

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/dl-teaching/interactives/activation-explorer.html

Compare activations and their **derivatives**, and see where gradients vanish.
Controls: activation type · input z · show derivative · compare saturation.

Interactive: activation functions

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↗ nipunbatra.github.io/dl-teaching/interactives/activation-explorer.html

Compare activations and their **derivatives**, and see where gradients vanish.
Controls: activation type · input z · show derivative · compare saturation.

Explore large positive and negative logits, where sigmoid's derivative becomes tiny.

Where a single weighted sum fails

A binary classifier draws one straight boundary

LINEAR SCORE

$$z = \boldsymbol{w}^\top \boldsymbol{x} + b$$

THRESHOLD

$$\hat{y} = 1 \text{ if } z \geq 0$$
$$\hat{y} = 0 \text{ if } z < 0$$

The boundary is where $z = 0$.

A binary classifier draws one straight boundary

V VISUAL

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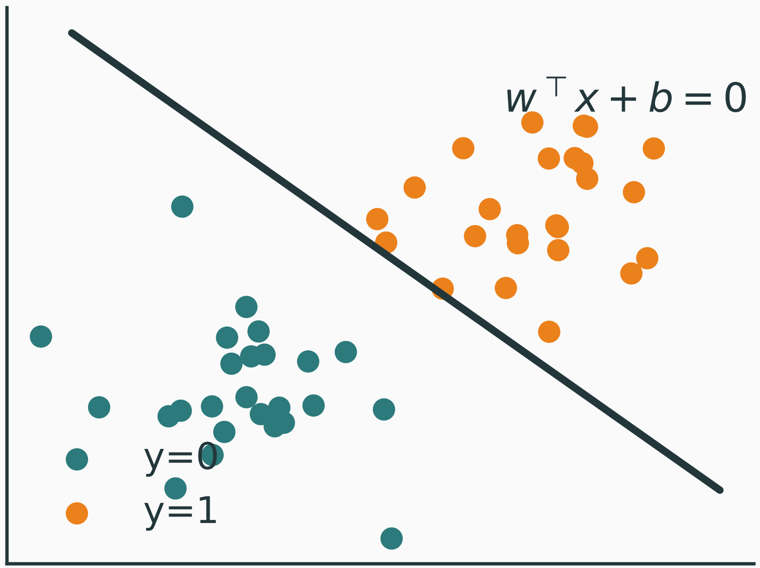

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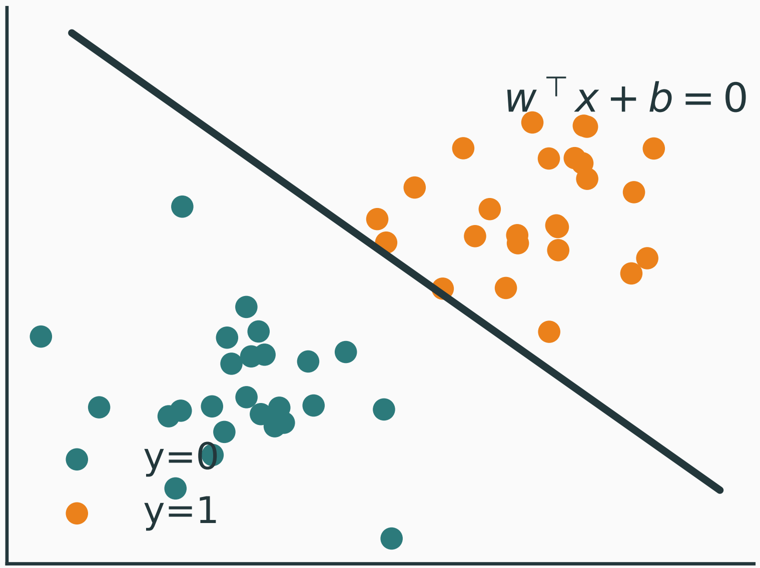

$w^\top x + b = 0$ is one line in 2-D, so binary linear classification works only when one line can separate the classes.

XOR is the smallest pattern that defeats one line

V VISUAL

CLASS A

$(0, 0)$ and $(1, 1)$

CLASS B

$(1, 0)$ and $(0, 1)$

Every proposed line leaves one same-class corner on the wrong side.

XOR is the smallest pattern that defeats one line

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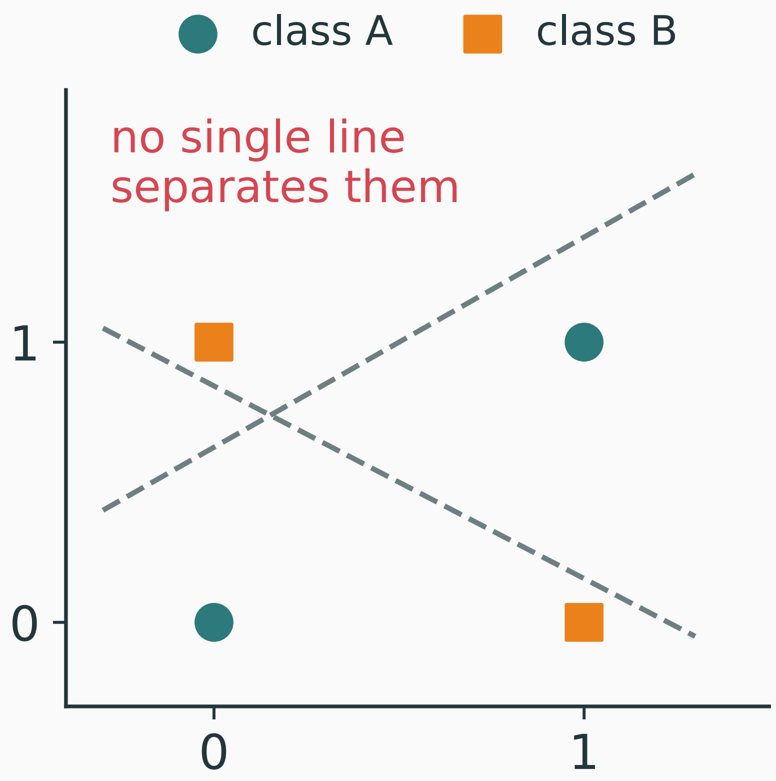

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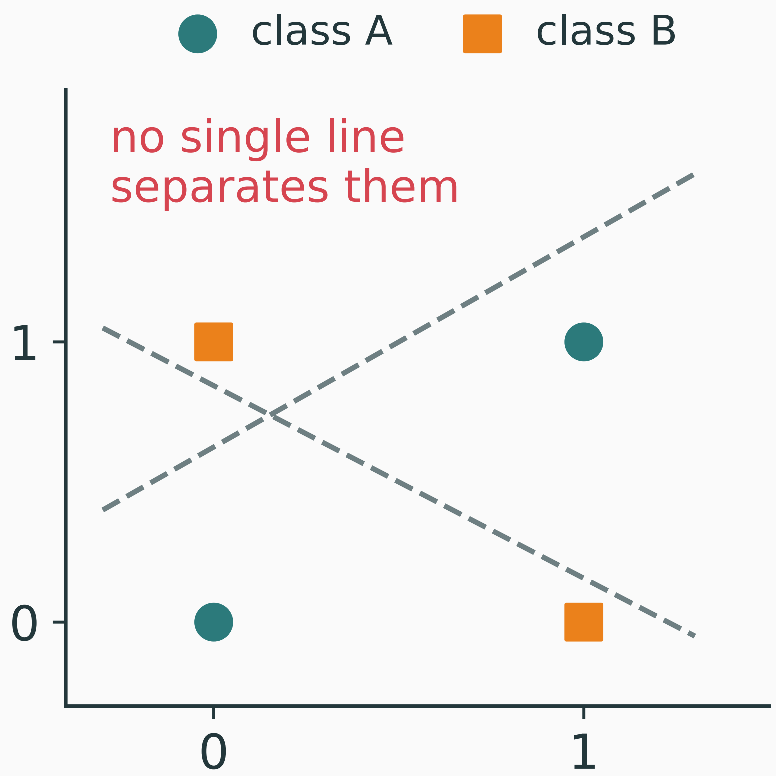

XOR requires a nonlinear feature transformation before a final straight separator can work.

Return to multi-class: several scores create straight-edged regions

V VISUAL

For each class k, compute one linear score $z_{k(\boldsymbol{x})} = \boldsymbol{w}_k^\top \boldsymbol{x} + b_k$.

Return to multi-class: several scores create straight-edged regions

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For each class k, compute one linear score $z_{k(\boldsymbol{x})} = \boldsymbol{w}_k^\top \boldsymbol{x} + b_k$. Predict the largest score: $\hat{y}(\boldsymbol{x}) = \arg\max_k z_{k(\boldsymbol{x})}$.

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Each pair of classes can exchange the lead only where their two scores tie.

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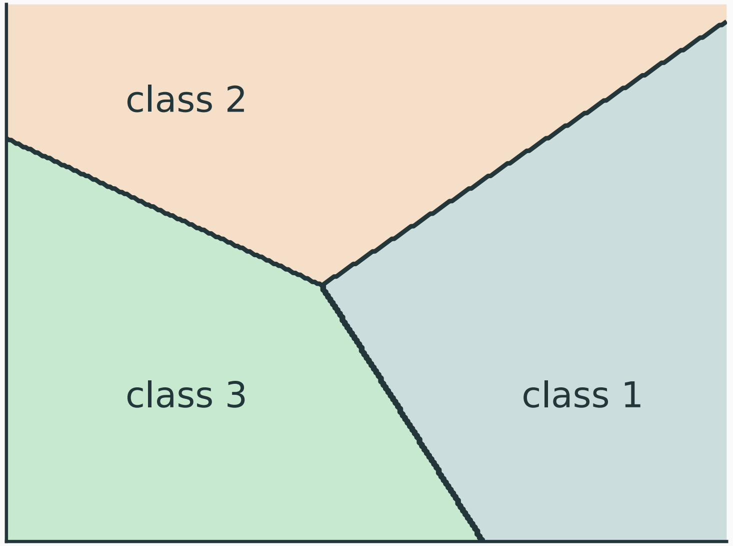
linear (polyhedral) regions



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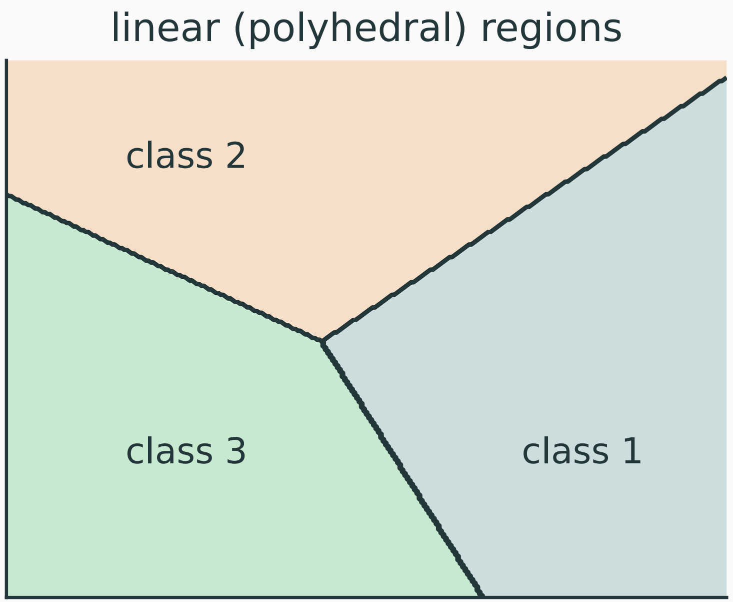
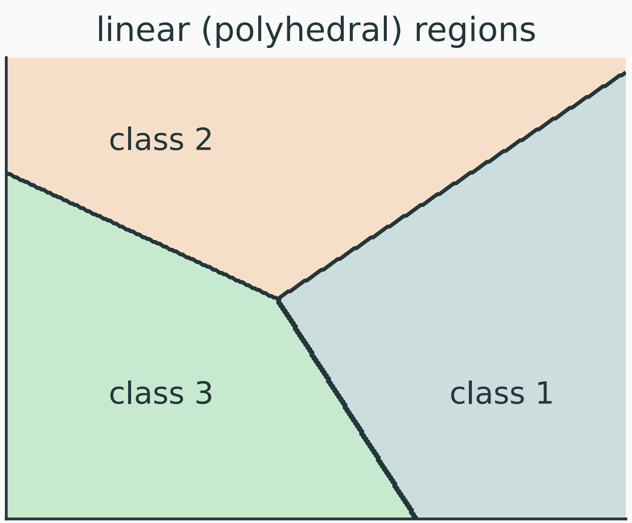


Multi-class regions can form polygons, but every edge still comes from equality between two linear scores.

Question: why are the multi-class boundaries straight?

Q QUESTION



three linear scores divide the input plane

The classifier predicts

$$\hat{y}(\boldsymbol{x}) = \arg\max_k p_{k(\boldsymbol{x})}.$$

When is a point on the boundary between predicted classes i and j?

A. Whenever $p_i = p_j$
B. When $p_i = p_j$ and both are maximal
C. Whenever $p_i + p_j = 1$
D. Only when all K probabilities are equal

Softmax preserves the winner

A ANSWER

SCORES

$$z_i = 2, \quad z_j = 1$$

$\rightarrow$

SOFTMAX PROBABILITIES

$$p_i \approx 0.731, \quad p_j \approx 0.269$$

Softmax preserves the winner

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$$\frac{p_i}{p_j} = \frac{\exp(z_i)}{\exp(z_j)} = \exp\left(z_i - z_j\right)$$

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EQUAL SCORES

$z_i = z_j \leftrightarrow p_i = p_j$

$\rightarrow$

LARGER SCORE

$z_i > z_j \leftrightarrow p_i > p_j$

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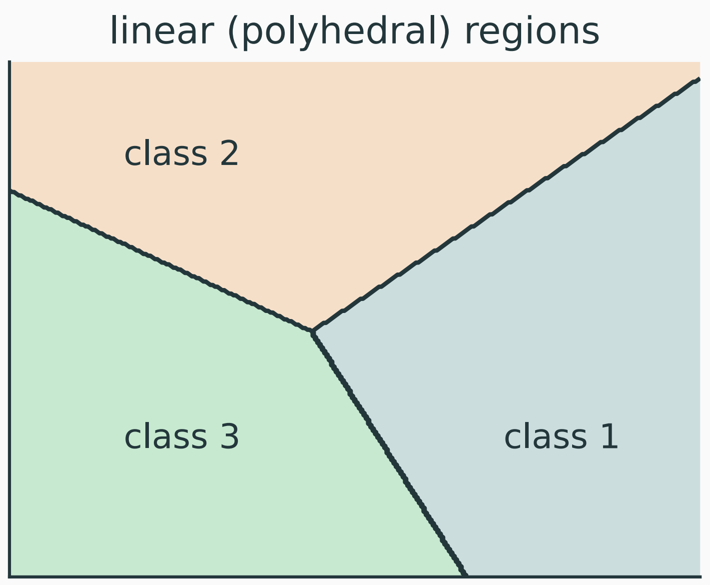
Softmax changes score scale, not score order: $\arg\max_k p_k = \arg\max_k z_k$**.**

Linear scores can tie only on a flat set

A ANSWER



Each coloured region is where one linear score is largest.

For class k, $z_{k(\boldsymbol{x})} = \boldsymbol{w}_k^\top \boldsymbol{x} + b_k$.

Linear scores can tie only on a flat set

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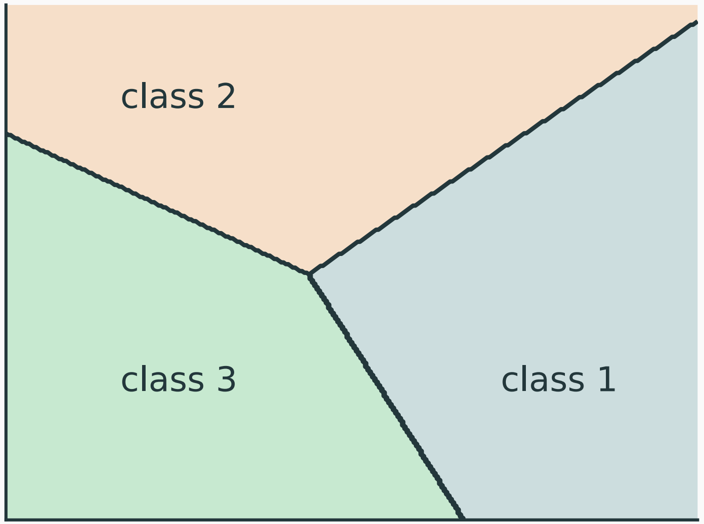

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For class k, $z_{k(\boldsymbol{x})} = \boldsymbol{w}_k^\top \boldsymbol{x} + b_k$. Two classes tie when $z_{i(\boldsymbol{x})} = z_{j(\boldsymbol{x})}$, so $\left(\boldsymbol{w}_i - \boldsymbol{w}_j\right)^\top \boldsymbol{x} + \left(b_i - b_j\right) = 0$.

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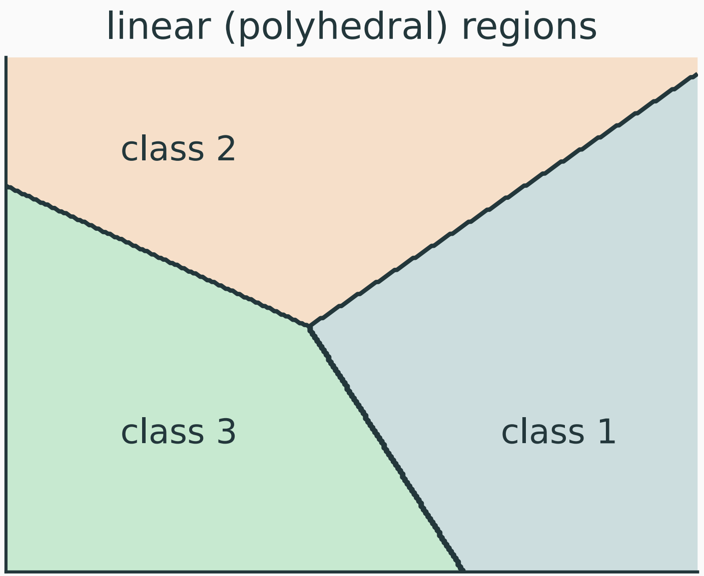

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This is a line in 2-D, a plane in 3-D, and a hyperplane in d dimensions.

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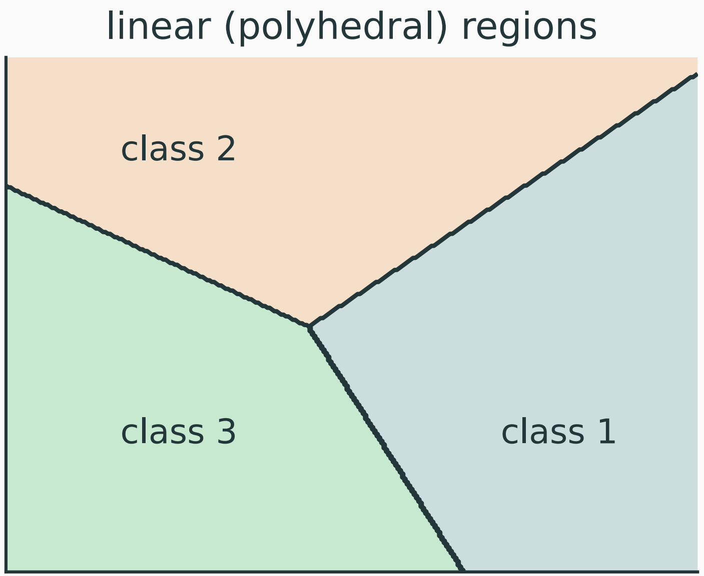

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This is a line in 2-D, a plane in 3-D, and a hyperplane in d dimensions.

Equal linear scores create a candidate straight boundary; softmax cannot bend it.

A concrete example: three linear scores compete

V VISUAL

CLASS 1 SCORE

$$z_1(x) = x$$

CLASS 2 SCORE

$$z_2(x) = -x$$

CLASS 3 SCORE

$$z_3(x) = 0.5$$

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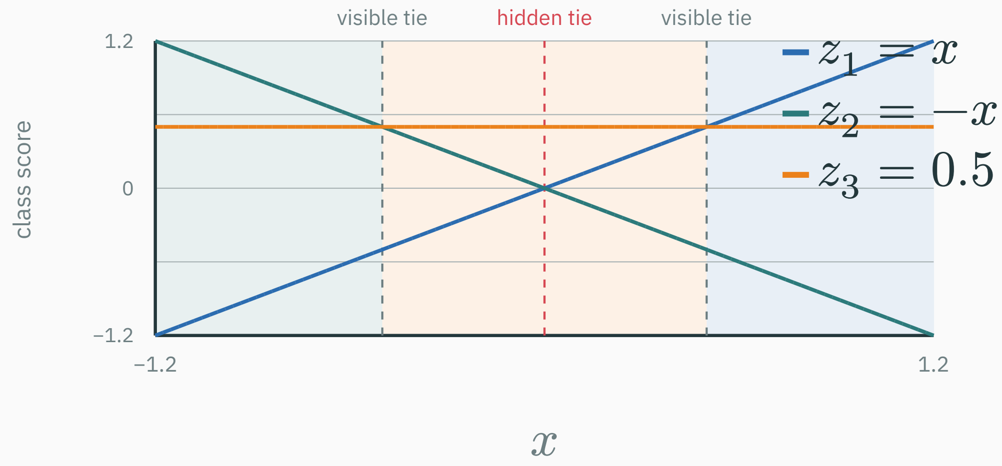

PREDICTED CLASS $= \arg\max_k z_{k(x)}$ · **boundaries at** $x = \pm 0.5$

2: $x < -.5$ | 3: $|x| < .5$ | 1: $x > .5$

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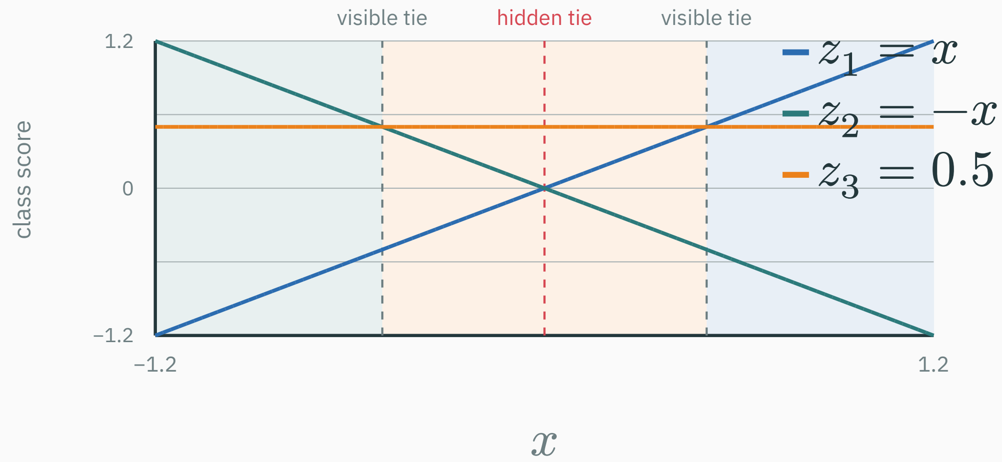

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2: $x < -.5$

3: $|x| < .5$

1: $x > .5$

The learned boundaries are $x = -0.5$ and $x = 0.5$; the class-1/class-2 tie at $x = 0$ remains hidden.

Evaluate five explicit inputs to see which ties are visible

D DERIVATION

Prediction $= \arg\max_k z_{k(x)}$; a boundary appears only when the tied scores are maximal.

x	$z_1 = x$	$z_2 = -x$	$z_3 = 0.5$	**what is visible?**
-1	-1	1	0.5	class 2 wins
-0.5	-0.5	0.5	0.5	**visible boundary 2/3**
0	0	0	0.5	**class 3 wins; 1/2 tie hidden**
0.5	0.5	-0.5	0.5	**visible boundary 1/3**
1	1	-1	0.5	class 1 wins

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0	0	0	0.5	**class 3 wins; 1/2 tie hidden**
0.5	0.5	-0.5	0.5	**visible boundary 1/3**
1	1	-1	0.5	class 1 wins

The candidate 1/2 tie occurs at $x = 0$, but the visible boundaries occur at $x = -0.5$ and $x = 0.5$.

Candidate boundary versus visible boundary

A ANSWER

CANDIDATE PAIRWISE TIE

$$\mathcal{T}_{ij} = \{\boldsymbol{x} : z_i = z_j\}$$

an entire hyperplane

keep only winners $\rightarrow$

VISIBLE CLASS BOUNDARY

$$\mathcal{B}_{ij} = \{\boldsymbol{x} : z_i = z_j \geq z_l \ \text{ for every } \ l\}$$

the top-scoring portion

Candidate boundary versus visible boundary

A ANSWER

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the top-scoring portion

In probability language: $p_i = p_j$ **and both are maximal**.

Candidate boundary versus visible boundary

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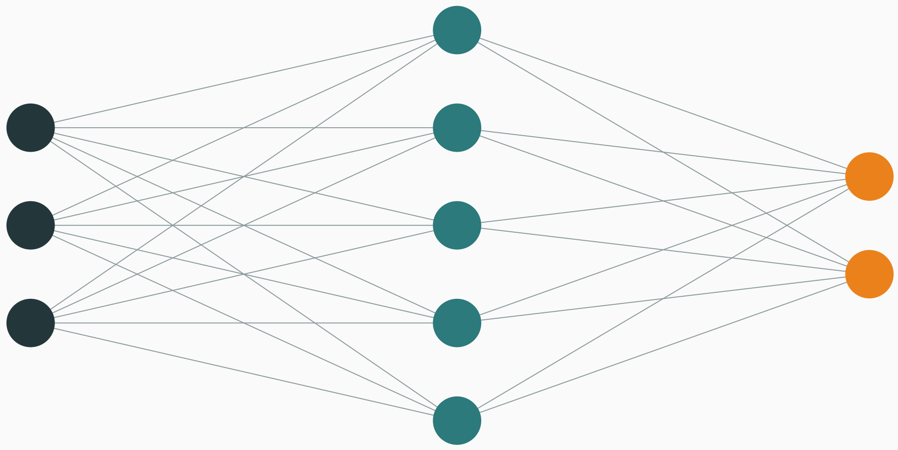
Correct: B. Multi-class linear classifiers have piecewise-straight visible boundaries.

The multilayer perceptron

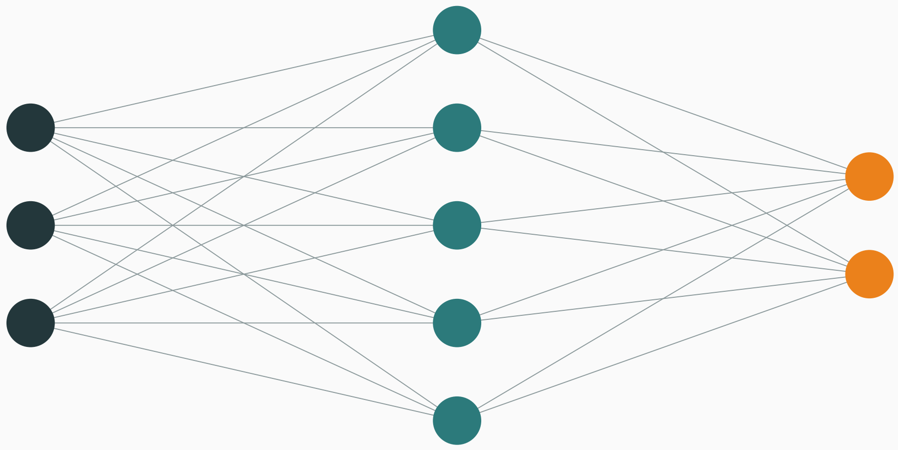
A hidden layer is a new column of learned features



3 **input features** 5 **learned features** 2 **class scores**

A hidden layer is a new column of learned features

V VISUAL



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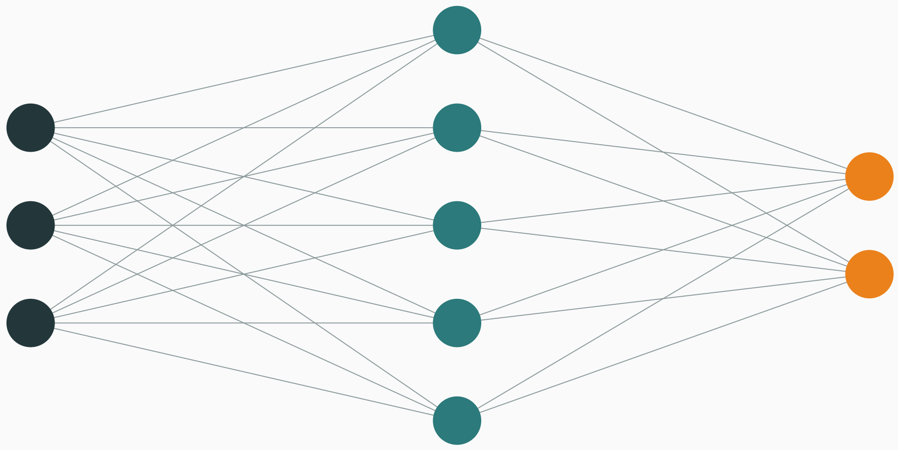

3 input features

5 learned features

2 class scores

$$\boldsymbol{h} = \varphi(\boldsymbol{W}_1\boldsymbol{x} + \boldsymbol{b}_1), \quad \boldsymbol{z} = \boldsymbol{W}_2\boldsymbol{h} + \boldsymbol{b}_2.$$

W_1 stores the $3 \times 5 = 15$ blue-to-teal edges; W_2 stores the $5 \times 2 = 10$ teal-to-orange edges.

General layer dimensions must agree

D DERIVATION

INPUT

$$\boldsymbol{x} \in \mathbb{R}^d$$

$\rightarrow$

HIDDEN LAYER

$$\boldsymbol{W}_1 \in \mathbb{R}^{m \times d}$$
$$\boldsymbol{b}_1 \in \mathbb{R}^{m}$$
$$\boldsymbol{h} \in \mathbb{R}^{m}$$

$\rightarrow$

OUTPUT LAYER

$$\boldsymbol{W}_2 \in \mathbb{R}^{K \times m}$$
$$\boldsymbol{b}_2 \in \mathbb{R}^{K}$$
$$\boldsymbol{z} \in \mathbb{R}^{K}$$

General layer dimensions must agree

D DERIVATION

INPUT

$\boldsymbol{x} \in \mathbb{R}^d$

$\rightarrow$

HIDDEN LAYER

$\boldsymbol{W}_1 \in \mathbb{R}^{m \times d}$

$\boldsymbol{b}_1 \in \mathbb{R}^m$

$\boldsymbol{h} \in \mathbb{R}^m$

$\rightarrow$

OUTPUT LAYER

$\boldsymbol{W}_2 \in \mathbb{R}^{K \times m}$

$\boldsymbol{b}_2 \in \mathbb{R}^K$

$\boldsymbol{z} \in \mathbb{R}^K$

$$(m \times d)(d \times 1) \to (m \times 1), \quad (K \times m)(m \times 1) \to (K \times 1)$$

General layer dimensions must agree

D DERIVATION

INPUT

$\boldsymbol{x} \in \mathbb{R}^d$

$\rightarrow$

HIDDEN LAYER

$\boldsymbol{W}_1 \in \mathbb{R}^{m \times d}$

$\boldsymbol{b}_1 \in \mathbb{R}^m$

$\boldsymbol{h} \in \mathbb{R}^m$

$\rightarrow$

OUTPUT LAYER

$\boldsymbol{W}_2 \in \mathbb{R}^{K \times m}$

$\boldsymbol{b}_2 \in \mathbb{R}^K$

$\boldsymbol{z} \in \mathbb{R}^K$

$$(m \times d)(d \times 1) \rightarrow (m \times 1), \quad (K \times m)(m \times 1) \rightarrow (K \times 1)$$

The inner dimensions cancel; the number of rows determines the next representation size.

One XOR network: keep these two features throughout

VISUAL

Both hidden units receive the same pair $\boldsymbol{x} = (x_1, x_2)$:

$$h_1 = \mathrm{ReLU}(x_1 + x_2), \quad h_2 = \mathrm{ReLU}(x_1 + x_2 - 1)$$

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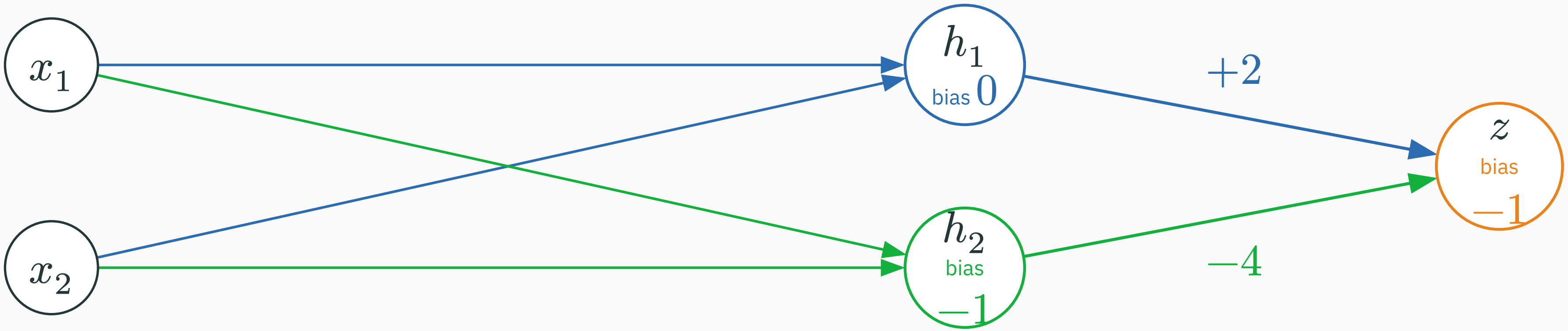
all four input weights $= +1$



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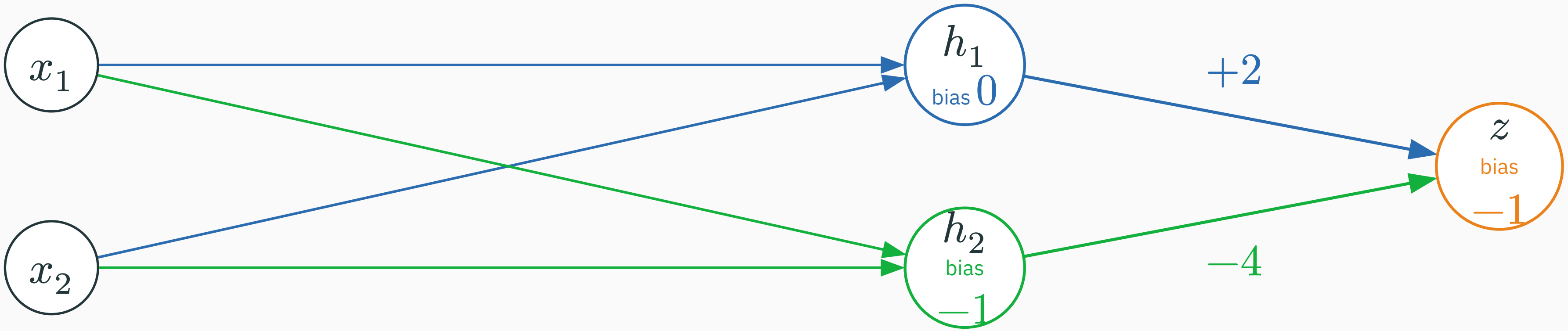

The output node computes $z = 2h_1 - 4h_2 - 1$; we keep this exact network throughout the XOR example.

Verify the XOR network on all four inputs

D DERIVATION

Input	$u_1 = x_1 + x_2$	h_1	$u_2 = x_1 + x_2 - 1$	h_2	$z = 2h_1 - 4h_2 - 1$	$\hat{y}$
$(0, 0)$	0	0	-1	0	-1	0
$(1, 0)$	1	1	0	0	1	1
$(0, 1)$	1	1	0	0	1	1
$(1, 1)$	2	2	1	1	-1	0

Verify the XOR network on all four inputs

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Input	$u_1 = x_1 + x_2$	h_1	$u_2 = x_1 + x_2 - 1$	h_2	$z = 2h_1 - 4h_2 - 1$	$\hat{y}$
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$(1,0)$	1	1	0	0	1	1
$(0,1)$	1	1	0	0	1	1
$(1,1)$	2	2	1	1	-1	0

$$\boldsymbol{W}_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \boldsymbol{b}_1 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{w}_{\text{out}} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}, \quad b_{\text{out}} = -1.$$

Verify the XOR network on all four inputs

D DERIVATION

Input	$u_1 = x_1 + x_2$	h_1	$u_2 = x_1 + x_2 - 1$	h_2	$z = 2h_1 - 4h_2 - 1$	$\hat{y}$
$(0,0)$	0	0	-1	0	-1	0
$(1,0)$	1	1	0	0	1	1
$(0,1)$	1	1	0	0	1	1
$(1,1)$	2	2	1	1	-1	0

$$\boldsymbol{W}_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \boldsymbol{b}_1 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{w}_{\text{out}} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}, \quad b_{\text{out}} = -1.$$

Predict 1 when $z \geq 0$. One ordinary output node correctly classifies all four XOR inputs.

First derive the feature functions for a general input

D DERIVATION

Let $x = \binom{x_1}{x_2}$. The weights and bias define a **function** before we substitute any particular input.

First derive the feature functions for a general input

D DERIVATION

Let $\boldsymbol{x} = \binom{x_1}{x_2}$. The weights and bias define a **function** before we substitute any particular input.

BLUE HIDDEN UNIT

$$\boldsymbol{w}_1 = \binom{1}{1}, \quad b_1 = 0$$

$$u_1(\boldsymbol{x}) = \boldsymbol{w}_1^\top \boldsymbol{x} + b_1$$
$$= 1x_1 + 1x_2 + 0 = x_1 + x_2$$

$$h_1(\boldsymbol{x})$$
$$= \mathrm{ReLU}(x_1 + x_2)$$

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GREEN HIDDEN UNIT

$$\boldsymbol{w}_2 = \binom{1}{1}, \quad b_2 = -1$$

$$u_2(\boldsymbol{x}) = \boldsymbol{w}_2^\top \boldsymbol{x} + b_2$$
$$= 1x_1 + 1x_2 - 1 = x_1 + x_2 - 1$$

$$h_2(\boldsymbol{x})$$
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First derive the feature functions for a general input

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These are exactly the two XOR features already shown in the node diagram; now we evaluate them at one input.

Check both hidden calculations in matrix form

D DERIVATION

$$\boldsymbol{u} = \boldsymbol{W}_1\boldsymbol{x} + \boldsymbol{b}_1$$

$$= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

$$\boxed{u_1 = 1(1) + 1(0) + 0 = 1}$$

$$\boxed{u_2 = 1(1) + 1(0) - 1 = 0}$$

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$$\boxed{u_1 = 1(1) + 1(0) + 0 = 1}$$

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$$\boldsymbol{h} = \mathrm{ReLU}(\boldsymbol{u}) = \begin{pmatrix} \max(0{,}1) \\ \max(0{,}0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Check both hidden calculations in matrix form

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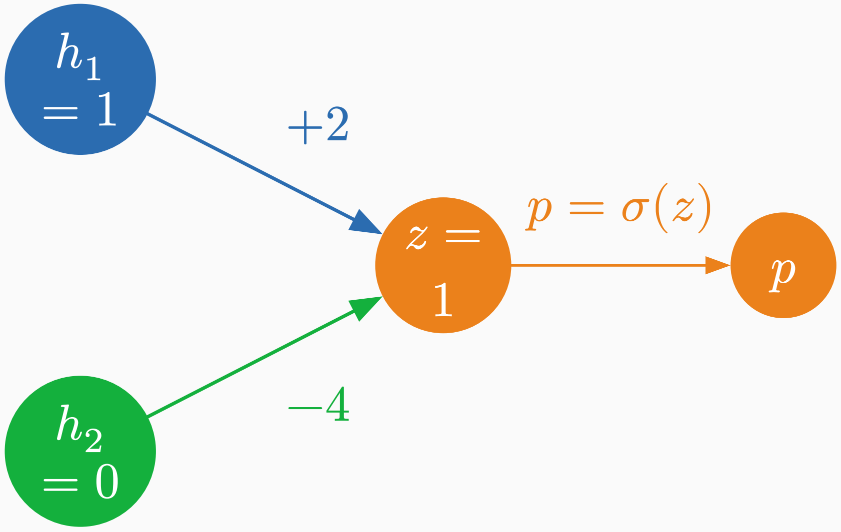
Rows of W_1 are neurons: each row computes one pre-activation, then ReLU acts elementwise.

Combine the two features, then compute probability and loss

D DERIVATION



Output logit

$$z = \begin{pmatrix} 2 & -4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} - 1$$

$$= 2(1) - 4(0) - 1 = 1.$$

Combine the two features, then compute probability and loss

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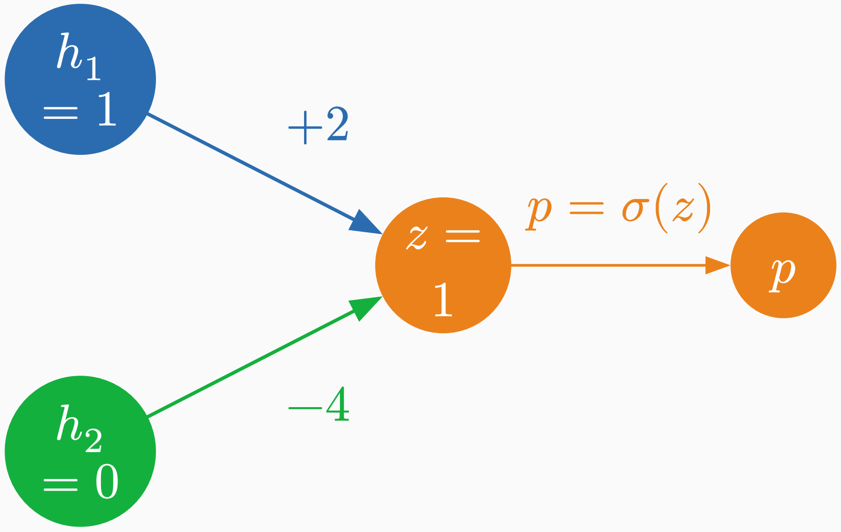

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$$= 2(1) - 4(0) - 1 = 1.$$

Probability

$$p = \sigma(1) = \frac{1}{1 + \exp(-1)} \approx 0.731.$$

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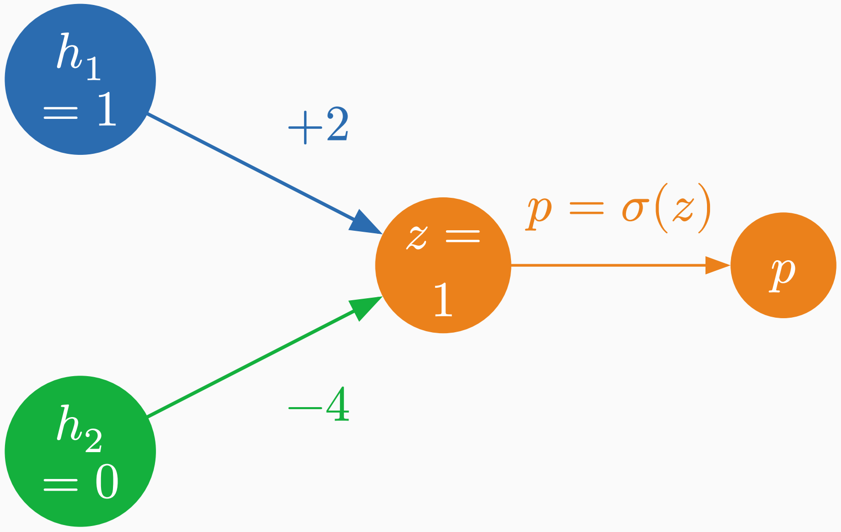

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Binary cross-entropy for $y = 1$

$$\mathcal{L} = -\log(0.731) \approx 0.313.$$

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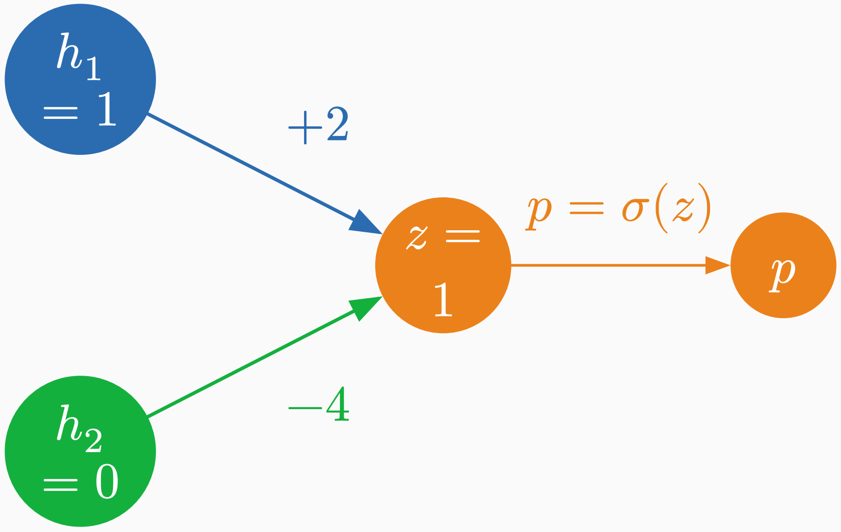

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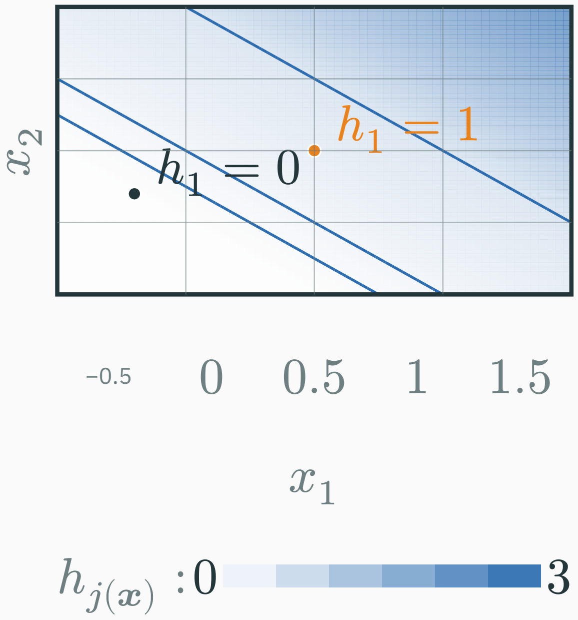
This is the same output node shown in the XOR network: weights $+2, -4$ and bias -1. Sigmoid only turns z into p.

Each ReLU is flat on one side of a line and rises on the other

V VISUAL

$$h_1(\boldsymbol{x}) = \mathrm{ReLU}(x_1 + x_2)$$

active above $x_1 + x_2 = 0$



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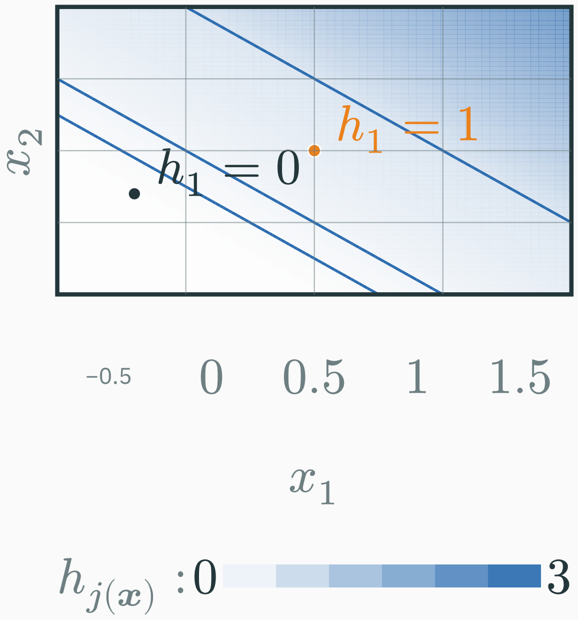

$$h_2(\boldsymbol{x}) = \mathrm{ReLU}(x_1 + x_2 - 1)$$

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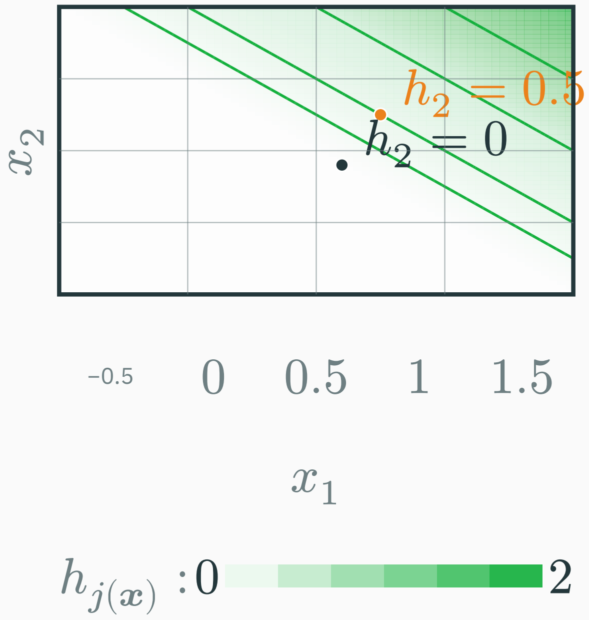

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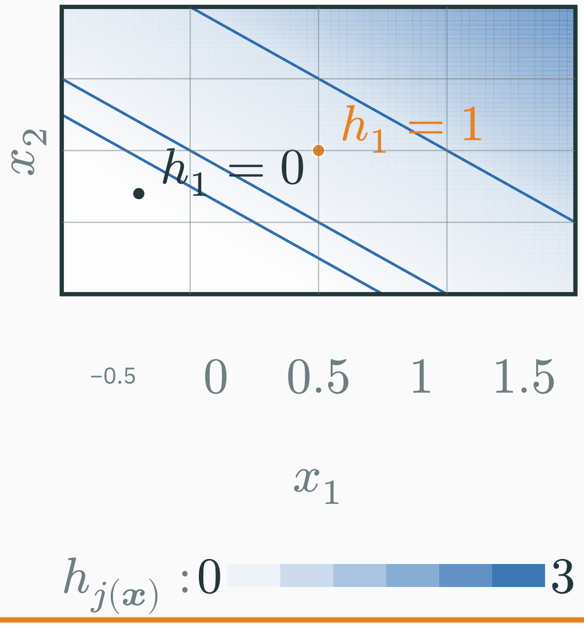

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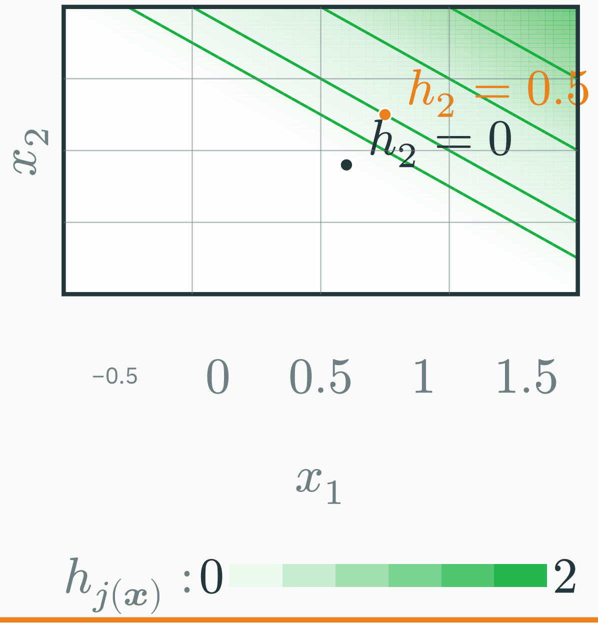


The grid is input space; each colour bar reports a hidden-unit activation value—not a probability.

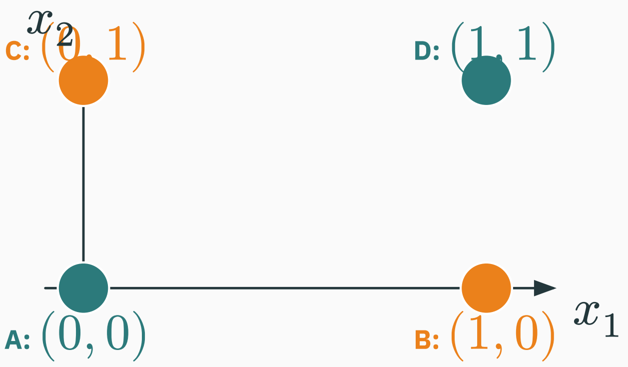
Compute each XOR point's new hidden-space coordinates

V VISUAL

Use the **same map** for every point: $T(x_1, x_2) = (h_1, h_2) = (\mathrm{ReLU}(x_1 + x_2), \mathrm{ReLU}(x_1 + x_2 - 1))$.

A: $(0,0) \to (0,0)$ **B:** $(1,0) \to (1,0)$ **C:** $(0,1) \to (1,0)$ **D:** $(1,1) \to (2,1)$

INPUT SPACE (x_1, x_2)



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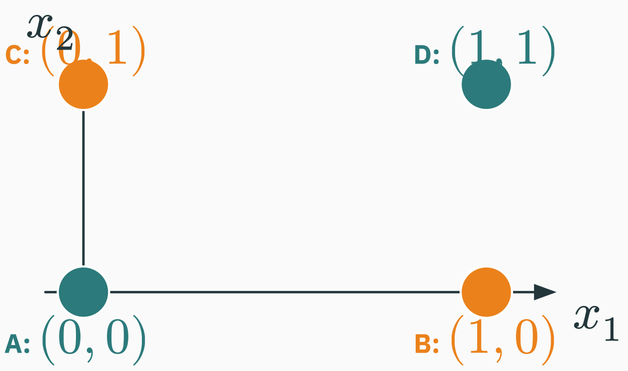

APPLY T

$\rightarrow$

B and C merge

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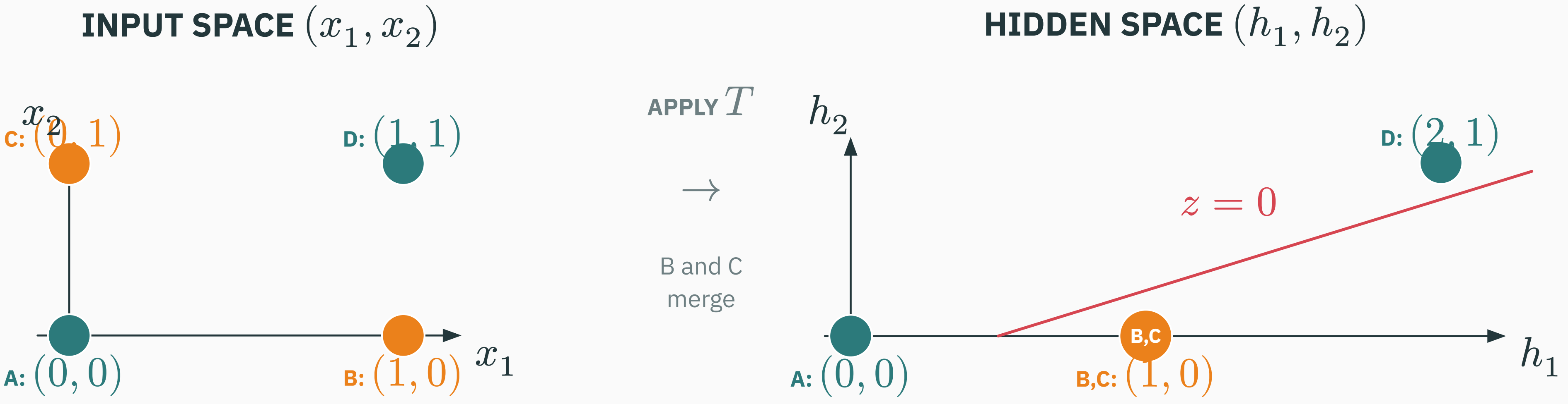

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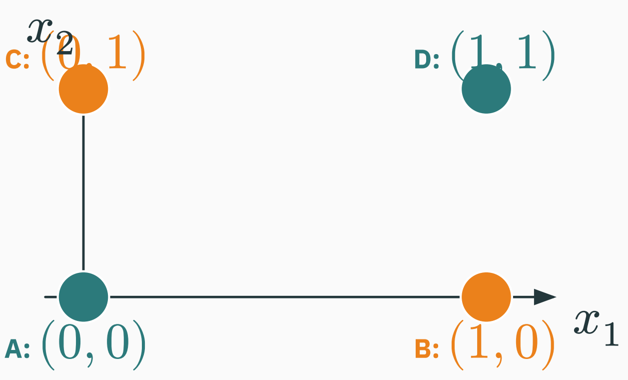
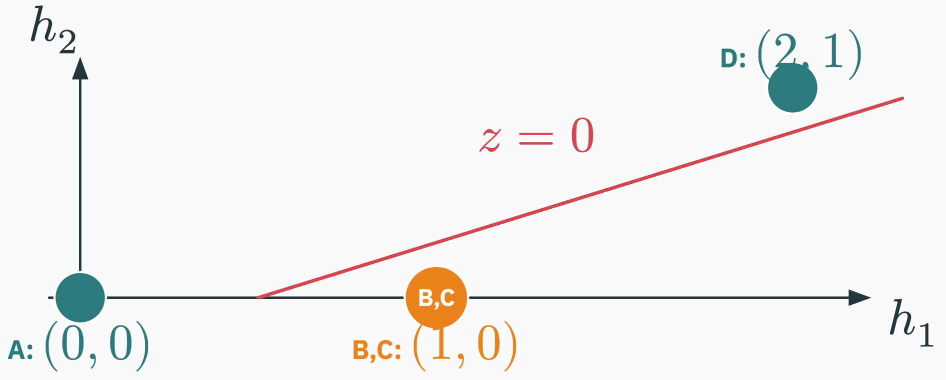


APPLY T

$\to$

B and C merge

HIDDEN SPACE (h_1, h_2)

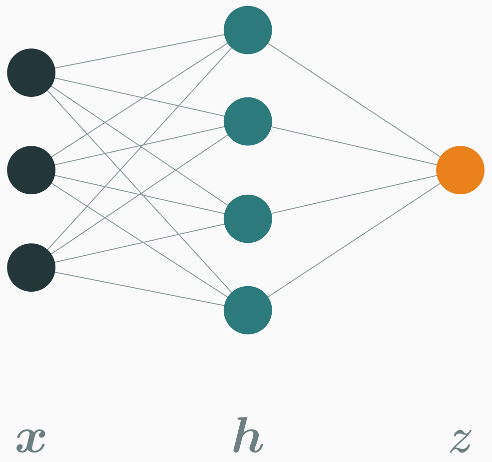


B and C both become $(1,0)$. Since $z = 2h_1 - 4h_2 - 1$, the boundary $z = 0 \leftrightarrow h_1 - 2h_2 = 0.5$ separates them from A and D.

Node view: binary uses one output; K-class uses K

V VISUAL

BINARY



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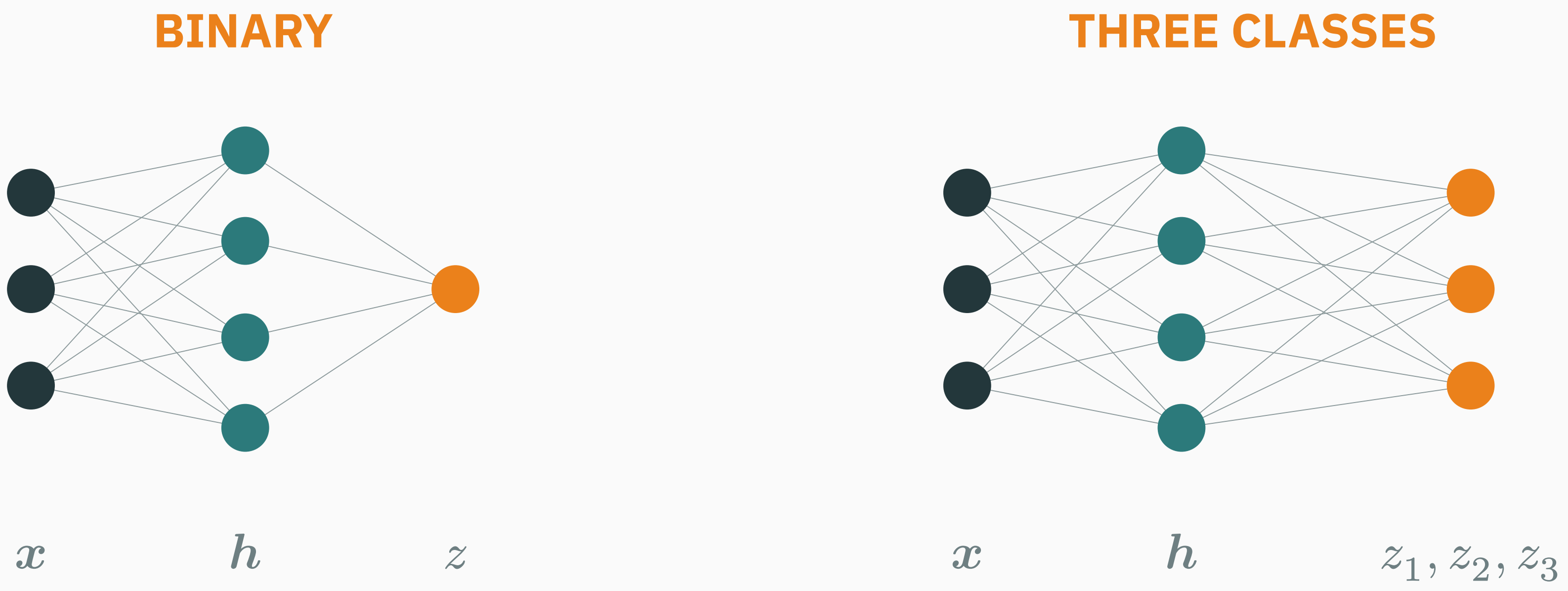
VISUAL



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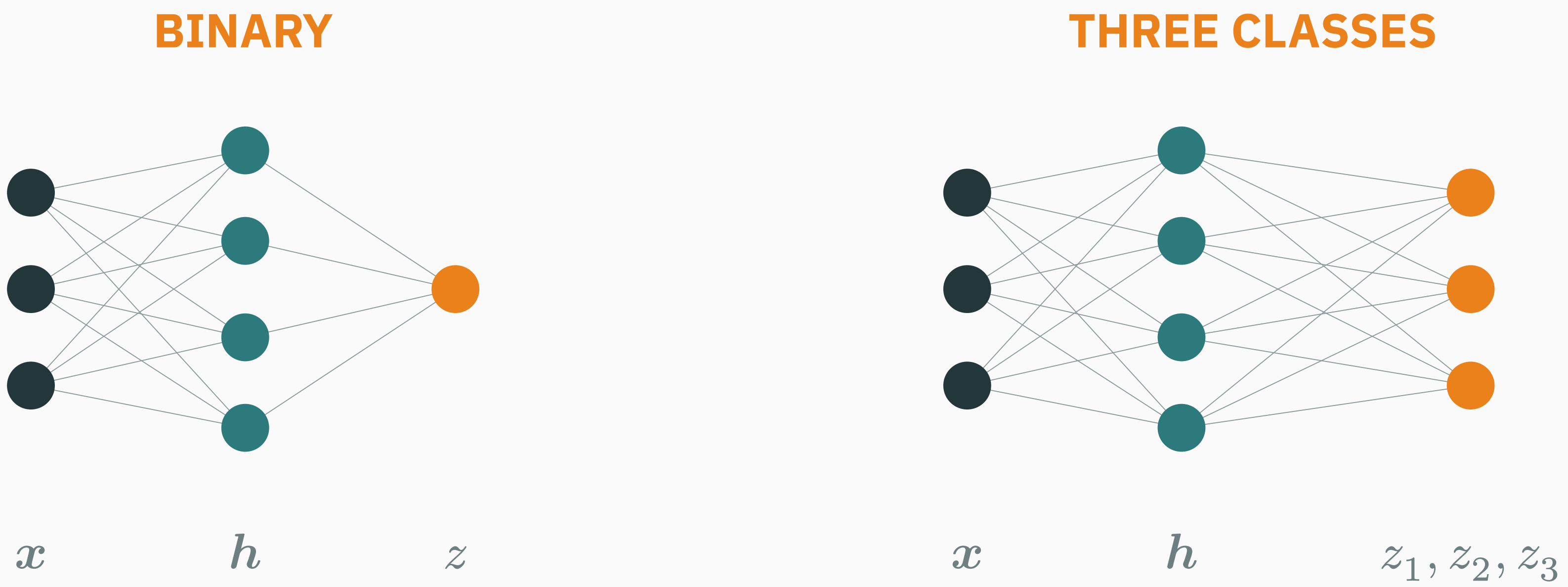
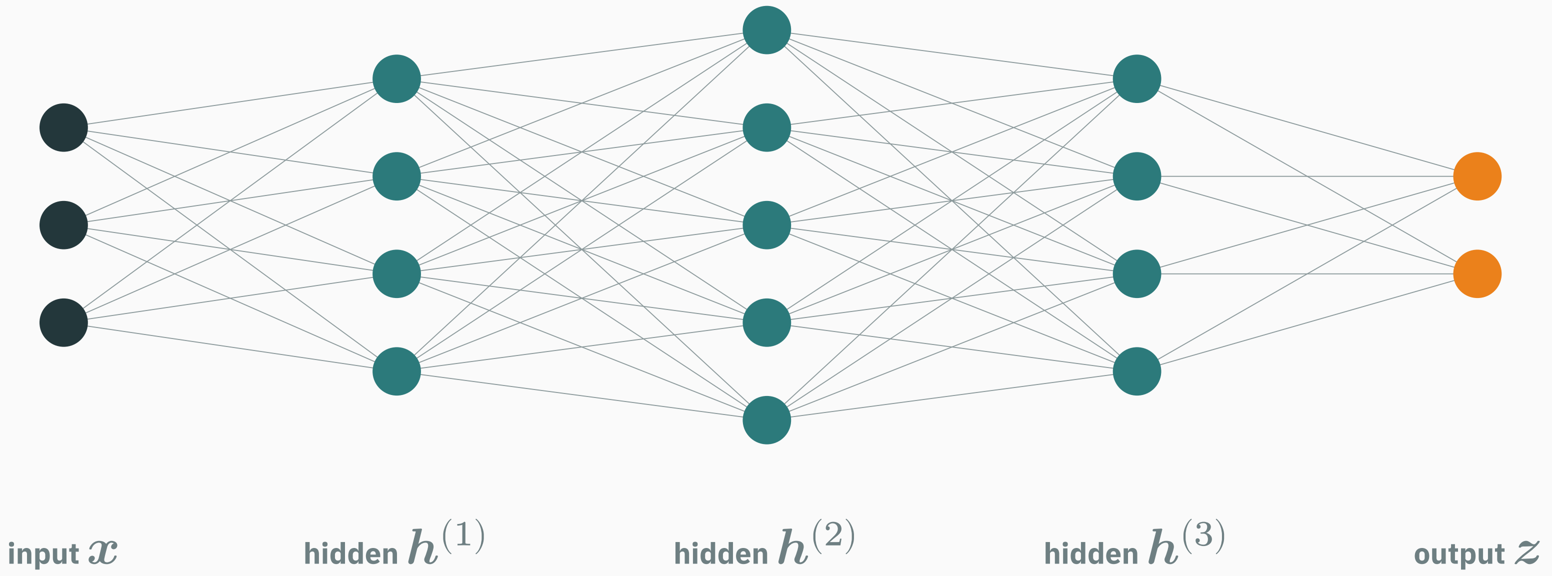


The hidden representation can stay the same; only the number and interpretation of output nodes changes.

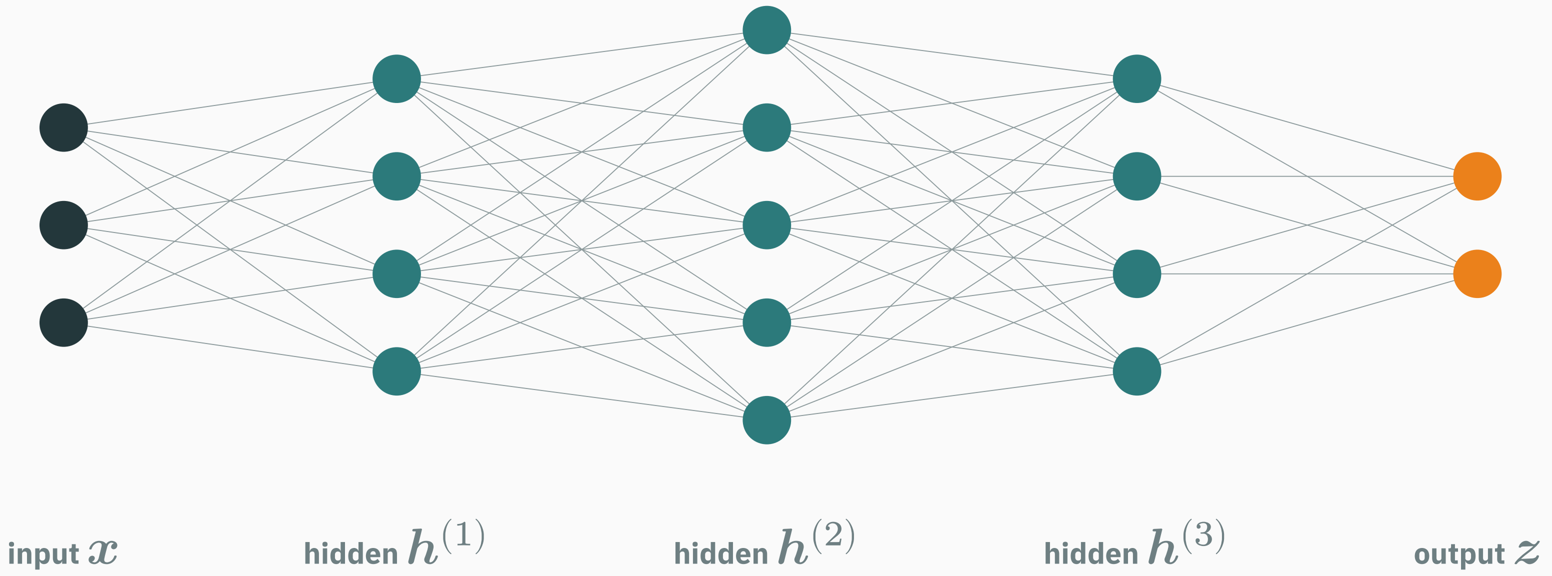
Node view: depth adds hidden columns

V VISUAL



Node view: depth adds hidden columns

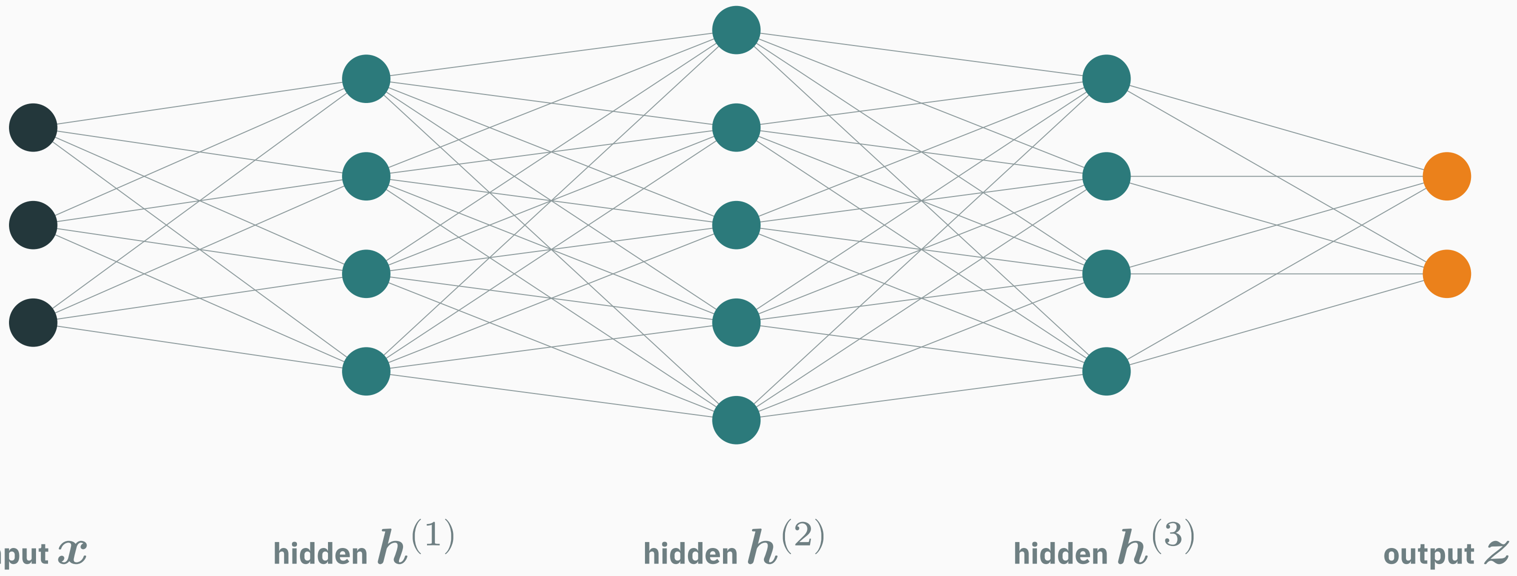
V VISUAL



Every node connects only to the next column; information reaches the output through successive learned transformations.

Node view: depth adds hidden columns

V VISUAL



Every node connects only to the next column; information reaches the output through successive learned transformations.

Depth is visible as the number of hidden columns, not the number of nodes within one column.

Universal approximation

Universal approximation asks one precise question

Q QUESTION

TARGET		REGION		TOLERANCE
continuous f	$\rightarrow$	closed, bounded D	$\rightarrow$	$\varepsilon > 0$

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Can some finite network g make $|f(x) - g(x)| < \varepsilon$ for **every** $x \in D$?

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With a suitable nonlinear activation, the theorem says: yes—such a finite one-hidden-layer network exists.

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With a suitable nonlinear activation, the theorem says: yes—such a finite one-hidden-layer network exists.

This is an existence claim about representational capacity; it does not promise that training will find the network.

A ReLU contributes one hinge

V VISUAL

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- $z < 0$: output is zero;
- $z > 0$: output grows linearly;
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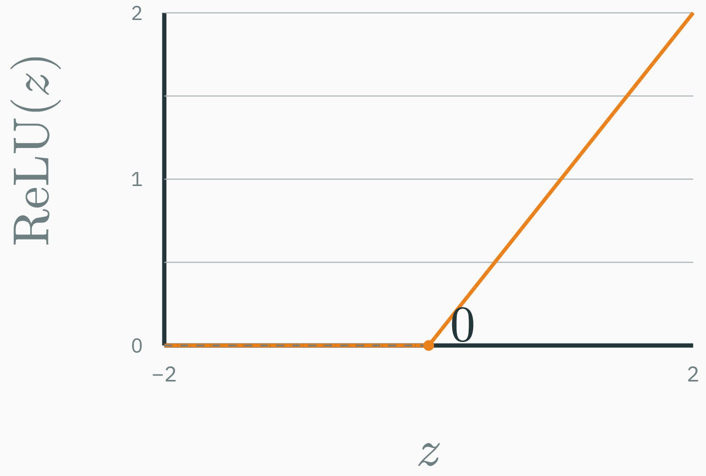

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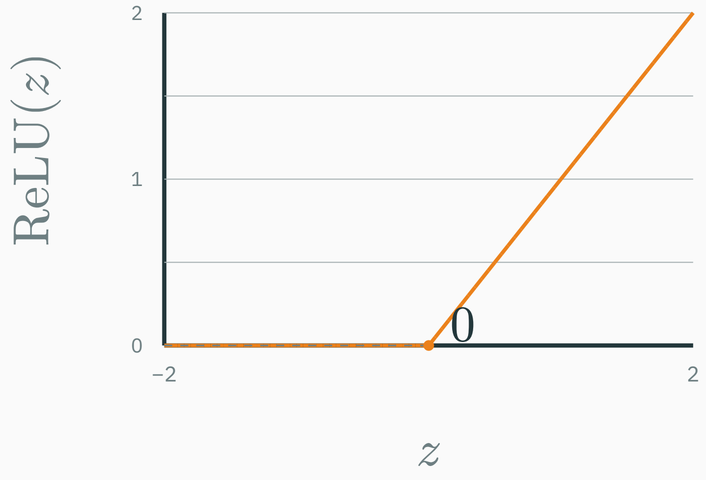
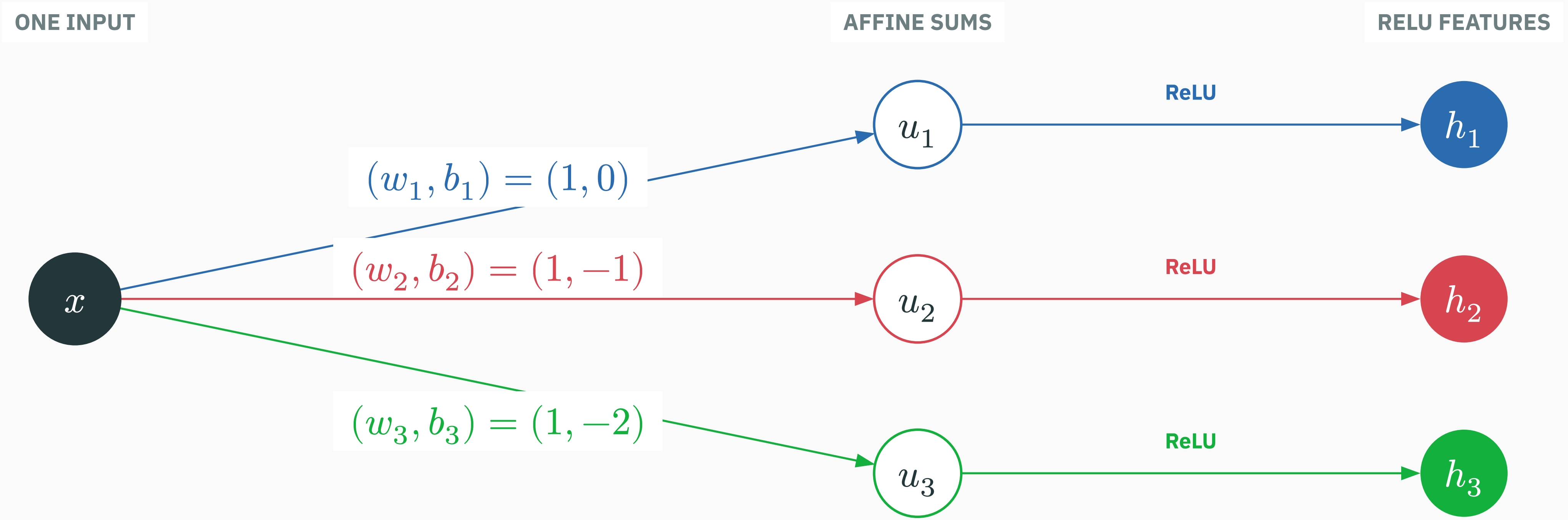


A hidden layer combines many shifted hinges into a piecewise-linear function.

One input becomes three symbolic hinge features

V VISUAL

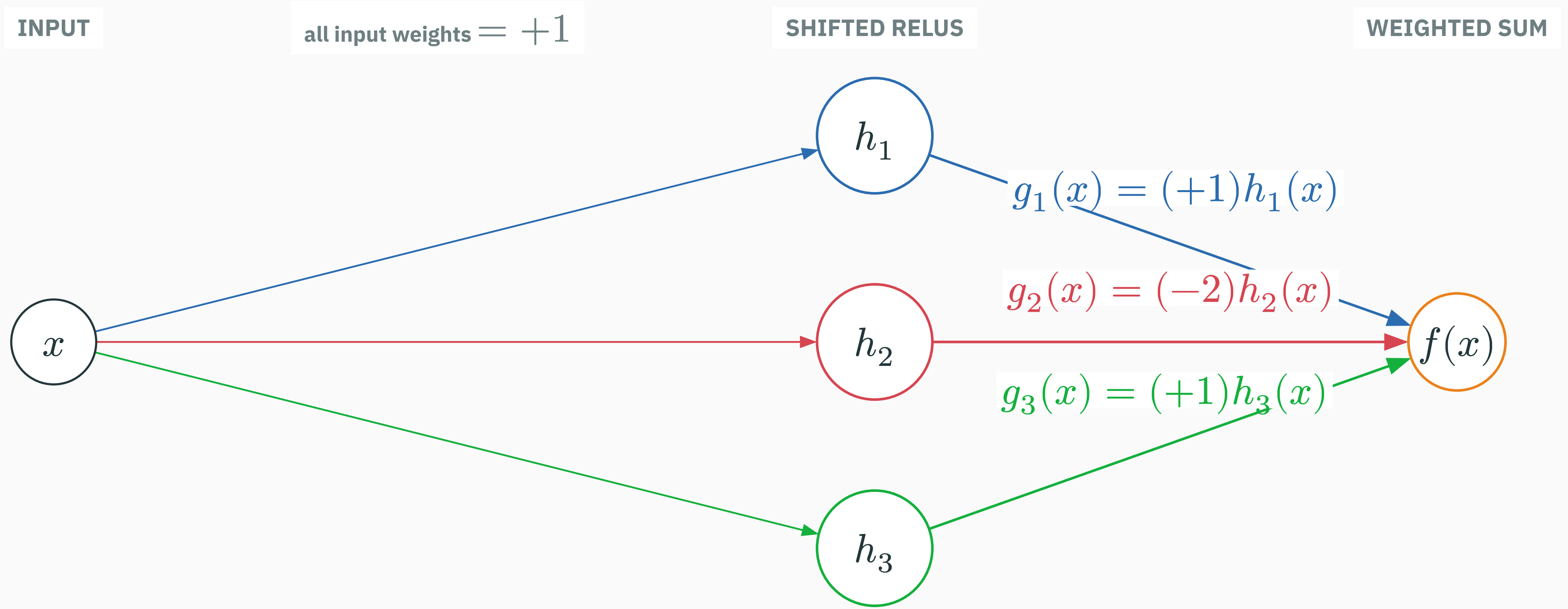
Every unit first computes $u_i = w_i x + b_i$, then applies $h_i = \mathrm{ReLU}(u_i)$.



The same x creates $h(x) = (\mathrm{ReLU}(x), \mathrm{ReLU}(x-1), \mathrm{ReLU}(x-2))$; the hinges are at $0, 1, 2$.

Output weights combine the three symbolic hinges

V VISUAL



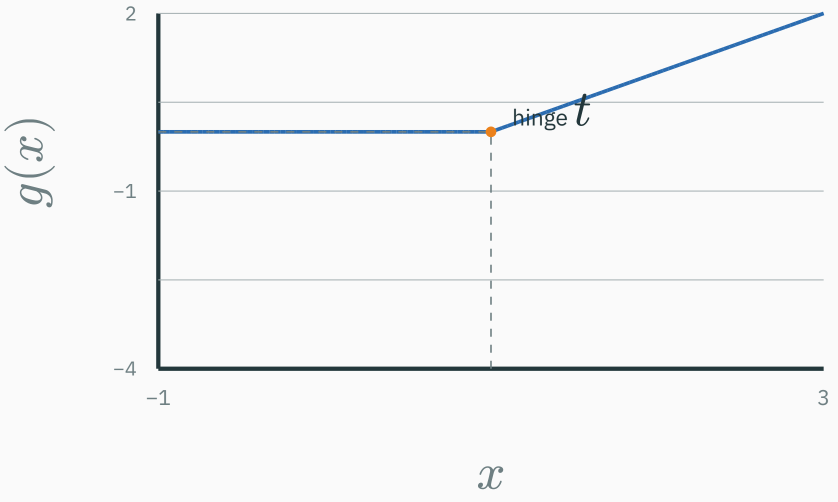
The output adds $f(x) = g_1(x) + g_2(x) + g_3(x)$—the theorem's "sum of hidden units" is literal.

Output weights choose each hinge's slope change

V VISUAL

Start with $h(x) = \mathrm{ReLU}(x - t)$; its output contribution is $g(x) = vh(x)$.

$v = +1 \quad \to \quad g(x) = +h(x)$



slope change $\Delta s = +1$

$v = -2 \quad \to \quad g(x) = -2h(x)$

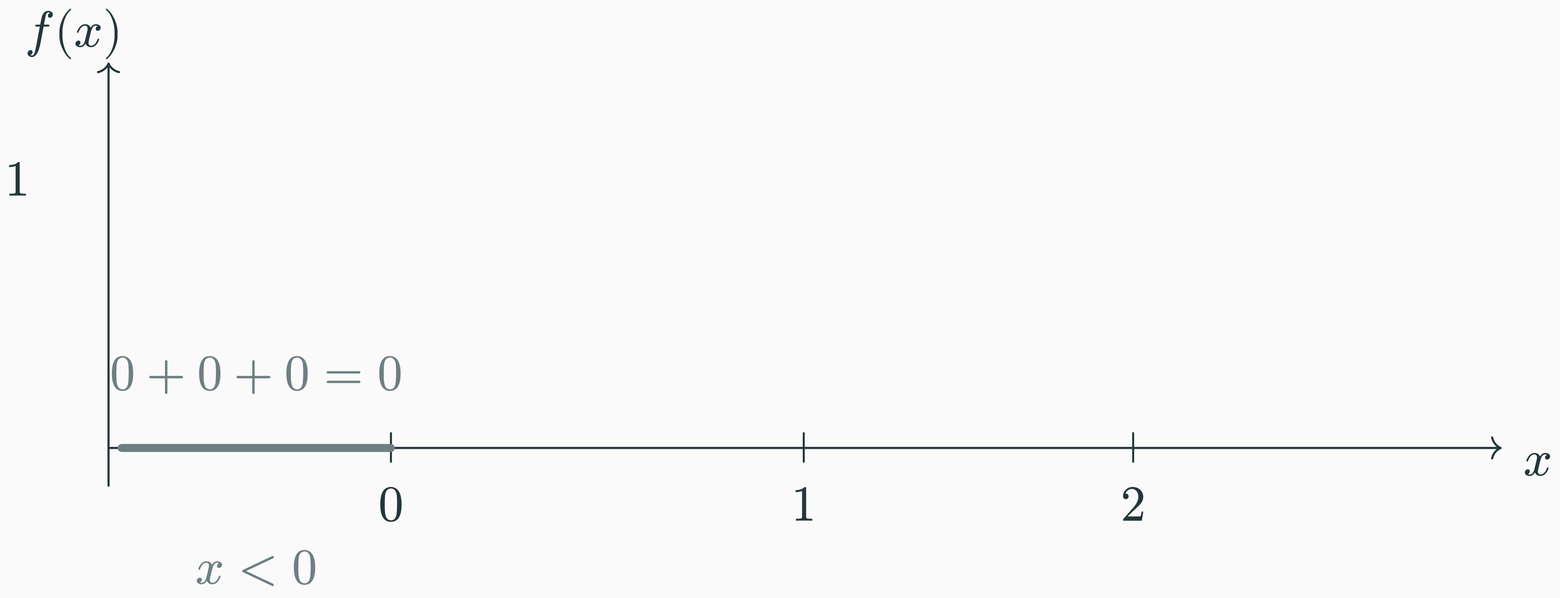
slope change $\Delta s = -2$

The hinge stays at t; the sign of v chooses the bend, and $|v|$ sets its strength.

Three hinges assemble a tent interval by interval

D DERIVATION

$$f(x) = h_1 - 2h_2 + h_3 = \mathrm{ReLU}(x) - 2\mathrm{ReLU}(x-1) + \mathrm{ReLU}(x-2).$$



Region 1: every hidden unit is off, so $f(x) = 0$.

Three hinges assemble a tent interval by interval

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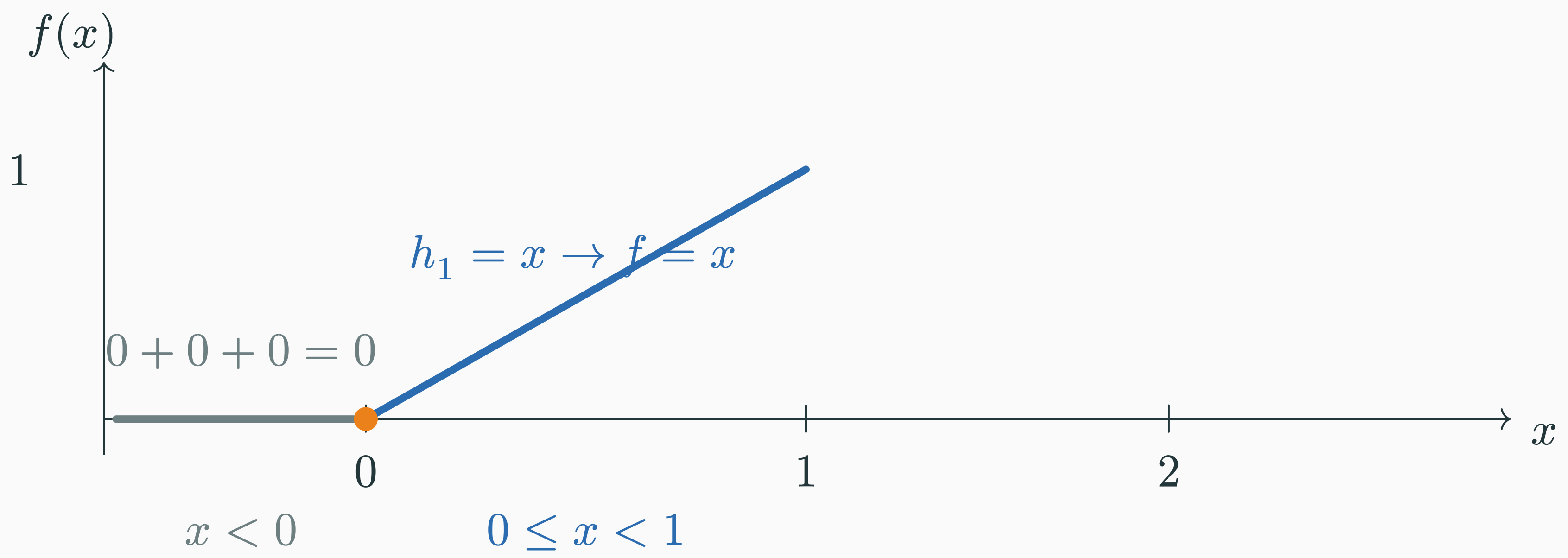

Region 2: $h_1 = x$ switches on, so the output rises with slope $+1$.

Three hinges assemble a tent interval by interval

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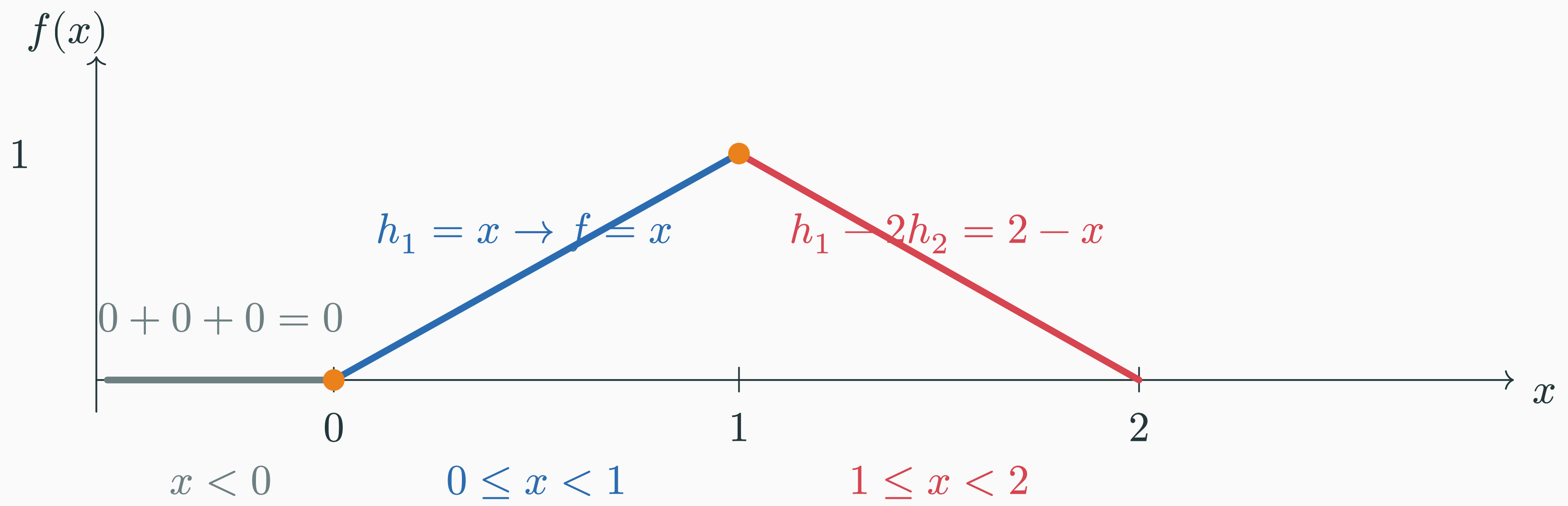

Region 3: $-2h_2$ switches on, changing the total slope from $+1$ to -1.

Three hinges assemble a tent interval by interval

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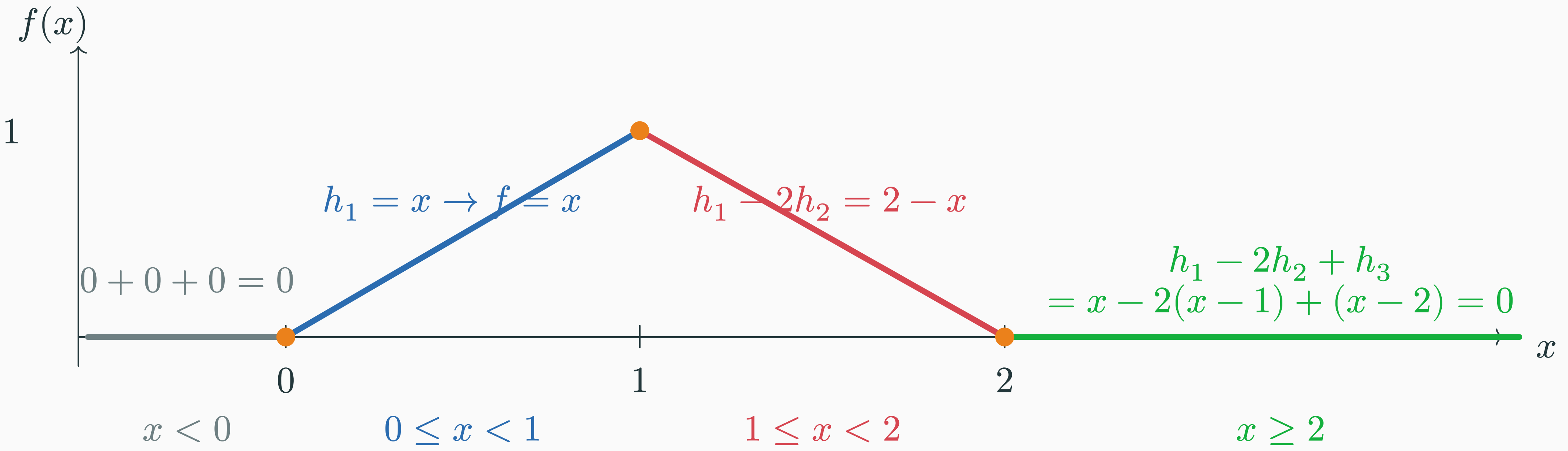

Region 4: h_3 switches on and cancels the remaining slope, so $f(x) = 0$.

Reading left to right: flat → rise → fall → flat—the weighted sum is a tent.

Learned ReLU units are the theorem's building blocks

D DERIVATION

For a one-dimensional input, a trained hidden unit has the form

$$h_{j(x)} = \mathrm{ReLU}\big(w_j x + b_j\big), \quad g(x) = c + \sum_j v_j h_{j(x)}.$$

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j	hinge t_j	w_j	b_j	output v_j	signed contribution $g_{j(x)}$
1	0	1	0	$+1$	$+\mathrm{ReLU}(x)$
2	1	1	-1	-2	$-2\mathrm{ReLU}(x-1)$
3	2	1	-2	$+1$	$+\mathrm{ReLU}(x-2)$

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Training learns hinge locations and signed contributions; universal approximation says enough such units can make the remaining gap arbitrarily small.

In many dimensions, a hinge becomes a hyperplane

V VISUAL

A unit reads the whole vector:
$h(\boldsymbol{x}) = \mathrm{ReLU}(\boldsymbol{w}^\top \boldsymbol{x} + b).$

In many dimensions, a hinge becomes a hyperplane

V VISUAL

A unit reads the whole vector: $h(\boldsymbol{x}) = \mathrm{ReLU}(\boldsymbol{w}^\top \boldsymbol{x} + b)$. The hinge is where the pre-activation is zero: $\boldsymbol{w}^\top \boldsymbol{x} + b = 0$.

> In 1-D this is a point. In 2-D it is a line. In 3-D it is a plane. In d dimensions it is a hyperplane.

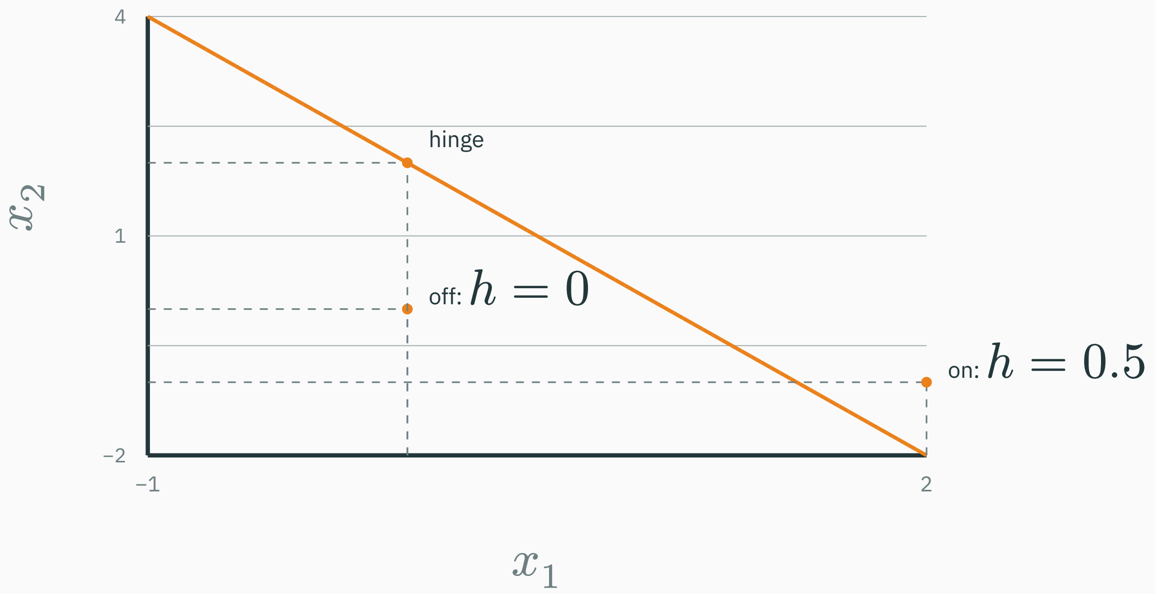
In many dimensions, a hinge becomes a hyperplane

V VISUAL

A unit reads the whole vector: $h(\boldsymbol{x}) = \mathrm{ReLU}(\boldsymbol{w}^\top \boldsymbol{x} + b)$. The hinge is where the pre-activation is zero: $\boldsymbol{w}^\top \boldsymbol{x} + b = 0$.

> In 1-D this is a point. In 2-D it is a line. In 3-D it is a plane. In d dimensions it is a hyperplane.

$$x_1 + 0.5x_2 - 1 = 0$$



In many dimensions, a hinge becomes a hyperplane

V VISUAL

A unit reads the whole vector: $h(\boldsymbol{x}) = \mathrm{ReLU}(\boldsymbol{w}^\top \boldsymbol{x} + b)$. The hinge is where the pre-activation is zero: $\boldsymbol{w}^\top \boldsymbol{x} + b = 0$.

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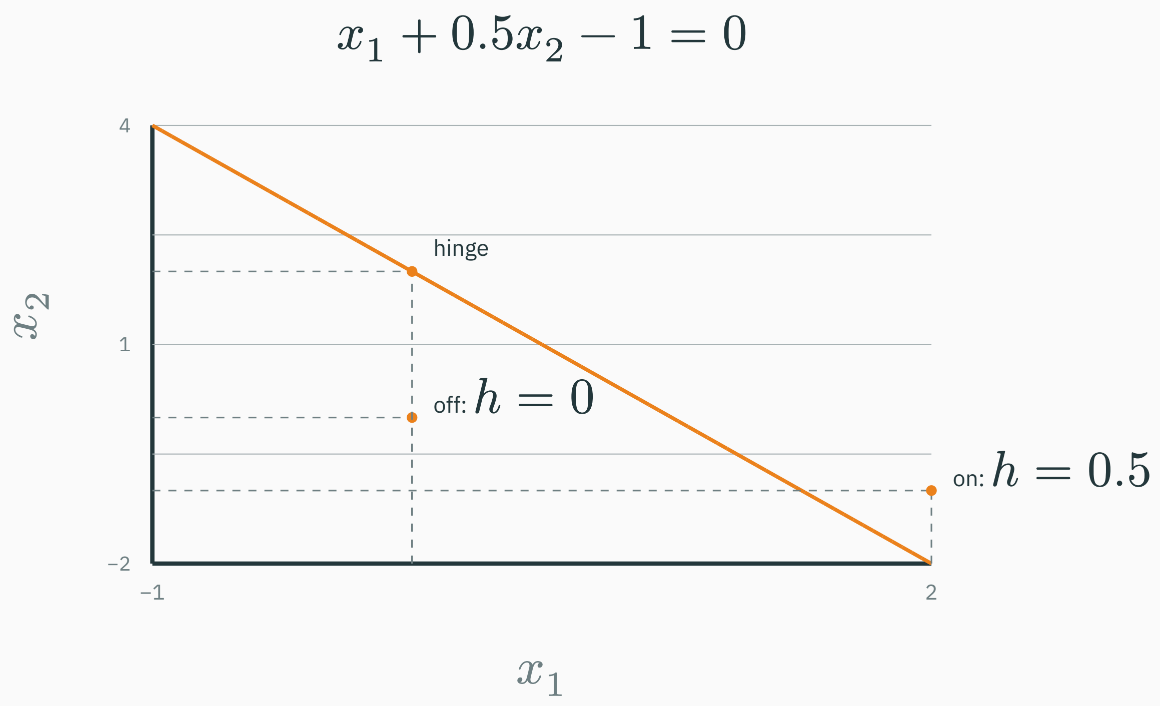
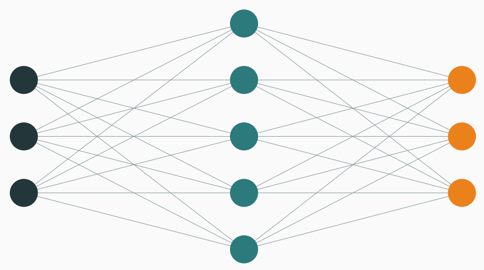


The one-dimensional breakpoint construction generalizes to learned, oriented folds of a multidimensional input space.

Beyond 1-D: approximate class scores, then classify



1 · Vector input ⟶ learned folds

$$h_{j(\boldsymbol{x})} = \mathrm{ReLU}\Big(\boldsymbol{w}_j^\top \boldsymbol{x} + b_j\Big).$$

2 · Features ⟶ continuous scores

$$z_{k(\boldsymbol{x})} = c_k + \sum_j v_{kj} h_{j(\boldsymbol{x})}.$$

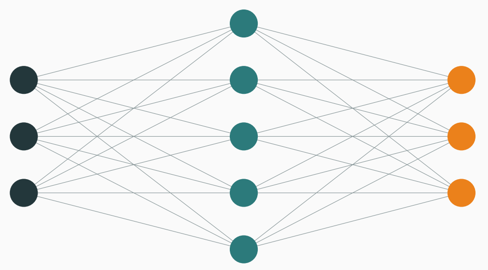
3 · Scores ⟶ classification

$$\boldsymbol{p} = \mathrm{softmax}(\boldsymbol{z}), \quad \hat{y} = \arg\max_k z_k.$$

A class changes where two top scores tie.

Beyond 1-D: approximate class scores, then classify

V VISUAL



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A class changes where two top scores tie.

On compact $D \subset \mathbb{R}^d$, approximate continuous logits; $\arg\max$ classifies wherever the winning score has a margin.

Interactive: place and combine bends

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/dl-teaching/interactives/relu-function-builder.html

Follow the same sequence as the slides: **one hinge** → **tent** → **smooth target** → **binary classifier**.

$$g_{j(x)} = v_j \; \mathrm{ReLU}(x - t_j)$$

WHERE IS THE BEND?

t_j

moves the hinge left or right

WHAT DOES THE BEND DO?

v_j

sets its direction and strength

Interactive: place and combine bends

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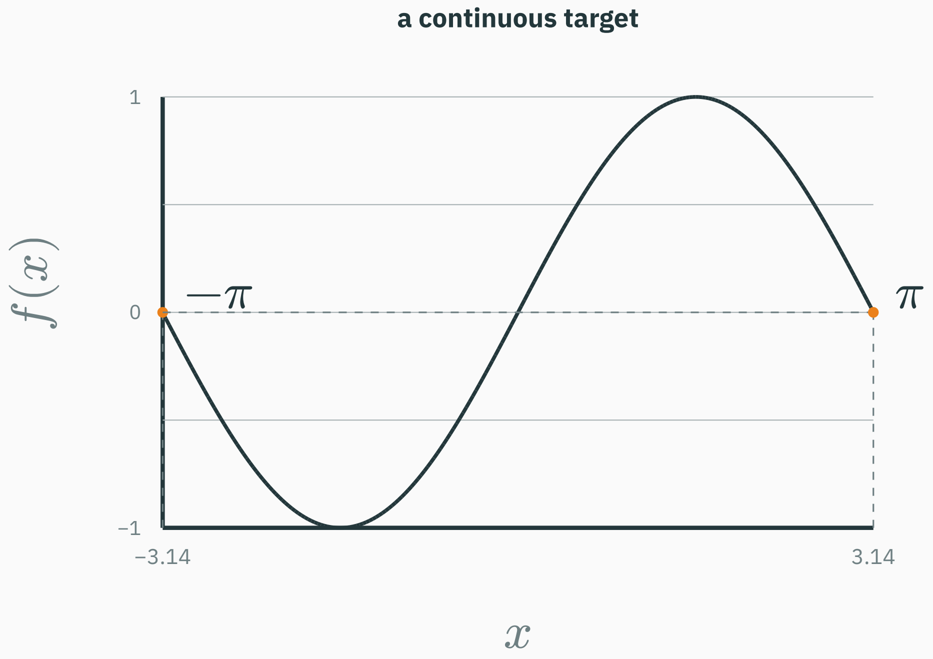
Add bends to trace a target; when their sum is a logit $z(x)$, the predicted class changes where $z = 0$.

First fix the target and the input region

V VISUAL



Continuous means that the target has no jumps.

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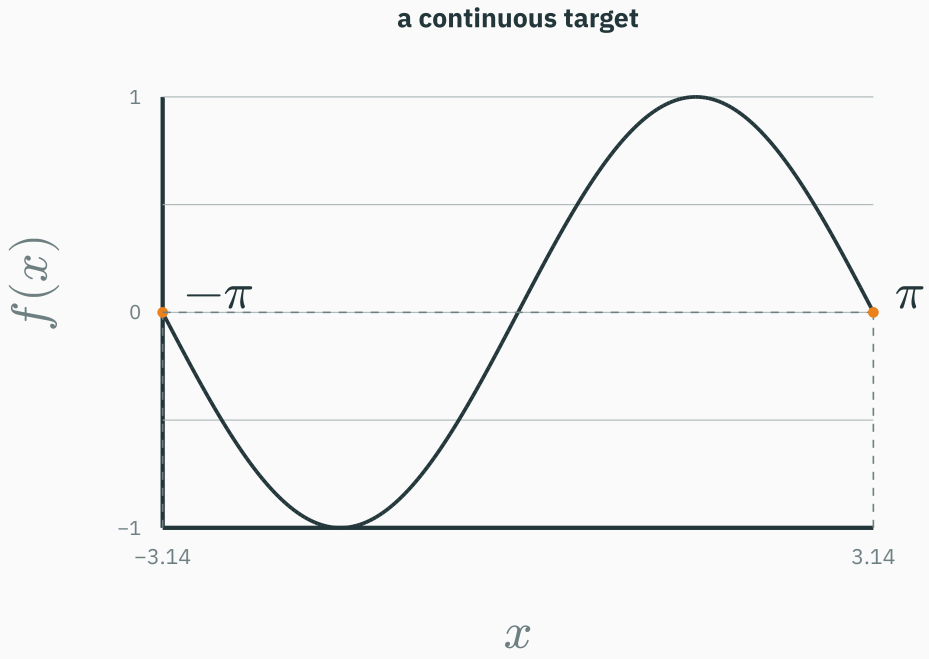

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In one dimension, a **compact domain** can be a closed, bounded interval such as

$$D = [-\pi, \pi].$$

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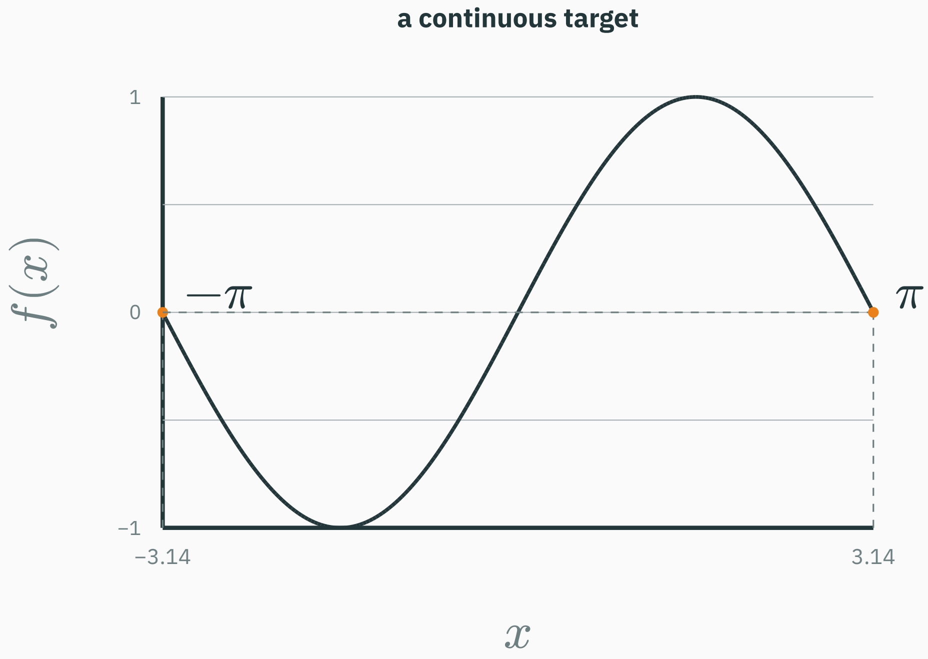
V VISUAL



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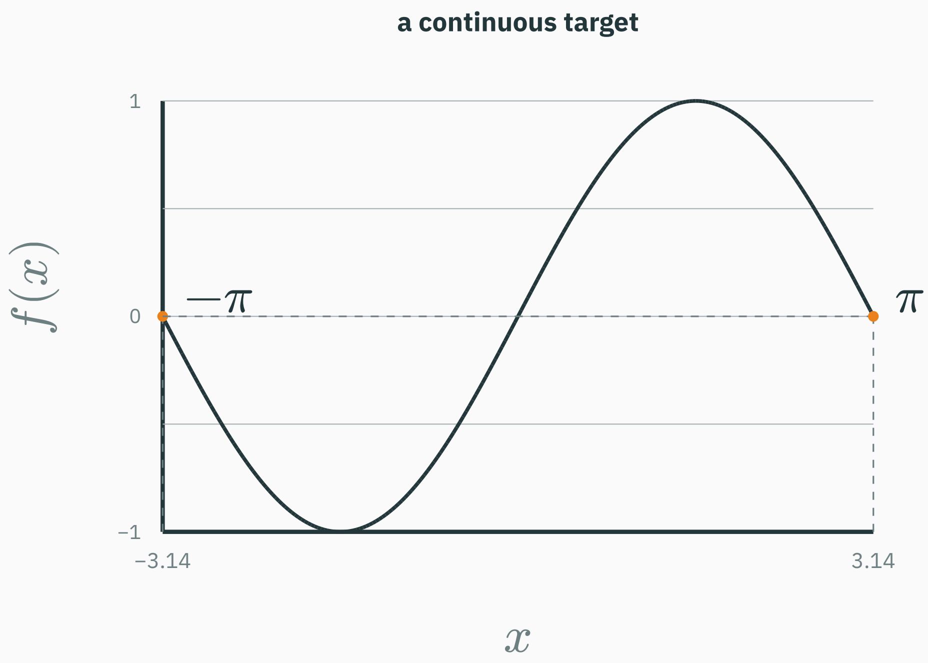

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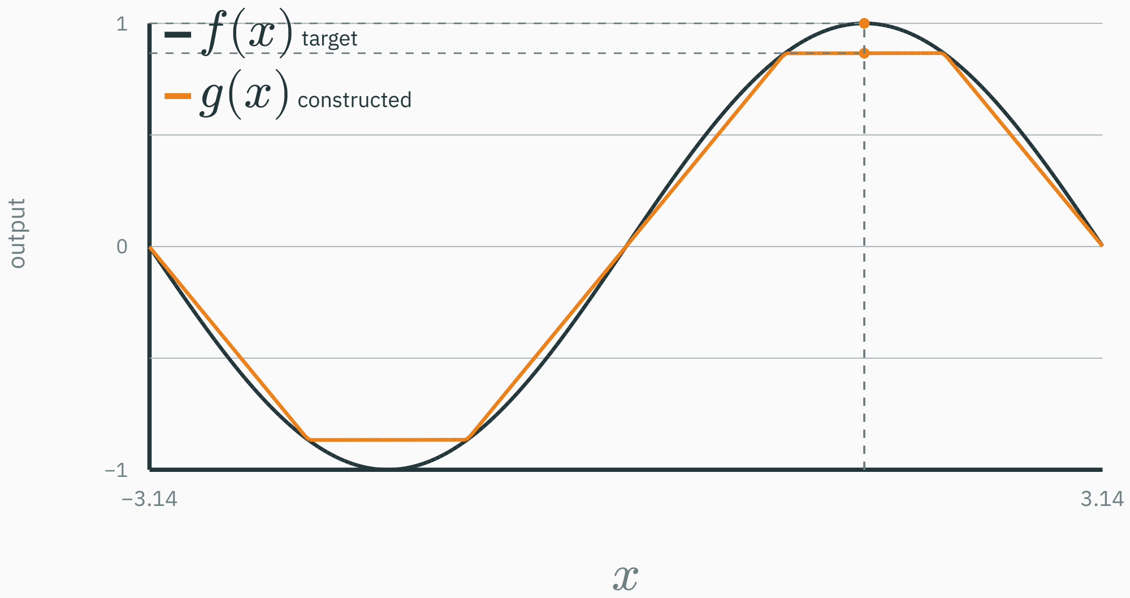
Approximation begins with one target function on one specified input region.

At one input, error is one vertical gap

V VISUAL

$$e(x_0) = |f(x_0) - g(x_0)|$$

vertical slice at $x_0 = 1.57$



Pointwise error is the vertical distance between the target and its constructed approximation at the same input.

Calculate the gap at $x_0 = 1.57$

D DERIVATION

$$e(x_0) = |f(x_0) - g(x_0)|$$

TARGET

$f(1.57) \approx 1.000$

CONSTRUCTED

$g(1.57) \approx 0.866$

ABSOLUTE DIFFERENCE

$|1.000 - 0.866|$

POINTWISE ERROR

$e(1.57) \approx 0.134$

One point is not enough: universal approximation asks for the largest gap over the whole region.

“sup” means the worst gap over the whole region

V VISUAL

$\sup_{x \in D} |f(x) - g(x)|$

sup means **supremum**: the smallest number at least as large as every pointwise error.

“sup” means the worst gap over the whole region

V VISUAL

$\sup_{x\in D}|f(x) - g(x)|$

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Here the error is continuous and $D = [-\pi, \pi]$ is closed and bounded, so a highest error exists. In this plot, read **sup** as **maximum**.

$\sup < \varepsilon$: no input in D crosses the tolerance.

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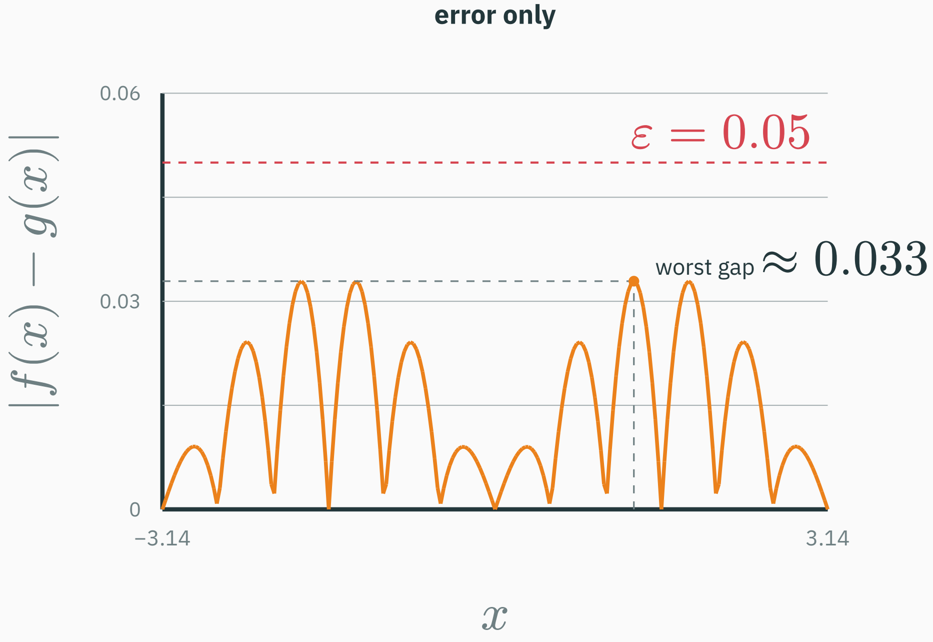


The entire orange error curve stays below $\varepsilon = 0.05$, so the uniform guarantee is satisfied.

More hinges reduce the largest approximation error

V VISUAL

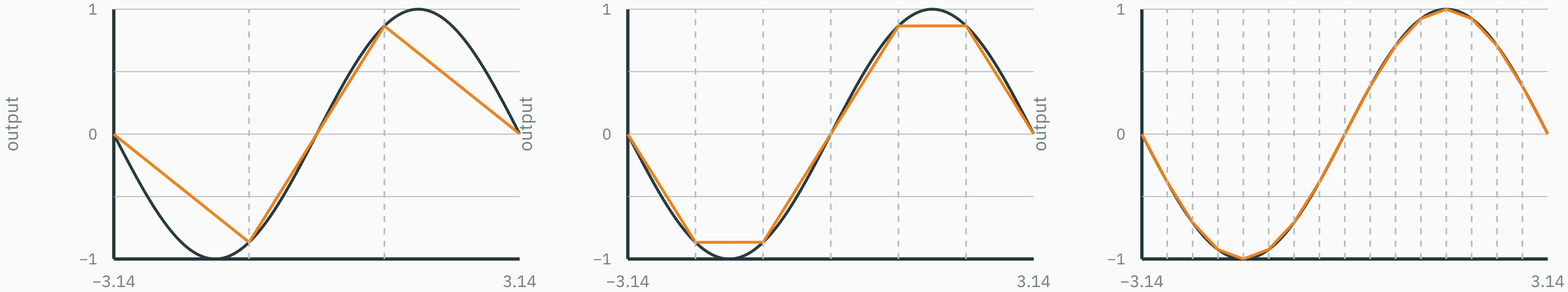
CONSTRUCTED INTERPOLANTS · FIXED KNOTS · NOT TRAINED solid: target $\sin x$ dashed: $g_{m(x)}$

Each g_m matches $\sin x$ at fixed knots and is linear between adjacent knots.

2 hinges $\varepsilon_m \approx 0.437$

5 hinges $\varepsilon_m \approx 0.134$

15 hinges $\varepsilon_m \approx 0.019$



More hinges shorten each linear segment, so the worst gap falls from 0.437 to 0.019.

Read the universal-approximation claim in order

D DERIVATION

1 · FIX TARGET

f and D

→

2 · DEMAND ACCURACY

choose any $\varepsilon > 0$

→

3 · CHOOSE NETWORK

some finite MLP g may depend on ε

→

4 · VERIFY

$\sup_{x \in D} |f(x) - g(x)| < \varepsilon$

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For every ε: we may request any positive tolerance.

There exists g: the network may change when the tolerance changes.

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This is a statement about representational capacity—not yet about how to find the network.

Existence does not guarantee learnability

Universal approximation does **not** promise:

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- that training will **find** that function;
- that **finite data** is enough;

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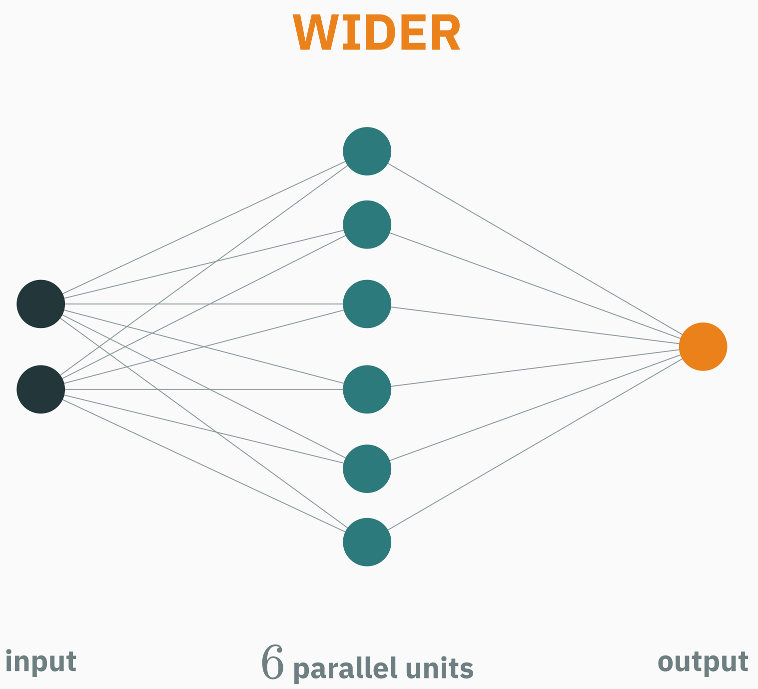
$$\boxed{\textbf{expressivity} \neq \textbf{trainability} \neq \textbf{generalization}}$$

Depth versus width

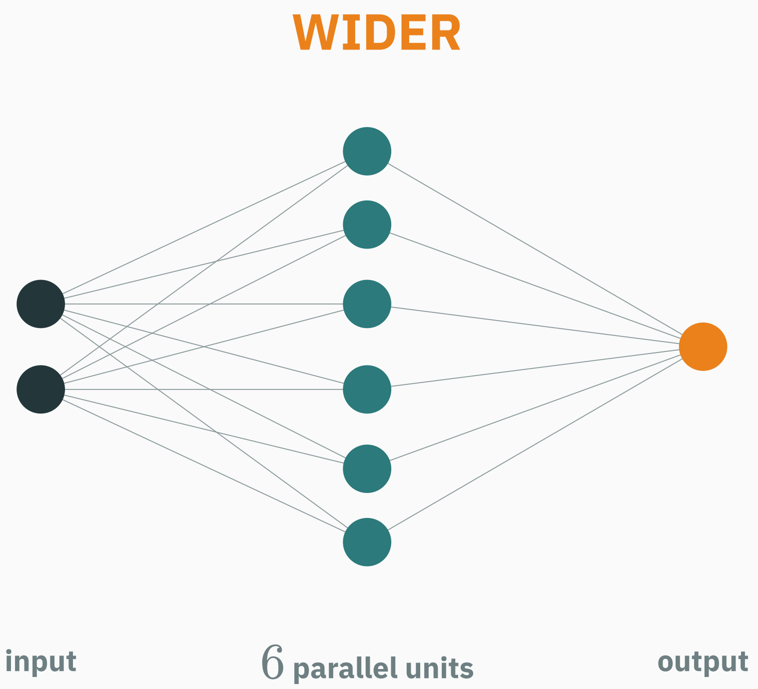
Node view: width adds rows; depth adds columns

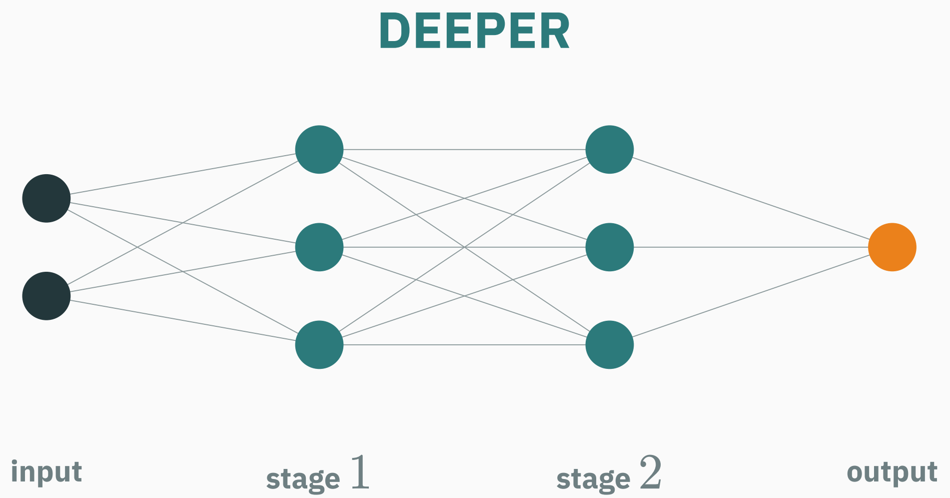
V VISUAL



Node view: width adds rows; depth adds columns

V VISUAL





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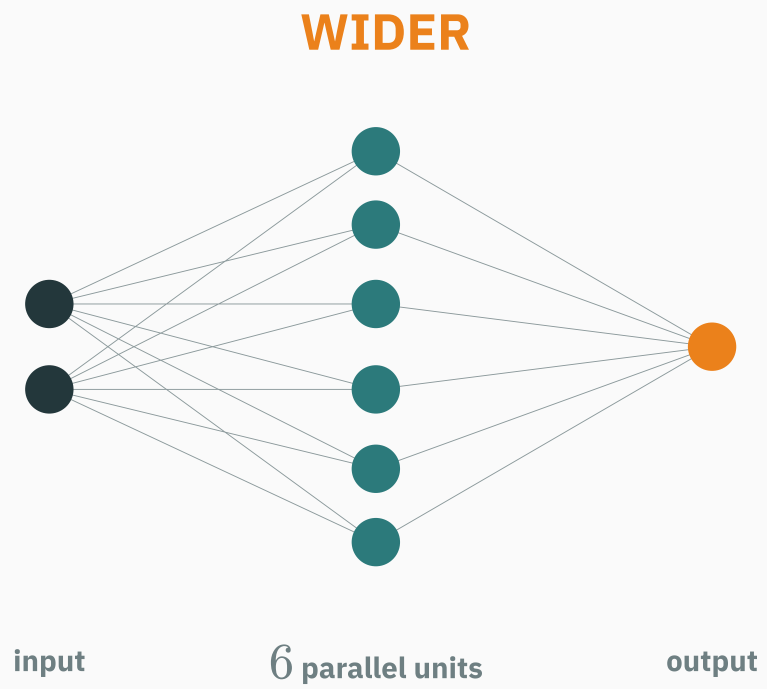
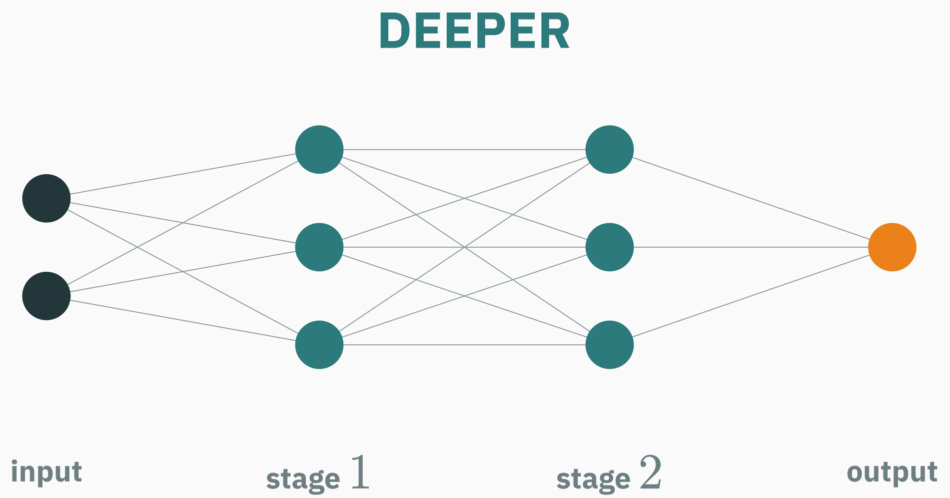
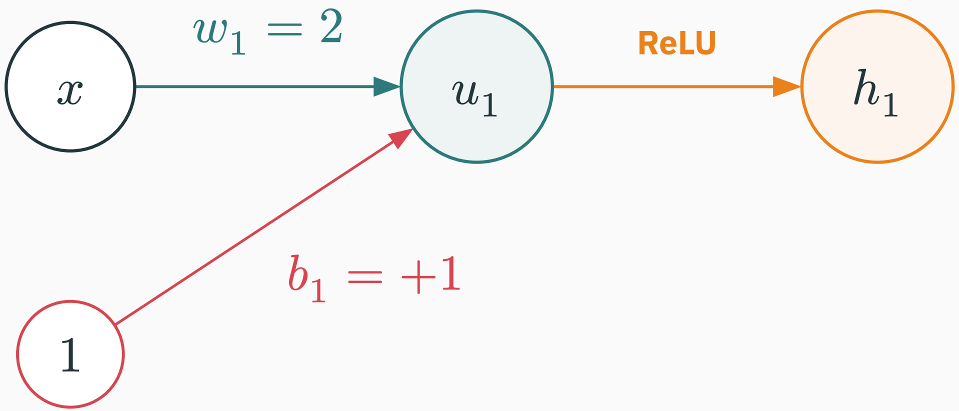




Count nodes vertically to read width; count hidden columns horizontally to read depth.

Layer 1 creates a hinge at $x = -0.5$

V VISUAL



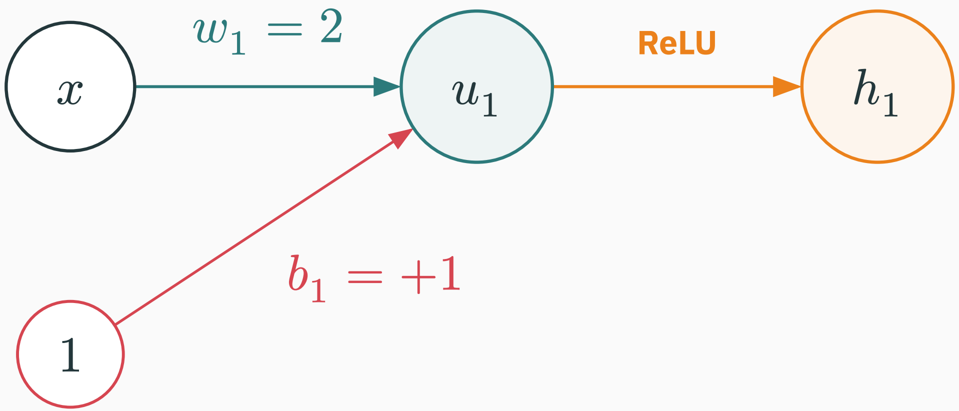
$$u_1 = 2x + 1$$
$$h_1 = f_1(x) = \mathrm{ReLU}(u_1)$$

At $x = 0.25$:

$$u_1 = 1.5 \rightarrow h_1 = 1.5$$

Layer 1 creates a hinge at $x = -0.5$

V VISUAL



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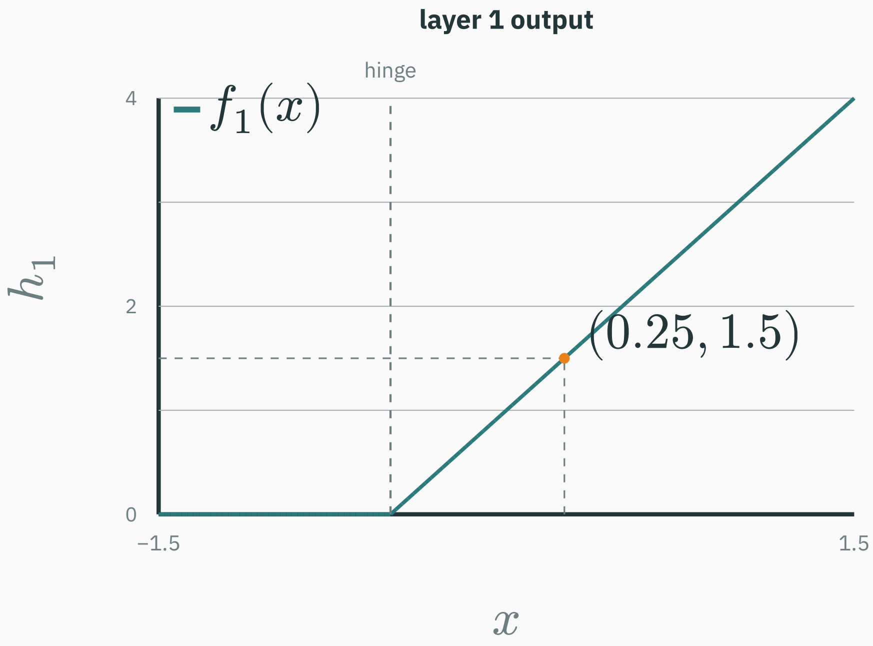

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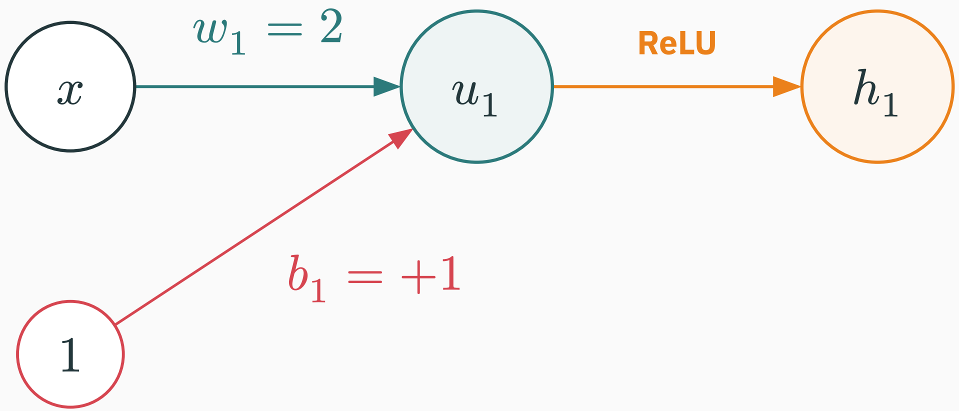

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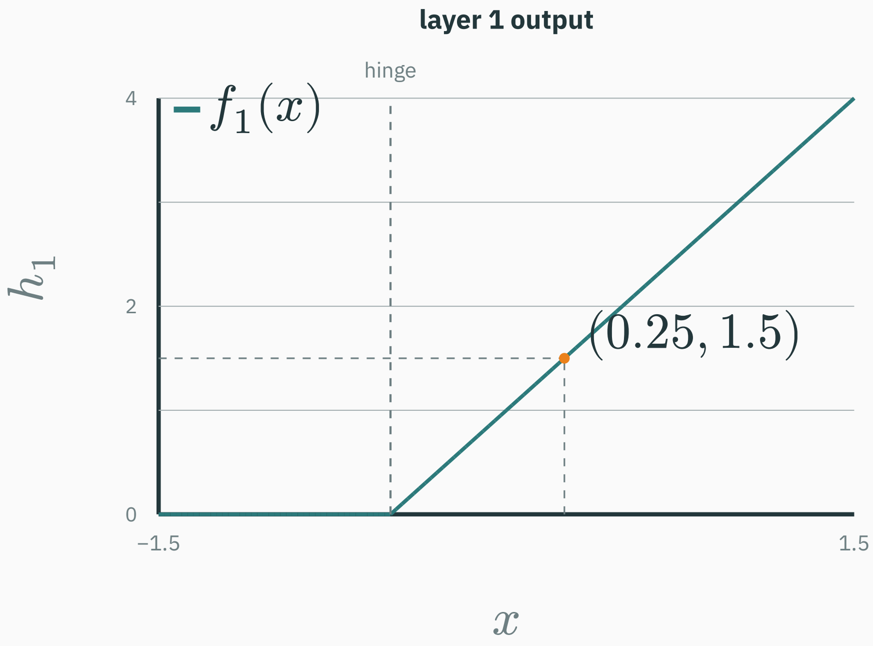
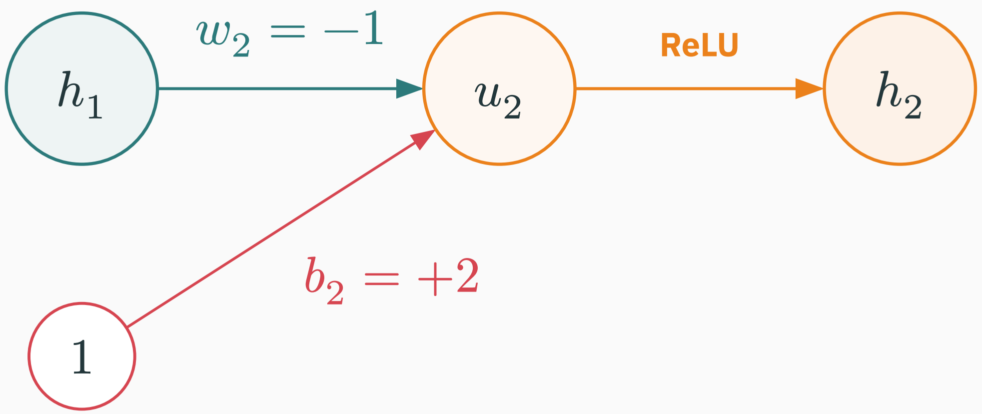


Layer 1 is flat until $2x + 1$ becomes positive; then its learned feature grows linearly.

The second layer transforms the learned feature

V VISUAL



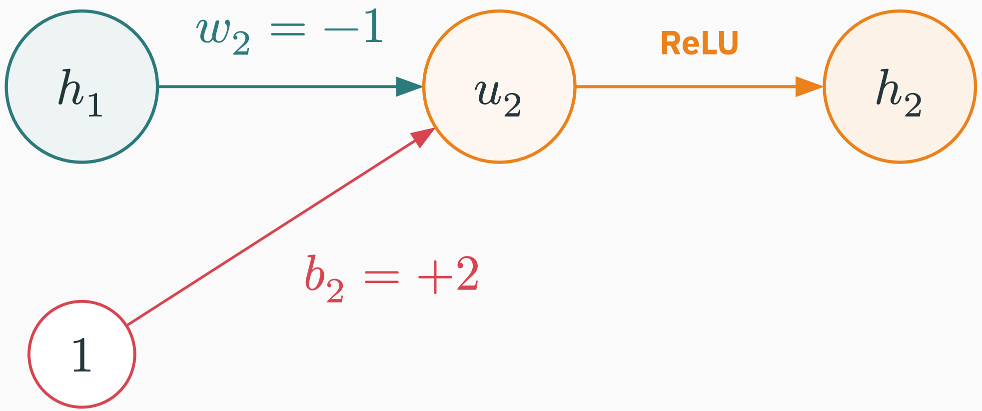
$$u_2 = 2 - h_1$$
$$h_2 = f_2(h_1) = \mathrm{ReLU}(u_2)$$

For $h_1 = 1.5$:

$$u_2 = 0.5 \to h_2 = 0.5$$

The second layer transforms the learned feature

VISUAL



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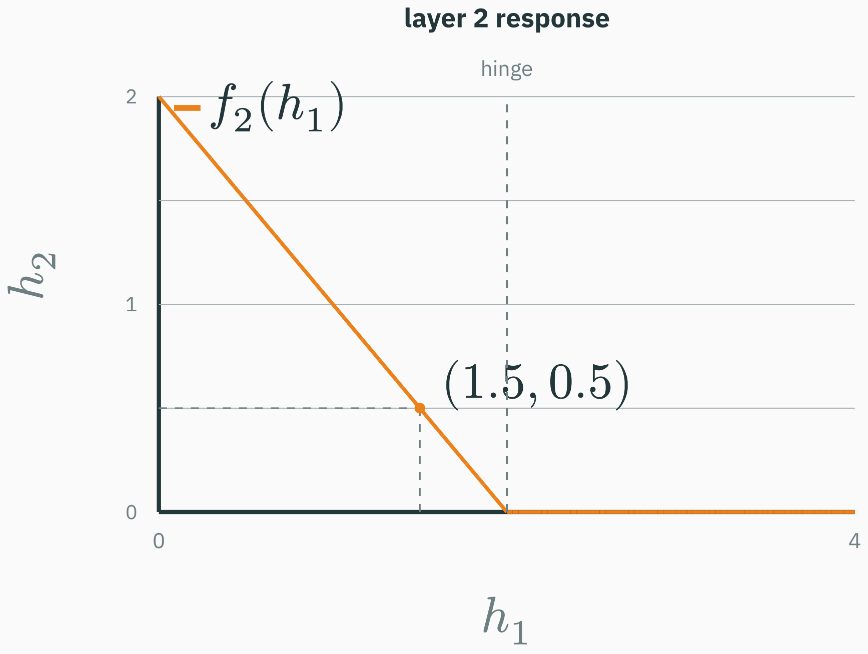

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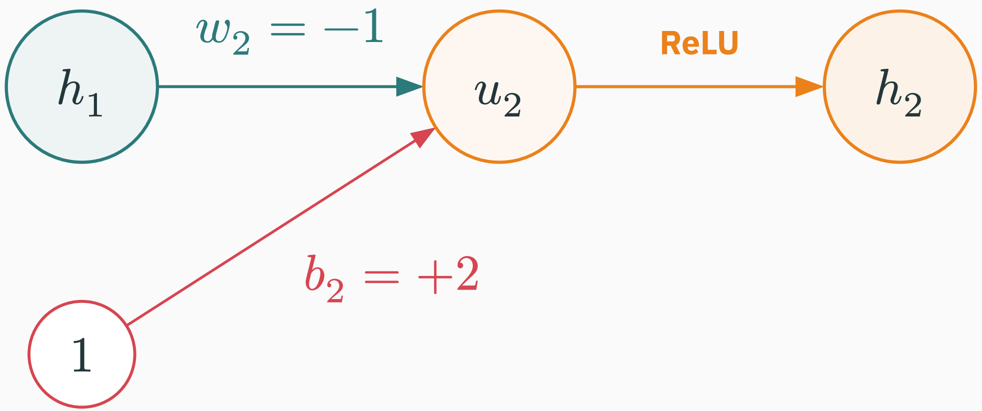
V VISUAL



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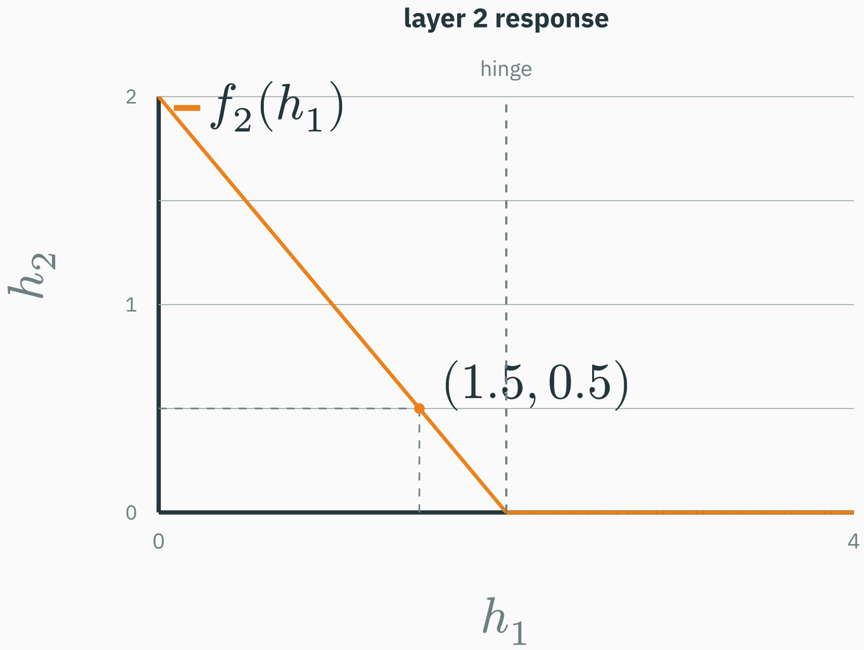


Layer 2 decreases as h_1 grows, then ReLU clips every negative result to zero.

Two layers create a clipped nonlinear ramp

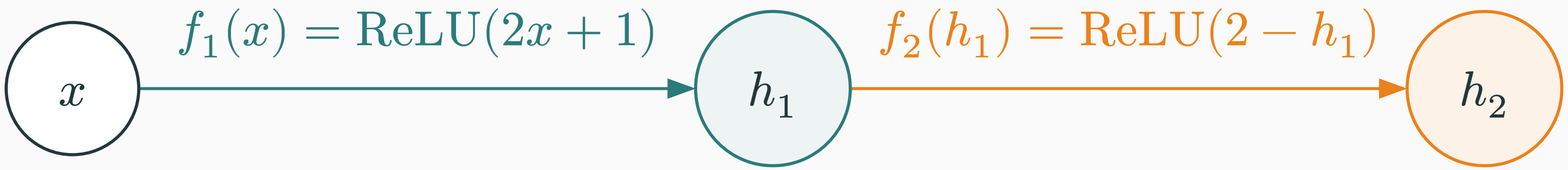
V VISUAL

$$x \xrightarrow{f_1(x) = \mathrm{ReLU}(2x+1)} h_1 \xrightarrow{f_2(h_1) = \mathrm{ReLU}(2-h_1)} h_2$$

$$h_2(x) = f_2(f_1(x)) = \mathrm{ReLU}(2 - \mathrm{ReLU}(2x+1))$$

Two layers create a clipped nonlinear ramp

V VISUAL



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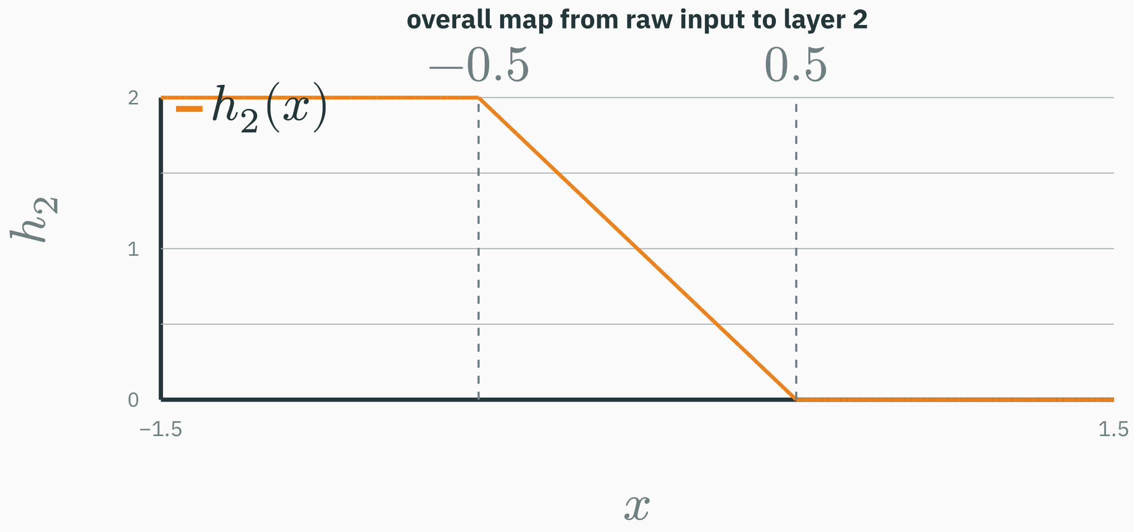

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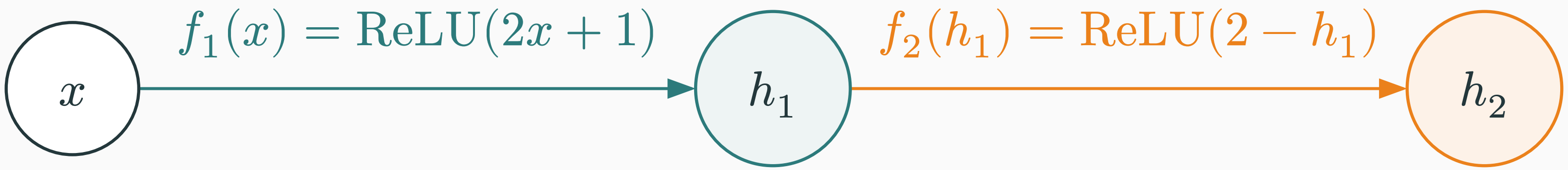

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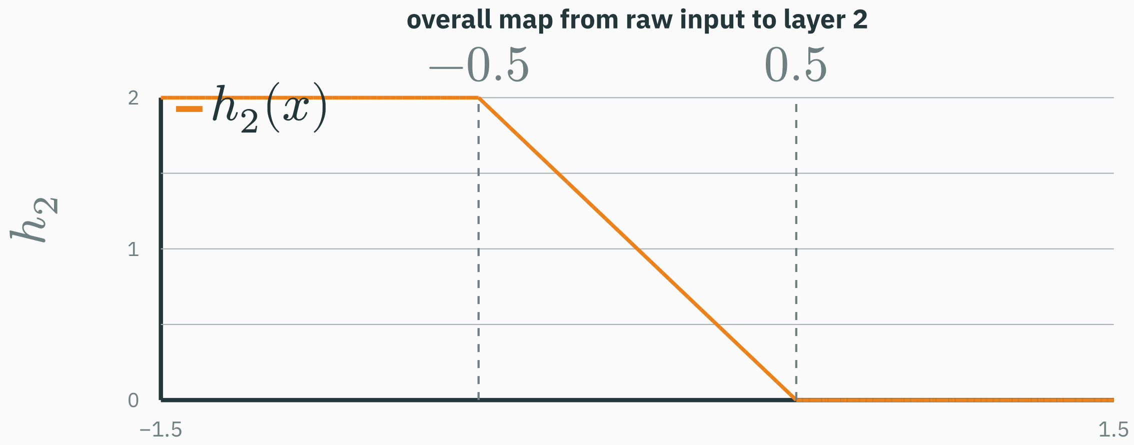
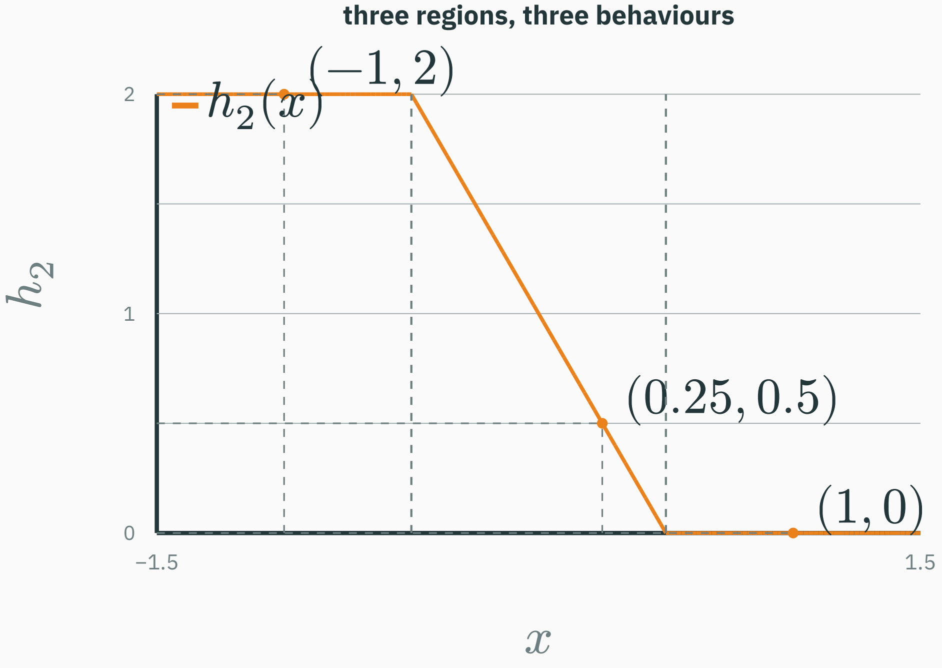


The first hinge turns on at $x = -0.5$; the second layer creates another bend at $x = 0.5$.

Three inputs reveal the complete composition

D DERIVATION



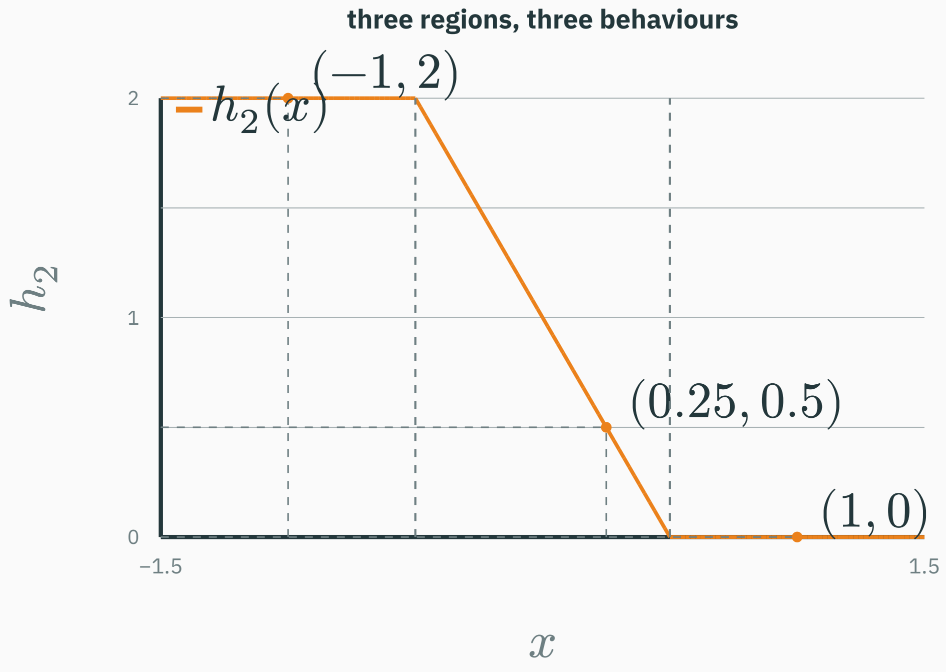
$x = -1$ $u_1 = -1 \to h_1 = 0$
$u_2 = 2 \to h_2 = 2$

$x = 0.25$ $u_1 = 1.5 \to h_1 = 1.5$
$u_2 = 0.5 \to h_2 = 0.5$

$x = 1$ $u_1 = 3 \to h_1 = 3$
$u_2 = -1 \to h_2 = 0$

Three inputs reveal the complete composition

D DERIVATION



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$u_2 = 0.5 \to h_2 = 0.5$

$x = 1$ $u_1 = 3 \to h_1 = 3$
$u_2 = -1 \to h_2 = 0$

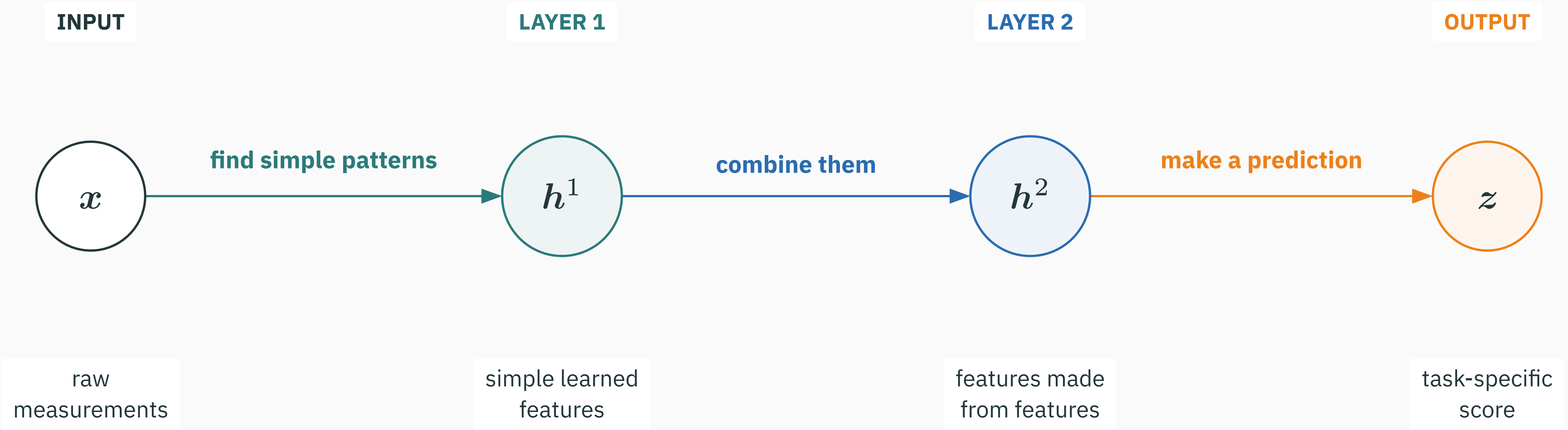
Depth means sequential reuse: layer 2 receives the learned value h_1 and transforms it again.

Depth builds features one stage at a time

V VISUAL

Each layer receives the representation produced immediately before it.

pixels $\rightarrow$ short strokes $\rightarrow$ closed loops $\rightarrow$ digit score



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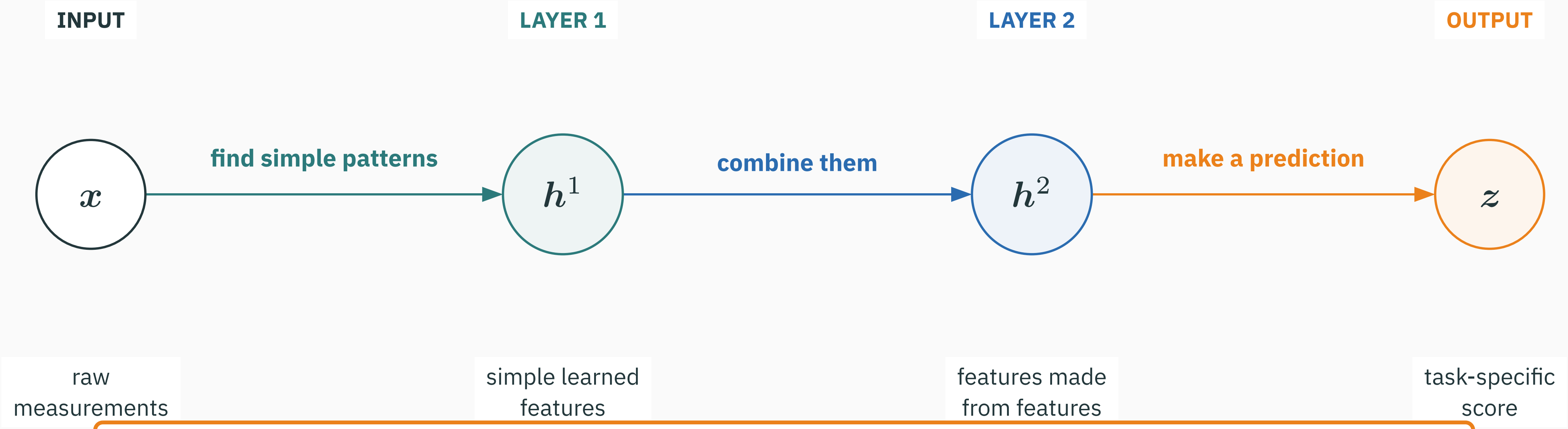
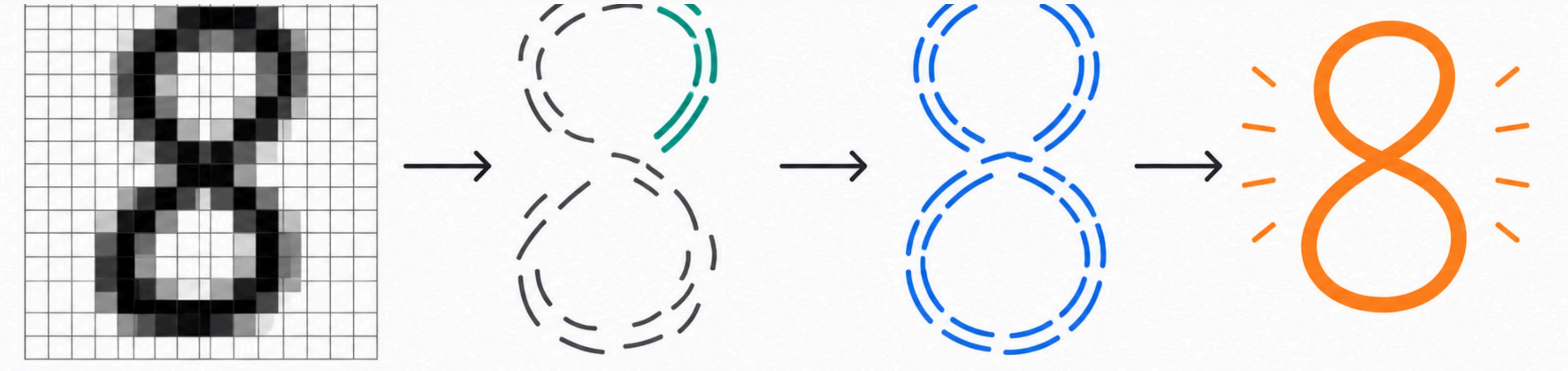


Later layers build on earlier features; they do not solve the whole task from raw input again.

Vision: depth builds strokes, then loops

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



INPUT
pixel intensities

LAYER 1
curved stroke

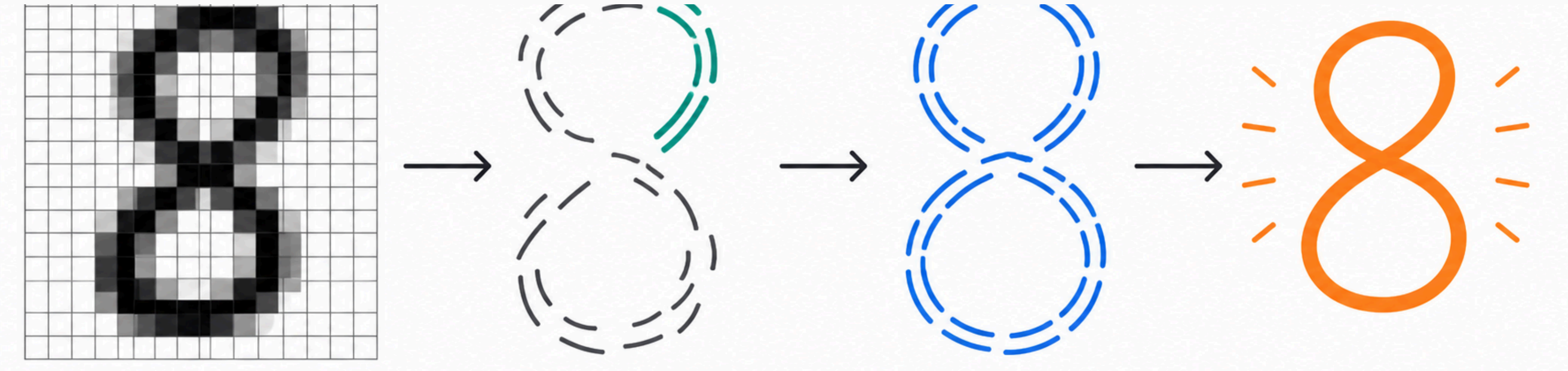
LAYER 2
two closed loops

OUTPUT
evidence for 8

Vision: depth builds strokes, then loops

V VISUAL

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



INPUT
pixel intensities

LAYER 1
curved stroke

LAYER 2
two closed loops

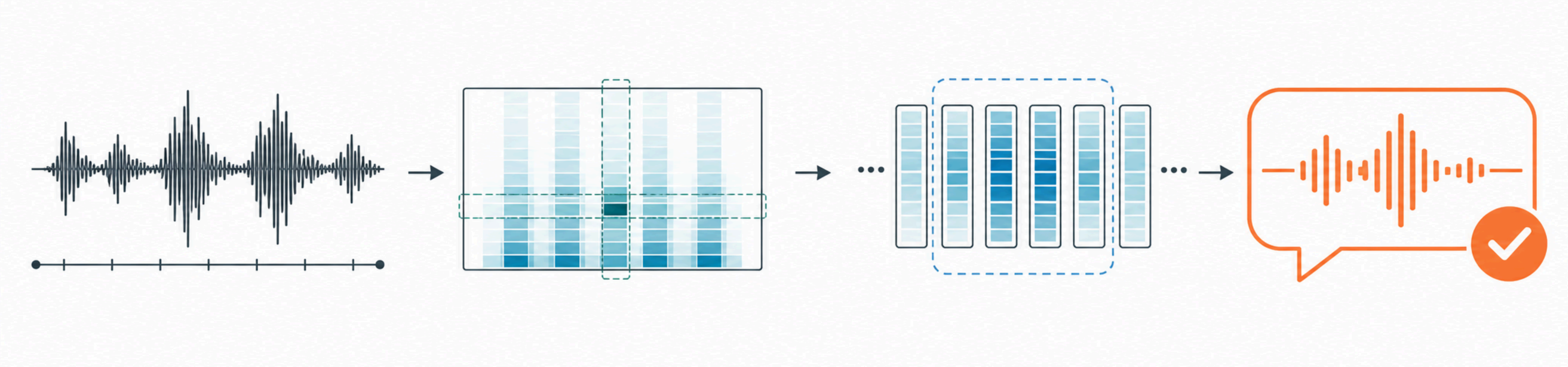
OUTPUT
evidence for 8

Early units respond to local edges; later units combine them into shapes that support the final class.

Audio: depth builds tones, then phonemes

V VISUAL

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



INPUT
waveform samples

LAYER 1
frequency + onset

LAYER 2
phoneme pattern

OUTPUT
word evidence

Audio: depth builds tones, then phonemes

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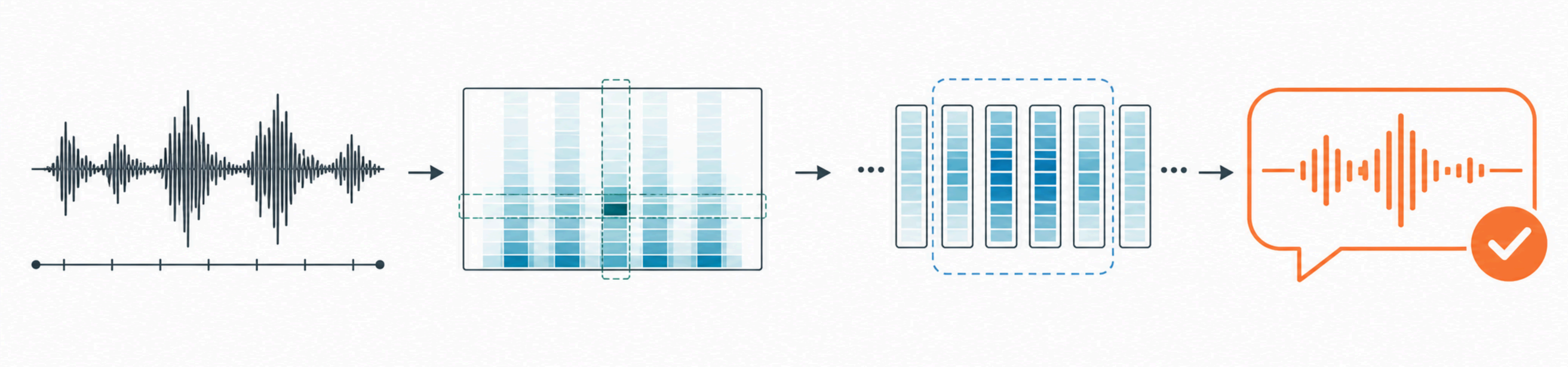

INPUT
waveform samples

LAYER 1
frequency + onset

LAYER 2
phoneme pattern

OUTPUT
word evidence

Early layers detect local frequency and timing cues; later layers combine them over longer time windows.

Text: depth combines words into phrase meaning

V VISUAL

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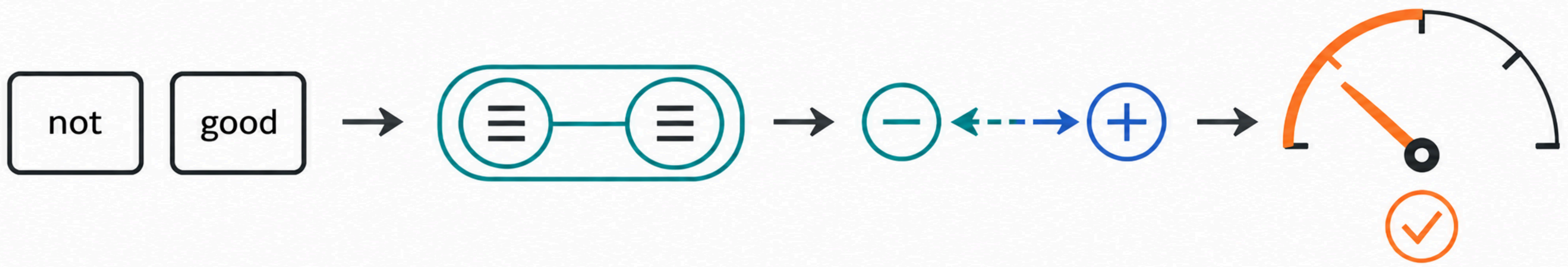


INPUT
tokens “not”, “good”

LAYER 1
ordered phrase

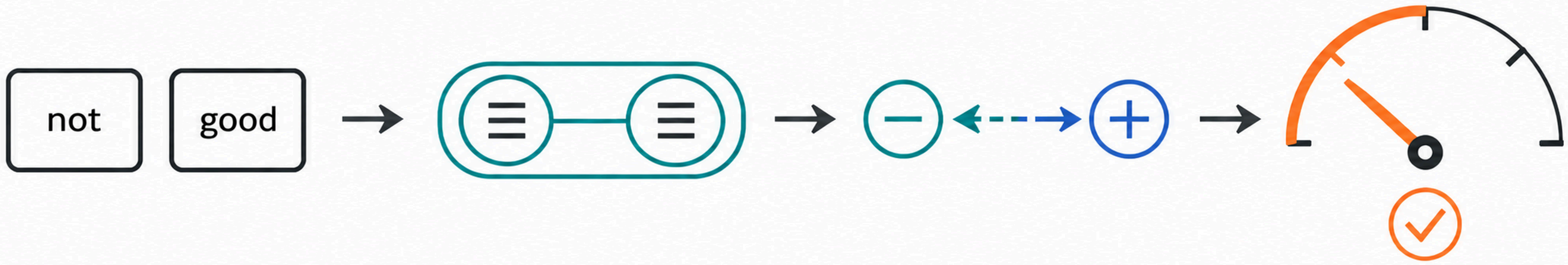
LAYER 2
negative evidence

OUTPUT
negative review

Text: depth combines words into phrase meaning

V VISUAL

CONCEPTUAL SCHEMATIC · illustrates a possible hierarchy; not measured model activations



INPUT
tokens “not”, “good”

LAYER 1
ordered phrase

LAYER 2
negative evidence

OUTPUT
negative review

A later unit can reverse the evidence from “good” after combining it with the preceding token “not”.

Choose width for variety; choose depth for composition

D DERIVATION

ADD WIDTH

- adds more features **in parallel**;
- helps when many patterns live at the same abstraction level;
- costs wider activations and weight matrices.

Example: add hinge locations to follow a detailed curve.

Choose width for variety; choose depth for composition

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ADD DEPTH

- adds more transformations **in sequence**;
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- costs longer paths and harder optimization.

Example: strokes $\to$ loops $\to$ digit identity.

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- adds more transformations **in sequence**;
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- costs longer paths and harder optimization.

Example: strokes $\to$ loops $\to$ digit identity.

There is no universally best shape. Compare models under a meaningful parameter, memory, and compute budget.

Choose width for variety; choose depth for composition

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Example: strokes $\to$ loops $\to$ digit identity.

Modern networks usually need both: enough width per stage and enough depth to compose stages.

Interactive: compare width and depth on the same task

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/mlp-decision-boundary

Train a linear classifier or a small MLP on **XOR / circles / spirals / moons**. Compare one versus two hidden layers, then adjust width while watching the learned boundary. Controls: dataset · linear/MLP · hidden width · one/two layers · activation · train/reset/resample.

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Hold the dataset fixed. First add width within one layer; then redistribute a similar number of hidden units across two layers. Which boundary appears sooner and which trains more reliably?

Interactive: compare width and depth on the same task

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Hold the dataset fixed. First add width within one layer; then redistribute a similar number of hidden units across two layers. Which boundary appears sooner and which trains more reliably?

The comparison is empirical: width and depth change both representational structure and optimization behaviour.

Connecting to deep-learning practice

Forward pass, step 1: flatten the image into a vector

V VISUAL

Start with a 2×2 toy image so that every number stays visible.

PIXEL GRID

$$\boldsymbol{X}_{\text{image}} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

four pixel intensities

read row by row $\longrightarrow$

INPUT VECTOR

$$\boldsymbol{x} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \in \mathbb{R}^4$$

the same four values

Forward pass, step 1: flatten the image into a vector

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REAL IMAGE

a 28×28 image becomes $\boldsymbol{x} \in \mathbb{R}^{784}$ because $28(28) = 784$

Forward pass, step 1: flatten the image into a vector

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the same four values

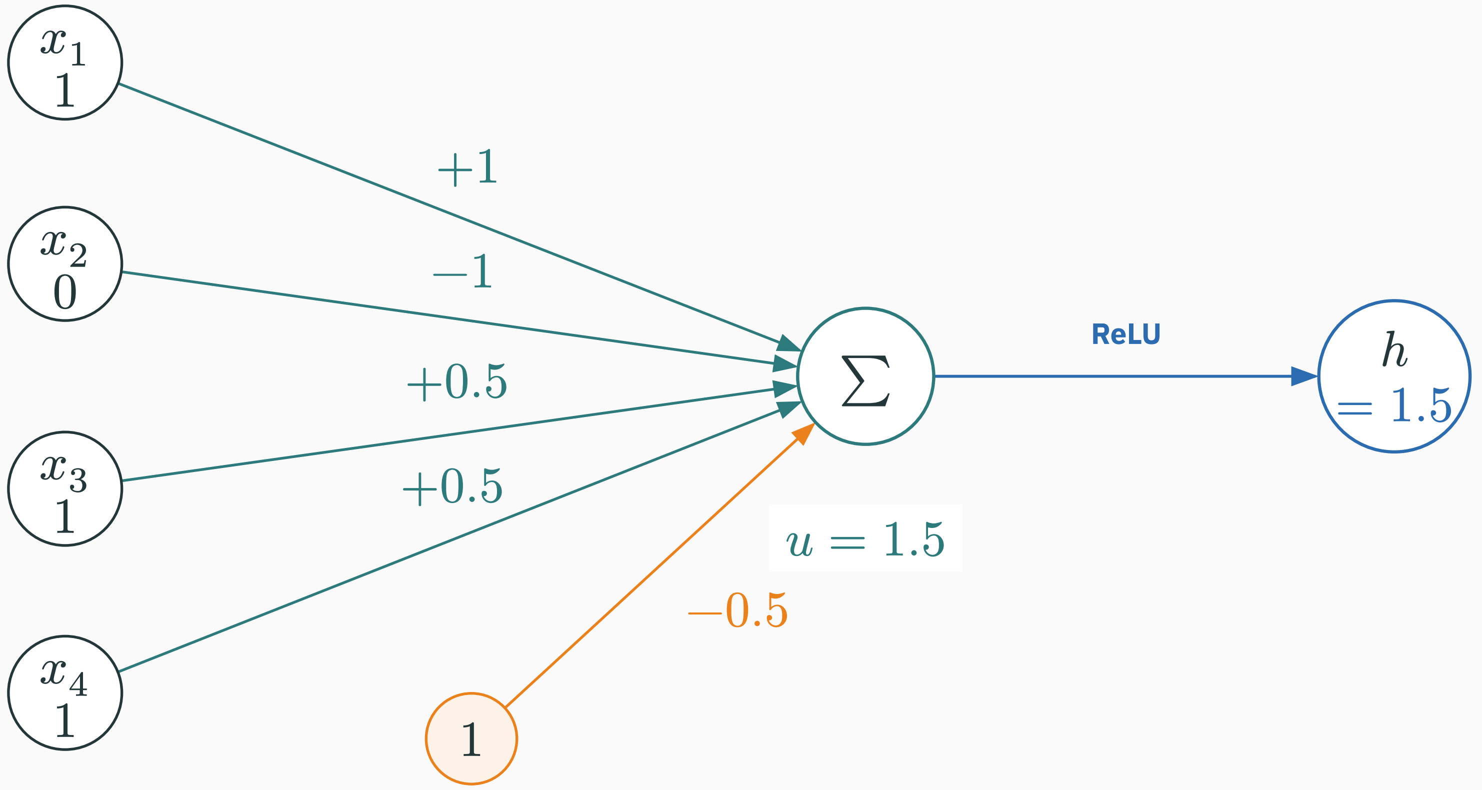
REAL IMAGE

a 28×28 image becomes $\boldsymbol{x} \in \mathbb{R}^{784}$ because $28(28) = 784$

Flattening changes the arrangement of the pixels—not their values or meaning.

Forward pass, step 2: one hidden unit tests one pattern

D DERIVATION



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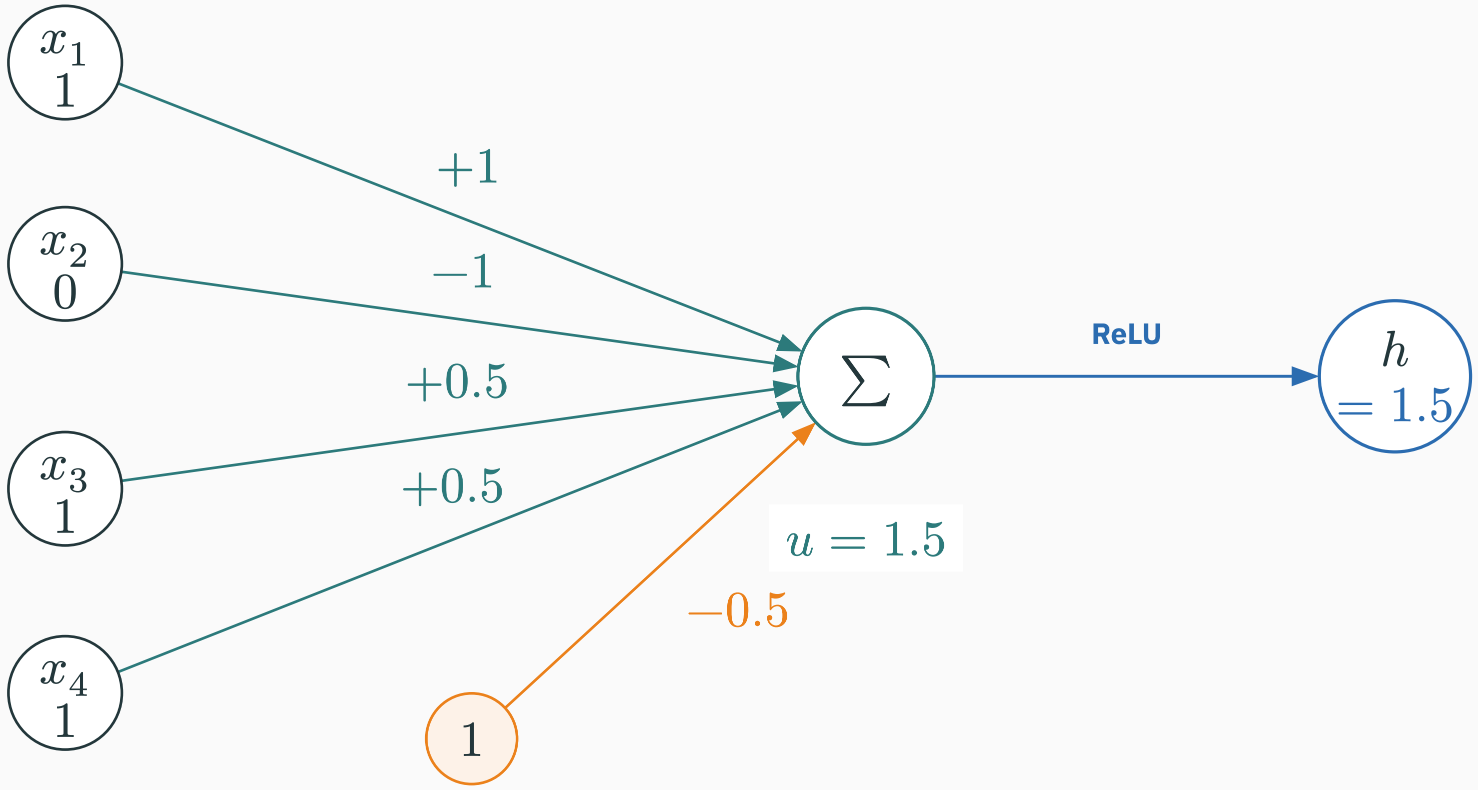
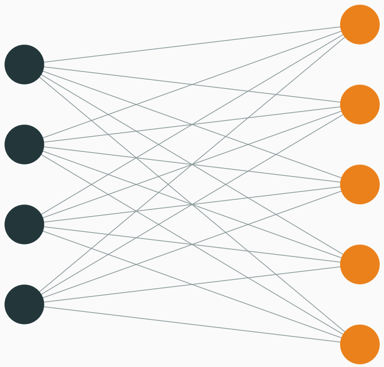


$u = 1(1) - 1(0) + 0.5(1) + 0.5(1) - 0.5 = 1.5$**, then** $h = \max(0, u) = 1.5$**. The edge weights define the pixel pattern this feature detects.**

Forward pass, step 3: 128 units test 128 patterns

V VISUAL



schematic: only a few nodes are drawn

Each receiving unit has its own row of weights:

$$u_j = \boldsymbol{w}_j^\top \boldsymbol{x} + b_j$$
$$h_j = \max(0, u_j)$$

$$\boldsymbol{u} = \boldsymbol{W}_1 \boldsymbol{x} + \boldsymbol{b}_1 \in \mathbb{R}^{128}$$
$$\boldsymbol{h} = \mathrm{ReLU}(\boldsymbol{u}) \in \mathbb{R}^{128}$$

Forward pass, step 3: 128 units test 128 patterns

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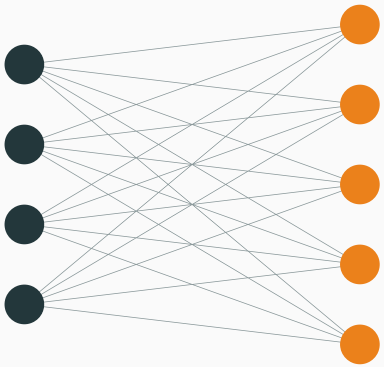

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schematic: only a few nodes are drawn

For one trained layer: $h_{17} = 2.1$ $\to$ curved-stroke unit **on**; $h_{83} = 0$ $\to$ another unit **off**.

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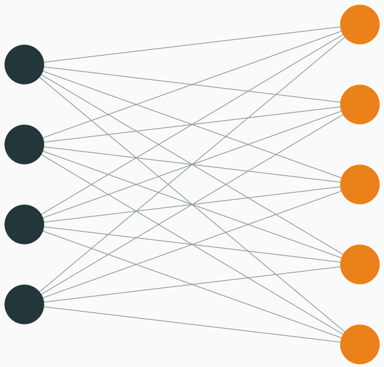

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The hidden vector is a list of 128 learned measurements of the same image.

Forward pass, step 4: hidden features become class scores

D DERIVATION

Use a three-class toy output so that every logit can be computed explicitly.

HIDDEN FEATURES

$$\boldsymbol{h} = (1.5, 0.5)^\top$$

$\rightarrow$

LINEAR OUTPUT LAYER

$$\boldsymbol{W}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad \boldsymbol{b}_2 = (0.5, 0, 0)^\top$$

$$\boldsymbol{z} = \boldsymbol{W}_2\boldsymbol{h} + \boldsymbol{b}_2 = (2.0, 0.5, -1.0)^\top$$

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$$z_A = 1(1.5) + 0(0.5) + 0.5 = 2.0$$

$$z_B = 0(1.5) + 1(0.5) + 0 = 0.5$$

$$z_C = -1(1.5) + 1(0.5) + 0 = -1.0$$

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Logits are unrestricted class scores: larger means preferred, but they are not probabilities yet.

Forward pass, step 5: softmax compares every score

D DERIVATION

LOGITS

$$\boldsymbol{z} = (2.0, 0.5, -1.0)^\top$$

$\rightarrow$

EXPONENTIATE AND ADD

$$\exp(\boldsymbol{z}) \approx (7.389, 1.649, 0.368)^\top$$

$$\text{total} = 9.406$$

$\rightarrow$

PROBABILITIES

$$\boldsymbol{p} \approx (0.786, 0.175, 0.039)^\top$$

$$\text{sum} = 1$$

Forward pass, step 5: softmax compares every score

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$$\text{sum} = 1$$

$$p_A = \frac{\exp(2.0)}{\exp(2.0) + \exp(0.5) + \exp(-1.0)} = \frac{7.389}{9.406} \approx 0.786$$

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$$p_A = \frac{\exp(2.0)}{\exp(2.0)+\exp(0.5)+\exp(-1.0)} = \frac{7.389}{9.406} \approx 0.786$$

Softmax preserves the score ordering while turning all three scores into one probability distribution.

Forward pass, step 6: the label selects the loss

D DERIVATION

Cross-entropy reads the probability assigned to the true class: $\mathcal{L} = -\log p_y$.

IF Y = A

$$\mathcal{L} = -\log(0.786) \approx 0.241$$

IF Y = B

$$\mathcal{L} = -\log(0.175) \approx 1.743$$

IF Y = C

$$\mathcal{L} = -\log(0.039) \approx 3.244$$

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High probability on the true class gives a small loss.

Confident probability on the wrong class gives a large loss.

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High probability on the true class gives a small loss.

Confident probability on the wrong class gives a large loss.

The forward pass ends with one scalar loss; it has evaluated the weights but has not updated them.

A minibatch stacks examples as rows

D DERIVATION

Apply the same hidden unit from step 2 to three inputs at once.

THREE INPUT ROWS

$$\boldsymbol{X} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

shape 3×4

$\rightarrow$

ONE FEATURE ACROSS THE BATCH

$$\boldsymbol{u} = \boldsymbol{X}\boldsymbol{w} + b\mathbf{1} = (1.5, -1, -0.5)^\top$$

$$\boldsymbol{h} = \mathrm{ReLU}(\boldsymbol{u}) = (1.5, 0, 0)^\top$$

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The same $\boldsymbol{w}$ and b are reused for row 1, row 2, and row 3.

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The same $\boldsymbol{w}$ and b are reused for row 1, row 2, and row 3.

Rows are different examples; columns are the features computed for those examples.

The full minibatch keeps the same network

D DERIVATION

For $n = 64$ images, all 128 hidden units and all 10 logits are computed together:

BATCH INPUT

$\boldsymbol{X} : 64 \times 784$

one image per row

$\rightarrow$

BATCH HIDDEN FEATURES

$\boldsymbol{H} = \mathrm{ReLU}(\boldsymbol{X}\boldsymbol{W}_1^\top + \mathbf{1}\boldsymbol{b}_1^\top)$

shape 64×128

$\rightarrow$

BATCH LOGITS

$\boldsymbol{Z} : 64 \times 10$

ten scores per row

The full minibatch keeps the same network

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BATCH LOGITS

$\boldsymbol{Z} : 64 \times 10$

ten scores per row

Shared across rows: weights and biases.

Recomputed for this batch: activations, probabilities, and losses.

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shape 64×128

$\rightarrow$

BATCH LOGITS

$\boldsymbol{Z} : 64 \times 10$

ten scores per row

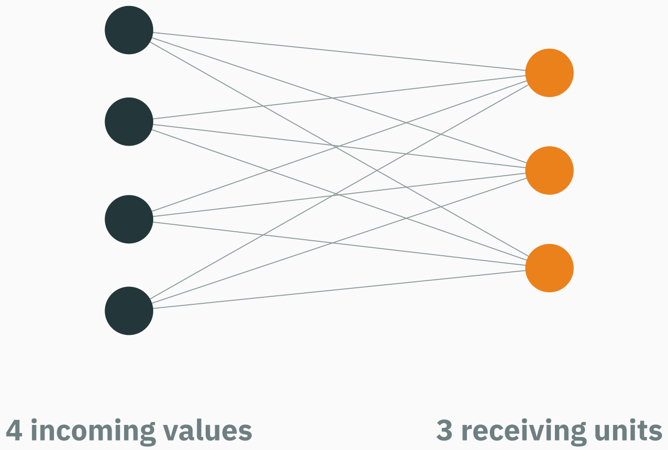
Shared across rows: weights and biases.

Recomputed for this batch: activations, probabilities, and losses.

Batching changes the array shapes, not the prediction rule: every row follows the same network.

A tiny layer reveals the parameter-count rule

V VISUAL



every connecting edge carries one weight

Weights

4 inputs $\times$ 3 units $= 12$

Biases

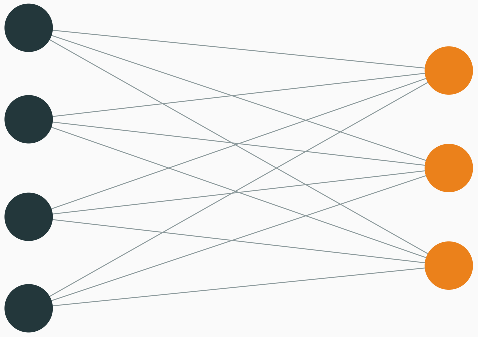
one per receiving unit $= 3$

Total

$12 + 3 = 15$ parameters

A tiny layer reveals the parameter-count rule

VISUAL



4 incoming values

3 receiving units

every connecting edge carries one weight

Weights

4 inputs $\times\ 3$ units $= 12$

Biases

one per receiving unit $= 3$

Total

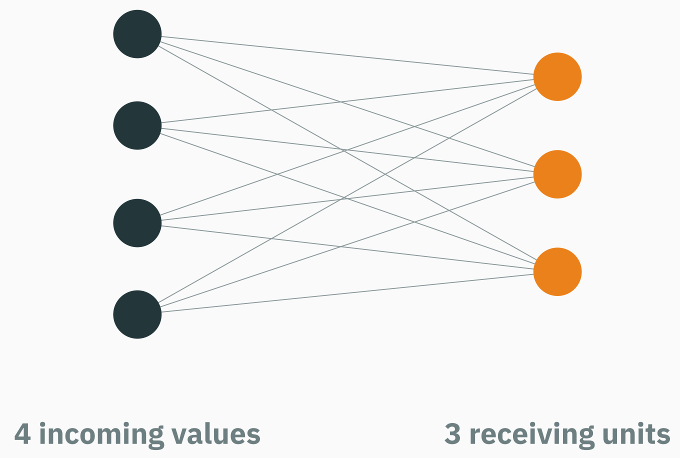
$12 + 3 = 15$ parameters

GENERAL RULE

$$(\text{output width} \times \text{input width}) + \text{output width}$$

A tiny layer reveals the parameter-count rule

V VISUAL



every connecting edge carries one weight

Weights

4 inputs $\times$ 3 units $= 12$

Biases

one per receiving unit $= 3$

Total

$12 + 3 = 15$ parameters

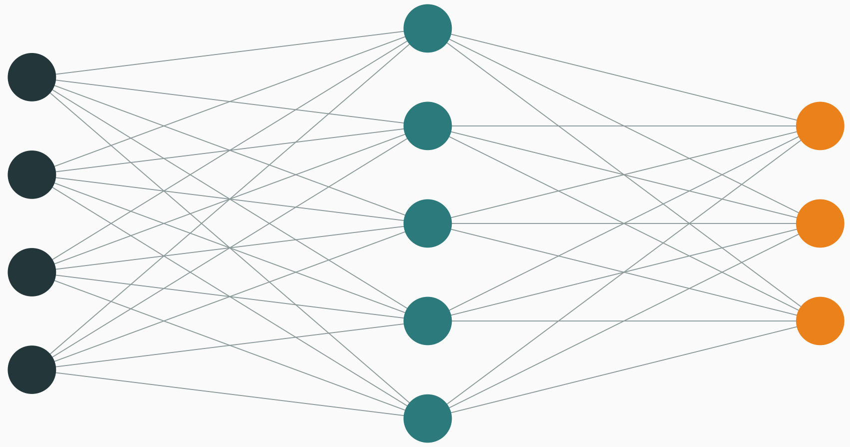
GENERAL RULE

$$(\text{output width} \times \text{input width}) + \text{output width}$$

Count the arrows, then add one bias for every receiving node.

Apply the same counting rule to both affine layers

D DERIVATION



784 **inputs** 128 **hidden units** 10 **outputs**

The drawing is schematic; the labels give the actual widths.

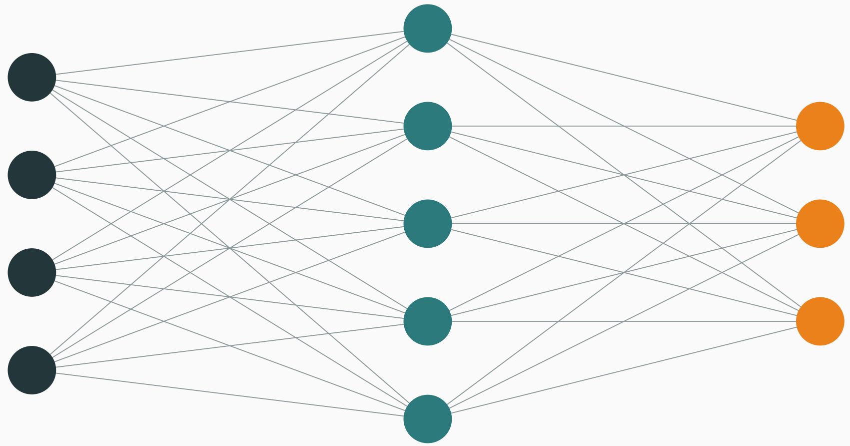
INPUT → HIDDEN

$$\boldsymbol{W}_1 : 128 \times 784 \quad \rightarrow \quad 128(784) = 100{,}352$$
$$\boldsymbol{b}_1 \in \mathbb{R}^{128} \quad \rightarrow \quad 128$$

Subtotal: $100{,}352 + 128 = 100{,}480$

Apply the same counting rule to both affine layers

D DERIVATION



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INPUT $\to$ HIDDEN

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$$\boldsymbol{b}_1 \in \mathbb{R}^{128} \quad \to \quad 128$$

Subtotal: $100{,}352 + 128 = 100{,}480$

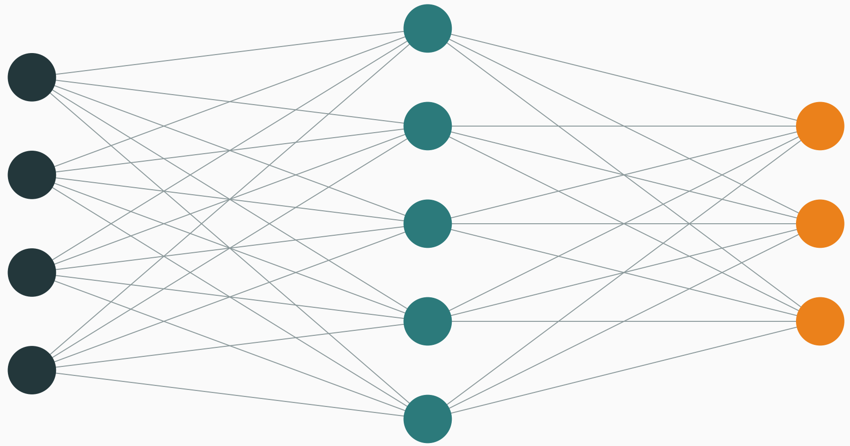
HIDDEN $\to$ OUTPUT

$$\boldsymbol{W}_2 : 10 \times 128 \quad \to \quad 10(128) = 1{,}280$$
$$\boldsymbol{b}_2 \in \mathbb{R}^{10} \quad \to \quad 10$$

Subtotal: $1{,}280 + 10 = 1{,}290$

Apply the same counting rule to both affine layers

D DERIVATION



The drawing is schematic; the labels give the actual widths.

INPUT → HIDDEN

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$$\boldsymbol{b}_1 \in \mathbb{R}^{128} \quad \to \quad 128$$

Subtotal: $100{,}352 + 128 = 100{,}480$

HIDDEN → OUTPUT

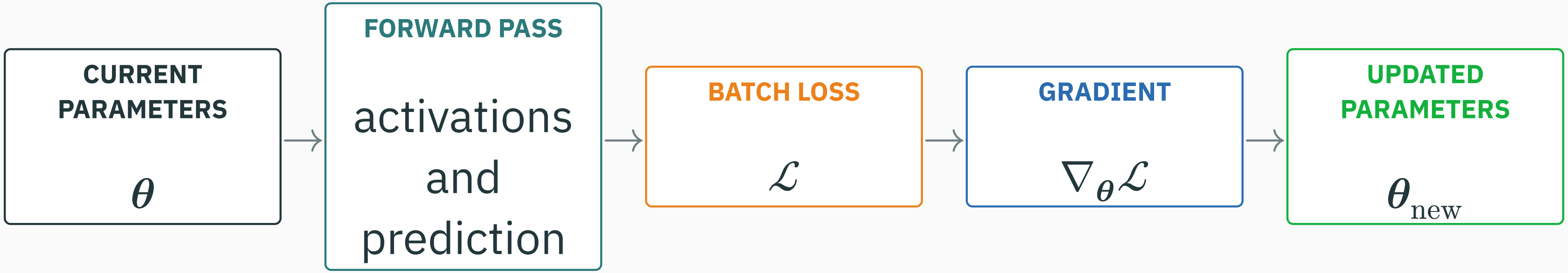
$$\boldsymbol{W}_2 : 10 \times 128 \quad \to \quad 10(128) = 1{,}280$$
$$\boldsymbol{b}_2 \in \mathbb{R}^{10} \quad \to \quad 10$$

Subtotal: $1{,}280 + 10 = 1{,}290$

Total trainable parameters: $100{,}480 + 1{,}290 = 101{,}770$.

Training changes parameters, not the architecture

D DERIVATION



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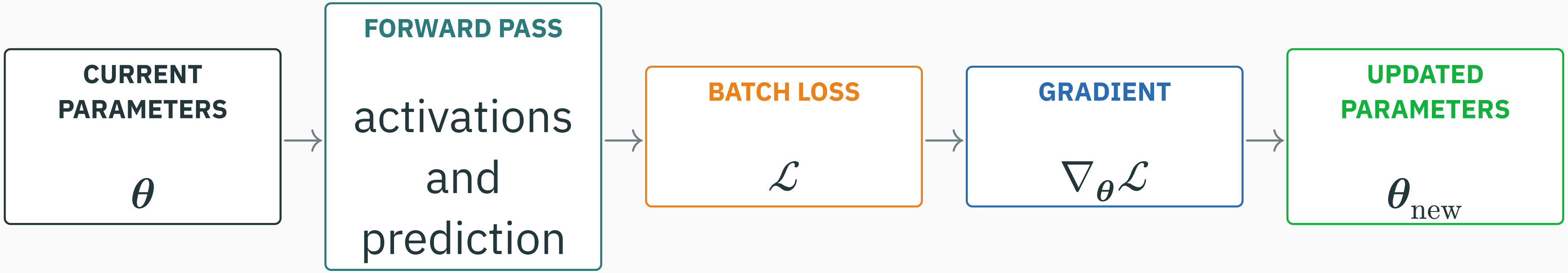


$$\boldsymbol{\theta}_{\text{new}} = \boldsymbol{\theta}_{\text{old}} - \eta \nabla_{\boldsymbol{\theta}}\mathcal{L}$$

Training changes parameters, not the architecture

D DERIVATION

CURRENT PARAMETERS $\boldsymbol{\theta}$ → **FORWARD PASS** activations and prediction → **BATCH LOSS** $\mathcal{L}$ → **GRADIENT** $\nabla_{\boldsymbol{\theta}}\mathcal{L}$ → **UPDATED PARAMETERS** $\boldsymbol{\theta}_{\text{new}}$

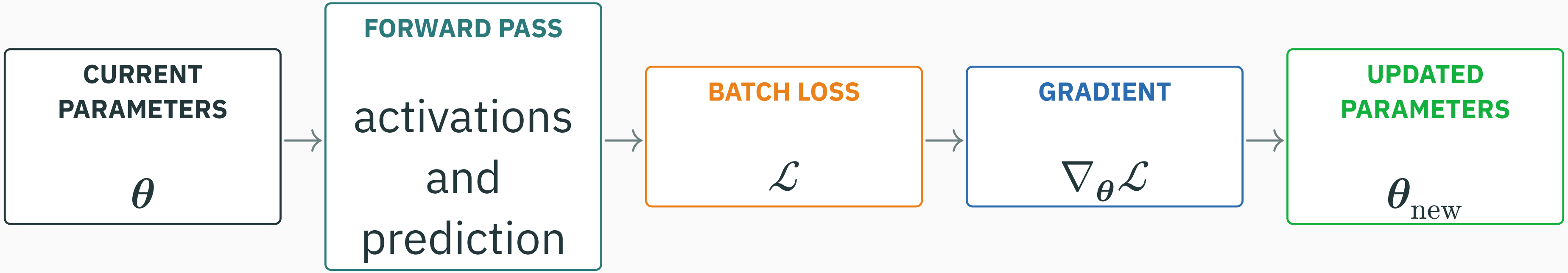
$$\boldsymbol{\theta}_{\text{new}} = \boldsymbol{\theta}_{\text{old}} - \eta \nabla_{\boldsymbol{\theta}}\mathcal{L}$$

Fixed: layer sizes, connectivity pattern, and activation choices.

Changed: the entries of weight matrices and bias vectors.

Training changes parameters, not the architecture

D DERIVATION



$$\boldsymbol{\theta}_{\text{new}} = \boldsymbol{\theta}_{\text{old}} - \eta \nabla_{\boldsymbol{\theta}}\mathcal{L}$$

Fixed: layer sizes, connectivity pattern, and activation choices.

Changed: the entries of weight matrices and bias vectors.

Forward propagation produces activations and loss; learning uses that loss to update the shared parameters.

One numerical parameter update

D DERIVATION

CURRENT VALUE

$$w_{\text{old}} = 0.8$$

→

GRADIENT AND STEP SIZE

$$\frac{\partial \mathcal{L}}{\partial w} = -0.3$$

$$\eta = 0.1$$

→

UPDATED VALUE

$$w_{\text{new}} = 0.8 - 0.1(-0.3) = 0.83$$

One numerical parameter update

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$\rightarrow$

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The negative gradient says increasing w locally decreases the loss, so the update moves w upward.

One numerical parameter update

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Lecture 3 will explain how backpropagation computes every component of $\nabla_\theta \mathcal{L}$ efficiently.

One numerical parameter update

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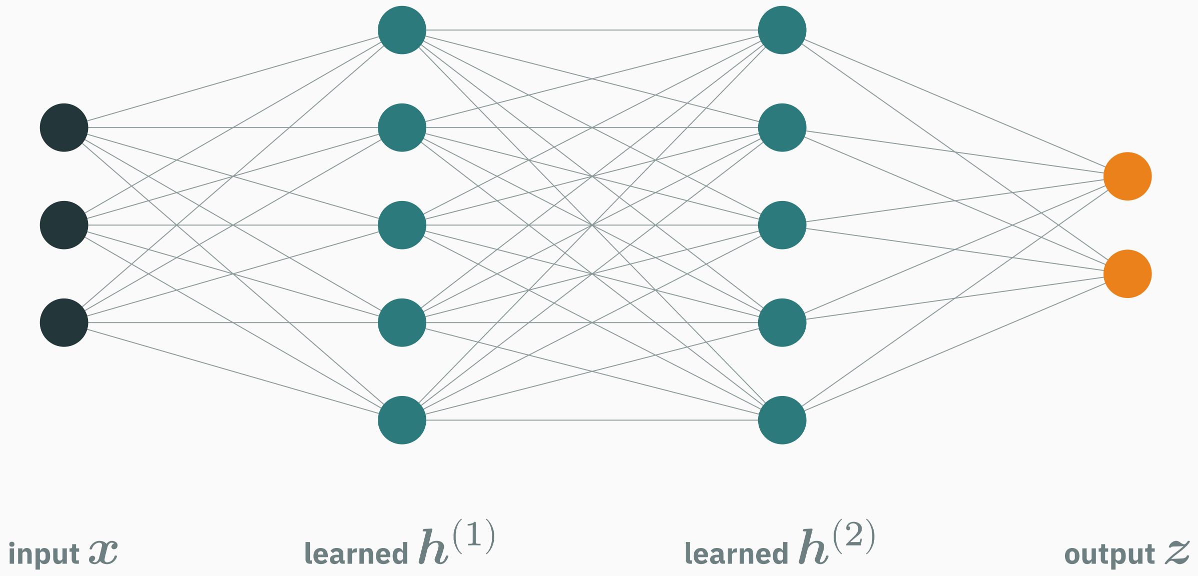
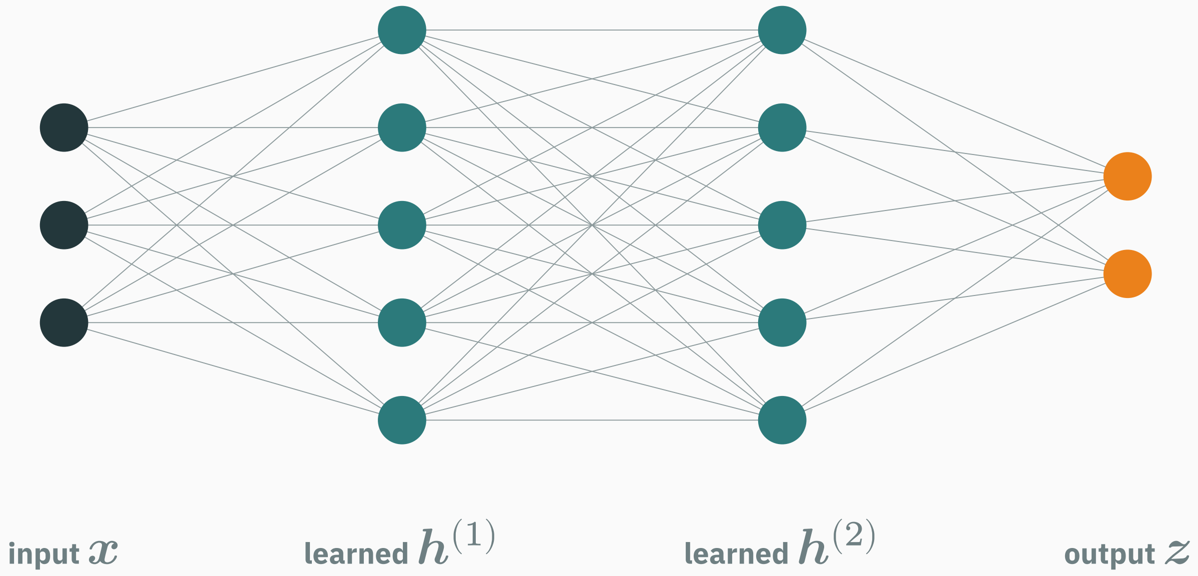
The forward pass evaluates the current network; the update changes the network for the next pass.

The decisive change is the learned representation



The final scores still feed the familiar Lecture 1 output model and loss.

The decisive change is the learned representation



The final scores still feed the familiar Lecture 1 output model and loss.

The output loss can stay the same while hidden layers make the representation nonlinear.

We now have $x \to f_{\boldsymbol{\theta}}(x) \to \mathcal{L}$.

Lecture 3: how do we compute $\nabla_{\boldsymbol{\theta}}\mathcal{L}$ efficiently?

computation graphs · the chain rule · backpropagation · automatic differentiation